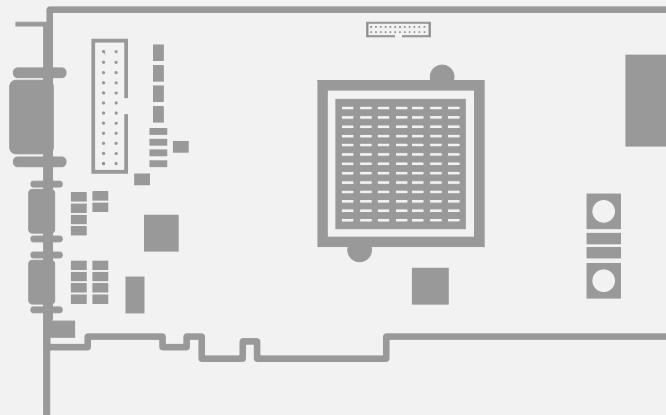


Grablink

Grablink Functional Guide

1191 Grablink Value
1621 Grablink Express
1622 Grablink Full

1623 Grablink DualBase
1624 Grablink Base
1626 Grablink Full XR



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1. About This Document

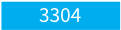
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1.1. Document Scope

This document describes the **functions** of all the products of the Grablink series together with their related products.

Grablink products

Product	S/N Prefix	Icon
1191 Grablink Value	GLV	
1621 Grablink Express	GEX	
1622 Grablink Full	FM1	
1623 Grablink DualBase	GDB	
1624 Grablink Base	GBA	
1626 Grablink Full XR	FXR	

Product	S/N Prefix	Icon
1625 DB25F I/O Adapter Cable	DBC	
3304 HD26F I/O Adapter Cable		

**NOTE**

The S/N prefix is a 3-letter string at the beginning of the card serial number.

**NOTE**

Icons are used in this document for tagging titles of card-specific content.

1.2. Document Revision History

Date	Version	Description
2019-07-24	17.0	MultiCam 6.17
2019-03-20	16.0	MultiCam 6.16
2018-08-10	15.1	First edition in MadCap Flare

1.3. Document Changes

MultiCam 6.17

New or revised topics:

- ["Configurations for Monochrome Pixels" on page 57](#)
- ["Configurations for Bayer CFA Pixels" on page 59](#)
- ["Configurations for RGB Pixels" on page 62](#)

2. Camera Link Interface

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2.1. Camera Link Configurations

Camera Link 2.1 Configurations

Configuration	Channel Link count (names)	Connectors count	Bit count	Ports count(names)
Lite	1 (X)	1	10	2 (A, B)*
Base	1 (X)	1	24	3 (A, B, C)
Medium	2 (X, Y)	2	48	6 (A to F)
Full	3 (X, Y, Z)	2	64	8 (A to H)
72-bit	3 (X, Y, Z)	2	72	9 (A to I)
80-bit	3 (X, Y, Z)	2	80	10 (A to J)


NOTE

(*) Up to 10 bits only

Supported Camera Link Configurations vs. Grablink Product

Configuration	Value	Express	Base	DualBase	Full	FullXR
Lite	-	-	OK	OK	-	-
Base	OK	OK	OK	OK	OK	OK
Medium	-	-	-	-	OK	OK
Full	-	-	-	-	OK	OK
72-bit	-	-	-	-	OK	OK
80-bit	-	-	-	-	OK	OK

2.2. Enable Signals

The Camera Link standard defines 4 enable signals:

- FVAL (Frame Valid) is defined HIGH for valid lines.
- LVAL (Line Valid) is defined HIGH for valid pixels.
- DVAL (Data Valid) is defined HIGH when data is valid.
- Spare has been defined for future use.

The Camera Link standard requires that enable signals are provided on Channel Link chips as follows:

Camera Link Configuration	FVAL	LVAL	DVAL	Spare
Lite	X	X	-	-
Base	X	X	X	X
Medium	X and Y	X and Y	X and Y	X and Y
Full	X, Y and Z	X, Y and Z	X, Y and Z	X, Y and Z
72-bit	X, Y and Z	X, Y and Z	X, Y and Z	X, Y and Z
80-bit	X only	X, Y and Z	-	-

Grablink Usage

On Grablink boards, the enable signals are used differently according to the type of acquisition:

Area-scan Image Acquisition

- The rising edge of the FVAL signal is used as a "Start-of-Frame"
- The rising edge of the LVAL signal is used as a "Start-of-Line"
- The DVAL signal can optionally be used as a clock qualifier
- The Spare signal is unused

Line-scan Image Acquisition

- The FVAL signal is unused
- The rising edge of the LVAL signal is used as a "Start-of-Line"
- The DVAL signal can optionally be used as a clock qualifier
- The Spare signal is unused

Raw Data Acquisition

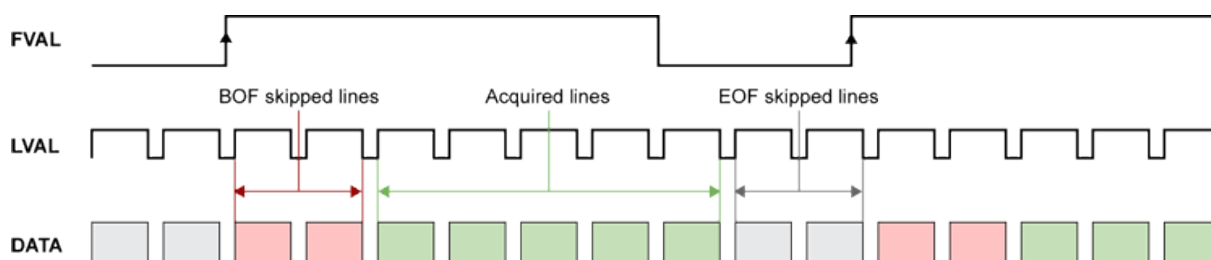
- The FVAL signal is used as a "Frame cover" signal surrounding the data to acquire
- The LVAL signal is unused
- The DVAL signal can optionally be used as a clock qualifier
- The Spare signal is unused

Enable Signals Timing Diagrams

FVAL Timing for Area-Scan Acquisition

When the Grablink board is configured for image acquisition from area-scan cameras:

- The rising edge of FVAL is used as a start of frame signal.
- The falling edge of FVAL is ignored.



FVAL timing diagram (area-scan acquisition)

If the acquisition trigger is armed when FVAL raises, the board:

1. Skips a pre-defined amount (**VsyncAft_Ln**) of video data lines at begin of frame (BOF),
2. Acquires a pre-defined amount (**Vactive_Ln**) of video data lines,
3. Terminates the image acquisition

The subsequent video data lines are all skipped until a new **FVAL** rising edge occurs.

Applicable limits on FVAL timing for Area-Scan Acquisition

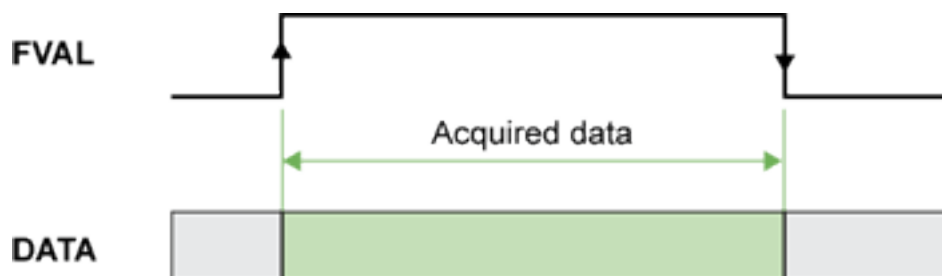
Parameter	Min	Typ	Max	Unit
FVAL high duration	64	Lines with active video	N/A	Clock cycle
FVAL low duration	64	Lines with blanked video	N/A	Clock cycle
FVAL rising edge to LVAL rising edge setup time	0	-	N/A	Clock cycle
Skipped lines at begin of frame (BOF)	0	0	255	LVAL cycle
Skipped lines at end of frame (EOF)	0	-	N/A	LVAL cycle

Refer to "[Camera Active Area Properties](#)" on page 42 for constraints on the value of **Vactive_Ln**.

FVAL Timing for Raw Data Acquisition

When the Grablink board is configured for raw data acquisition:

- The rising edge of FVAL starts the data acquisition.
- The falling edge of FVAL stops the data acquisition.



FVAL timing diagram (area-scan acquisition mode)

If the acquisition trigger is armed when FVAL rises, the board acquires all data until FVAL goes down.

You can configure the frame grabber to use the DVAL signal as a clock qualifier but this feature is rarely used. By default, the configuration ignores the DVAL signal. When DVAL is enabled, the data acquisition is inhibited for clock cycles where DVAL = 0.

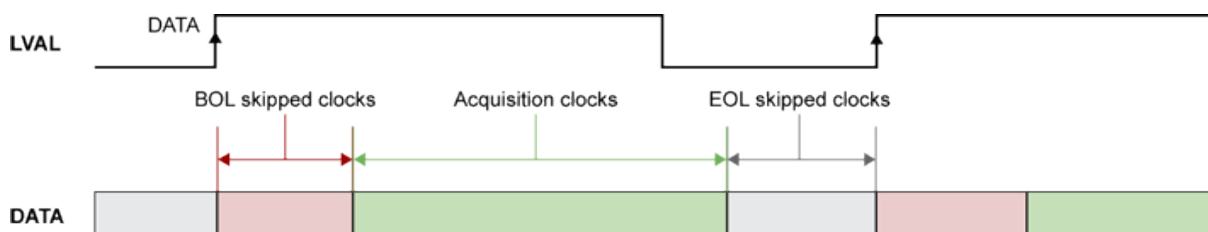
Applicable limits on FVAL timing for Raw Data Acquisition

Parameter	Min	Typ	Max	Unit
FVAL high duration	1	-	N/A	Clock cycle
FVAL low duration	1	-	N/A	Clock cycle

LVAL Timing Diagram

When the Grablink board is configured for image acquisition from an area-scan camera or from a line-scan camera:

- The rising edge of LVAL is actually the reference for the horizontal timing
- The falling edge of LVAL is ignored



LVAL timing diagram

If, at the rising edge of LVAL, the acquisition of the next line is enabled, the board

1. Skips a pre-defined amount (**HsyncAft_Tk**) of camera clock cycles at begin of line (BOL)
2. Acquires a pre-defined amount (**Hactive_Px**) pixel data
3. Terminates the line acquisition

The subsequent clock cycles are skipped until a new LVAL rising edge occurs.

You can configure the frame grabber to use the DVAL signal as a clock qualifier but this feature is rarely used. Therefore, the default configuration ignores the DVAL signal.

Applicable limits on LVAL timing for Area-Scan and Line-Scan Acquisitions

Parameter	Min	Typ	Max	Unit
LVAL high duration	1	# clocks with active video	N/A	Qualified clock cycle
LVAL low duration	1	# clocks with blanked video	N/A	Qualified clock cycle
Skipped clocks at begin of line (BOL)	0	0	255	Qualified clock cycle
Skipped clocks at end of line (EOL)	0	-	N/A	Qualified clock cycle
EOL skipped clocks	Others: 0	-	N/A	Qualified clock cycle

Refer to "Camera Active Area Properties" on page 42 for constraints on the value of **Hactive_Px**.

A qualified clock cycle is defined as:

- Any clock cycle when the frame grabber is configured to ignore DVAL.
- A clock cycle with DVAL = 1 when the frame grabber is configured to use DVAL.

When DVAL is enabled:

- The data acquisition is inhibited for clock cycles where DVAL = 0. Only the data corresponding to DVAL = 1 are effectively acquired.
- The BOL skipped clock counter is not incremented for clock cycles where DVAL = 0. In other words, **HsyncAft_Tk** specifies the amount number of "clock cycles with DVAL = 1" to skip at begin of line.

Enable Signals Configuration

The enumerated parameter **FvalMode** specifies the usage of the **FVAL** downstream signal; possible values are:

Value	Meaning
FA	The rising edge of the FVAL signal is used as a Start of Frame; the falling edge is irrelevant.
FN	The FVAL signal is ignored by the board.
FC	Raw grabbing; start/stop recording on FVAL rising/falling edges

The enumerated parameter **LvalMode** specifies the usage of the **LVAL** downstream signal; possible values are:

Value	Meaning
LA	The rising edge of the LVAL signal is used as a Start of Line; the falling edge is irrelevant.
LN	The LVAL signal is ignored by the board.

The enumerated parameter **DvalMode** specifies the usage of the **DVAL** downstream signal; possible values are:

Value	Meaning
DG	The DVAL signal is a clock qualifier, only the data transmitted during clock cycles with DVAL = 1 are captured by the board.
DN	The DVAL signal is ignored by the board.

The MultiCam channel must be configured for the appropriate camera operation modes by assigning the appropriate values to the **FvalMode**, **LvalMode**, and **DvalMode** parameters.

Configuring Enable Signals for Area-Scan Cameras

The following combinations of values of **FvalMode**, **LvalMode**, and **DvalMode** parameters are allowed for image acquisition from area-scan cameras (**Imaging = AREA**):

- **FA-LA-DN** and **FA-LA-DG** are the recommended configurations for capturing **fixed size images**. The leading (rising) edges of the FVAL and LVAL signals are used as vertical and horizontal synchronization events. Their trailing (falling) edges are ignored. The DVAL can optionally be used as a DATA qualifier.
- **FC-LA-DN** and **FC-LA-DG** are the recommended configurations for cameras delivering images having a **variable amount of fixed size lines**. The FVAL signal is used as a "Frame Cover" signal that surrounds the lines to be acquired. The leading (rising) edge of the LVAL signal is used as a horizontal synchronization event. Its trailing edge is ignored. The DVAL can optionally be used as a DATA qualifier.

Configuring Enable Signals for Line-Scan Cameras

The following combinations of values of **FvalMode**, **LvalMode**, and **DvalMode** parameters are recommended for image acquisition from line-scan (**Imaging = LINE**) and TDI line-scan cameras (**Imaging = TDI**):

- **FN-LA-DN** and **FN-LA-DG** are the recommended configurations for capturing fixed size lines. The FVAL signal is ignored. The leading (rising) edge of the LVAL signals is used as a start-of-line synchronization events. The trailing (falling) edge of LVAL is ignored. The DVAL can optionally be used as a DATA qualifier.

Configuring Enable Signals for Raw Data Acquisition

The following combinations of values of **FvalMode**, **LvalMode**, and **DvalMode** parameters are recommended for raw data acquisition from digital devices:

- **FC-LN-DN** and **FC-LN-DG** are the recommended configurations for digital devices delivering raw data or camera devices delivering irregularly structured images. The FVAL signal is used as a "Frame Cover" signal that surrounds the data to be acquired. The LVAL signal is ignored. The DVAL can optionally be used as a DATA qualifier.

2.3. Video Data Signals

Bit Assignments

Bit Assignments

Grablink products comply with the following bit assignment tables of section 4 of the Camera Link 2.1 Specification:

- Table 4-1: 8-Bit Modes, Base/Medium/Full
- Table 4-2: 8-Bit Modes, 80-bit
- Table 4-3: 10-Bit Modes, Base/Medium/Full
- Table 4-4: 10-Bit Modes, 80-bit
- Table 4-5: 12-Bit Modes, Base/Medium/Full
- Table 4-6: 14-Bit Modes, Base/Medium/Full/72-bit
- Table 4-7: 16-Bit Modes, Base/Medium/Full
- Table 4-8: 16-Bit Mode, 80-bit
- Table 4-9: Lite Modes

2.4. Upstream Control Lines

According to the Camera Link standard, four LVDS signals are reserved for general-purpose camera control.

They are defined as camera inputs and frame grabber outputs. Camera manufacturers can define these signals to meet their needs for a particular product. The signals are:

- Camera Control 1 (CC1)
- Camera Control 2 (CC2)
- Camera Control 3 (CC3)
- Camera Control 4 (CC4)

Each control line can be configured as:

Reset signal

A transition (either rising or falling edge) on this line resets the camera. This action initiates either a new exposure/readout, or a readout of a frame for an area-scan camera or a line for a line-scan camera.

Expose control signal

The leading edge (either rising or falling edge) on this line initiates a new exposure.

The trailing edge terminates the exposure and initiates the readout of a frame for an area-scan camera or a line for a line-scan camera.

The pulse width is actually the exposure time.

GPIO

Usually, CC1 is used for reset or expose signal for asynchronous reset cameras.

This board does not provide HDRIVE, VDRIVE signals. Usually, Camera Link cameras do not need genlocking.

2.5. Serial Communication

This board supports the asynchronous full-duplex serial communication between the camera and the frame grabber.

Serial Link for the Lite Camera Link Configuration

For cameras using the Lite configuration, the downstream serial communication link doesn't use a dedicated line pair but, instead, is embedded in the Channel Link.

Serial Link for the Base, Medium, Full, or 80-bit Camera Link Configurations

For cameras using the Base, Medium, Full, or 80-bit configurations, two differential line pairs of the first camera cable are dedicated to the serial communication, one for each direction:

- SerTFG— Downstream (camera to frame grabber) serial communication link
- SerTC— Upstream (frame grabber to camera) serial communication link

Serial Link Functionalities

The application software controls the serial communication channel through the standardized API defined by the Camera Link standard.

Alternatively, it can also be controlled using virtual COM ports. Therefore, the application must set the appropriate values to the parameters **SerialControlA** and **SerialControlB** respectively.

Supported baud rates

600, 1200, 1800, 2400, 3600, 4800, 7200, 9600 (Default), 14400, 19200, 28800, 38400, 57600, 115200 and 230400.

3. Cameras

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3.1. Camera Classes

In MultiCam, cameras are classified according to four Channel parameters: **Imaging**, **Spectrum**, **DataLink**, and **ColorMethod**.

Imaging Parameter

The enumerated parameter **Imaging** specifies the geometry of the camera sensor; possible values are:

Value	Description
AREA	The camera is an area-scan model. An area-scan camera is based on a 2D imaging sensor; it delivers individual image frames composed of a fixed amount of lines, each containing a fixed amount of pixels.
LINE	The camera is a line-scan model. A line-scan camera is based on 1D imaging sensor; it delivers individual image lines composed of a fixed amount of pixels.
TDI	The camera is a TDI line-scan model. A particular type of line-scan camera using the Time Delayed Integration technology (TDI). It is based on a 2D imaging sensor having a small amount of lines; like any other line-scan camera, it delivers individual image lines composed of a fixed amount of pixels.

**NOTE**

The board's behavior is the same when **Imaging** is **TDI** instead of **LINE**; however TDI cameras exhibit less operational modes since they lack an electronic shutter.

Spectrum Parameter

The enumerated parameter **Spectrum** specifies the spectral sensitivity of the camera; possible values are:

Value	Description
BW	The camera delivers a monochrome image obtained from an image sensor operating in the human visible domain of the light spectrum.
IR	The camera delivers a monochrome image obtained from an image sensor operating in the infra-red domain of the light spectrum.
COLOR	The camera delivers a color image.

**NOTE**

The board's behavior is exactly the same when **Spectrum** is **IR** instead of **BW**.

DataLink Parameter

The enumerated parameter **DataLink** specifies the data transfer method used by the camera; the unique possible value is:

Value	Description
CAMERALINK	The camera delivers a digital video signal complying to the Camera Link standard.

ColorMethod Parameter

The enumerated parameter **ColorMethod** specifies the color analysis method used by the camera; possible values are:

Value	Description
NONE	The camera delivers a monochrome image.
BAYER	The camera uses a single imaging sensor coated with a Bayer Color Filter Array and delivers the raw Bayer CFA data as a single video data stream embedding the RGB information.
PRISM	The camera uses a wavelength-separating prism to feed three distinct imaging sensors. The color information is available as three R, G, B video data streams.
RGB	The camera uses a coated sensor and an internal processor to reconstruct the full color information. The color information is available as three R, G, B video data streams.
TRILINEAR	The camera uses three parallel sensing linear arrays of pixels exhibiting different wavelength sensitivities. The color information is available as three R, G, B video data streams.

Notes

- This board provides a limited support of TRILINEAR cameras since the scan-delay compensation function is not available.
- The board's behavior is the same when **ColorMethod** is **TRILINEAR** instead of **RGB**.
- The board's behavior is exactly the same when **ColorMethod** is **PRISM** instead of **RGB**.

Supported Camera Classes

Grablink products support the following combinations of values:

Imaging	Spectrum	ColorMethod	Camera Class
AREA	BW	NONE	Monochrome area-scan cameras – Visible spectrum
	IR	NONE	Monochrome area-scan cameras – Infra-red spectrum
	COLOR	BAYER	Color area-scan cameras –Raw BAYER data output
		RGB	Color area-scan cameras – RGB data output
		PRISM	3-CCD Color area-scan cameras – RGB data output
LINE	BW	NONE	Monochrome line-scan cameras – Visible spectrum
	IR	NONE	Monochrome line-scan cameras – Infra-red spectrum
	COLOR	BAYER	Color line-scan cameras –Raw BAYER data output
		RGB	Color line-scan cameras – RGB data output
		PRISM	3-CCD Color line-scan cameras – RGB data output
		TRILINEAR	Color line-scan cameras – 1 tri-linear sensor – RGB data output
TDI	BW	NONE	Monochrome Time-Delay-Integration line-scan cameras – Visible spectrum
	IR	NONE	Monochrome Time-Delay-Integration line-scan cameras – Infrared spectrum
	COLOR	PRISM	3-CCD Color Time-Delay-Integration line-scan cameras – 3 sensors with filters – RGB data output

3.2. Camera Operation Modes

The classification of the operation modes of the industrial cameras is based on the following MultiCam parameters: **Expose** and **Readout**.

Expose

The enumerated parameter **Expose** specifies the camera exposure principle used by the camera; possible values are:

Value	Description
INTCTL	The line or frame exposure condition is totally controlled by the camera. The exposure duration is set through camera configuration settings.
PLSTRG	The line or frame exposure condition starts upon receiving a pulse from the frame grabber.
WIDTH	The duration of a pulse issued by the frame grabber determines the line or frame exposure condition.
INTPRM	The exposure is permanent.

Readout

The enumerated parameter **Readout** specifies the camera readout principle used by the camera; possible values are

Value	Description
INTCTL	The frame read-out condition is totally controlled by the camera. The read-out duration is set through camera configuration settings.

Supported Area-Scan Camera Operation Modes

Grablink products support the following combinations of values for area-scan cameras (**Imaging** = **AREA**):

CamConfig	Expose	Readout	Description
PxxSC	INTCTL	INTCTL	Synchronous progressive scan, camera-controlled exposure
PxxRC	PLSTRG	INTCTL	Asynchronous progressive scan, camera-controlled exposure
PxxRG	WIDTH	INTCTL	Asynchronous progressive scan, grabber-controlled exposure

Supported Line-Scan Camera Operation Modes

Grablink products support the following combinations of values for (non-TDI) line-scan cameras (Imaging = LINE):

CamConfig	Expose	Readout	Description
LxxxxSP	INTPRM	INTCTL	Free-running or camera-controlled line rate Permanent exposure or disabled electronic shutter No control line
LxxxxSC	INTCTL	INTCTL	Free-running or camera-controlled line rate Camera-controlled exposure with electronic shutter Single control line
LxxxxRP	INTPRM	PLSTRG	Grabber-controlled camera line rate and exposure Permanent exposure or disabled electronic shutter Single control line
LxxxxRC	PLSTRG	INTCTL	Grabber-controlled camera line rate camera controlled exposure Single control line
LxxxxRG	WIDTH	INTCTL	Grabber-controlled camera rate and exposure Camera-controlled exposure with electronic shutter Single control line
LxxxxRG2	PLSTRG	PLSTRG	Grabber-controlled camera rate and exposure Camera-controlled exposure with electronic shutter Two control lines

Supported TDI Line-Scan Camera Operation Modes

Grablink products support the following combinations of values for TDI line-scan cameras (Imaging = TDI):

CamConfig	Expose	Readout	Description
LxxxxSP	INTPRM	INTCTL	Free-running, permanent exposure
LxxxxRP	INTPRM	PLSTRG	Grabber-controlled line-scanning, permanent exposure

3.3. Camera Configurations

MultiCam embeds predefined camera parameters settings for each available camera operation mode.

The user selects the by assigning a value to the **Camera** and **CamConfig** parameters. For all Grablink products:

- **Camera** must be set to **MyCameraLink**
- **CamConfig** must be set to any of the following value :

CamConfig	Camera Class	Camera Operation Mode	CamFile template
PxxSC	Area-scan	Synchronous progressive scan Camera-controlled exposure	MyCameraLink_ PxxSC.cam
PxxRC	Area-scan	Asynchronous progressive scan Camera-controlled exposure	MyCameraLink_ PxxRC.cam
PxxRG	Area-scan	Asynchronous progressive scan Grabber-controlled exposure	MyCameraLink_ PxxRG.cam
LxxxxSC	Line-scan	Free-running or camera-controlled line rate Camera-controlled exposure with electronic shutter No control line	N/A
LxxxxRC	Line-scan	Grabber-controlled camera line rate Camera controlled exposure Single control line	MyCameraLink_ LxxxxRC.cam
LxxxxRG	Line-scan	Grabber-controlled camera rate and exposure Camera-controlled exposure with electronic shutter Single control line	MyCameraLink_ LxxxxRG.cam
LxxxxSP	(TDI) Line-scan	Free-running or camera-controlled line rate Permanent exposure or disabled electronic shutter No control line	MyCameraLink_ LxxxxSP.cam
LxxxxRP	(TDI) Line-scan	Grabber-controlled camera line rate and exposure Permanent exposure or disabled electronic shutter Single control line	MyCameraLink_ LxxxxRP.cam

CamFile templates are delivered for most of the camera operation modes.

**NOTE**

A **CamFile** is a text file gathering all the relevant Channel parameters.

3.4. Camera Tap Properties

TapConfiguration Parameter

The enumerated parameter **TapConfiguration** declares the Camera Link tap configuration used by the camera.

Refer to ["Supported Tap Configurations" on page 31](#) for an exhaustive list of configurations supported by Grablink products.

TapGeometry Parameter

The **tap geometry** is a Euresys proprietary [taxonomy](#) that describes, with a standardized name, the geometrical properties characterizing the different taps of a multi-tap camera.

The enumerated parameter **TapGeometry** declares the Camera Link tap geometry used by the camera.

Refer to ["Supported Tap Geometries" on page 40](#) for an exhaustive list of valid combinations of **TapConfiguration** and **TapGeometry** values.

Image Reconstruction

For most **TapConfiguration** and **TapGeometry** values, the frame grabber is capable to re-arrange the data in the destination surface.

Refer to ["Image Reconstruction" on page 49](#) for more information.

TapConfiguration Glossary

Naming Convention

A tap configuration is designated by:

`<Config>_<TapCount>T<BitDepth>(B<TimeSlots>)`

<Config>

Designates the Camera Link configuration as follows:

Camera Link Configuration name	<Config> value
Lite	LITE
Base	BASE
Medium	MEDIUM
Full	FULL
72-bit	DECA
80-bit	DECA

<TapCount>

Total number of pixel taps. Values range: 1 to 10.

<BitDepth>

Number of bits per tap. Values list: {8, 10, 12, 14, 16, 24, 30, 36, 42, 48}.

<TimeSlots>

Number of consecutive time slots required to transfer one pixel data. Values list: {2, 3}

The field and the letter B are omitted when a single time slot is sufficient to deliver all the pixel data.

Examples

BASE_1T8: Base Camera Link configuration, 1 tap, 8-bit pixel data

BASE_1T24: Base Camera Link configuration, 1 tap, 24-bit pixel data (likely RGB)

DECA_8T10: 80-bit Camera Link configuration, 8 taps, 10-bit pixel data

DECA_8T30B3: 80-bit Camera Link configuration, 8 taps, 30-bit pixel data (likely RGB), 3 time slots

Supported Tap Configurations

This topic lists all the Camera Link tap configurations (a.k.a. modes) defined in the section 4 of version 2.1 of the Camera Link standard.

The tap configurations are grouped by bit-depth then pixel type. Within a table, entries are sorted by increasing number of taps.



NOTE

Tap Configuration is a Euresys proprietary [taxonomy](#) that integrates, the channel link configuration, the number of taps and the pixel bit depth.

For each entry, it specifies:

- CL2.1 Name:** The name of the configuration as written in the section 4 of version 2.1 of the Camera Link standard
- Euresys Name:** The Euresys name of the configuration, i.e. the value of the TapConfiguration parameter.
- Compatible products:** The list of Grablink products supporting that configuration. An empty cell indicates that the configuration is not supported.

8-bit Tap Configurations

Monochrome 8-bit Tap Configurations

CL2.1 Name	Euresys Name	Compatible products
Lite	LITE_1T8	Base DualBase
Base 1 tap	BASE_1T8	Value Express Base DualBase Full FullXR
Base 2 taps	BASE_2T8	Value Express Base DualBase Full FullXR
Base 3 taps	BASE_3T8	Base DualBase Full FullXR
Medium 4 taps	MEDIUM_4T8	Full FullXR
Medium 5 taps	-	-
Medium 6 taps	MEDIUM_6T8	Full FullXR
Full 7 taps	-	-
Full 8 taps	FULL_8T8	Full FullXR
Full 9 taps	DECA_9T8	Full FullXR
80-bit 10 taps	DECA_10T8	Full FullXR

RGB 8-bit Tap Configurations

CL2.1 Name	Euresys Name	Compatible products
Base 1 tap	BASE_1T24	Value Express Base DualBase Full FullXR
Medium 2 taps	MEDIUM_2T24	Full FullXR
Full 3 taps	DECA_3T24	Full FullXR
80-bit 10 taps	-	-

RGBI 8-bit Tap Configurations

CL2.1 Name	Euresys Name	Compatible products
Medium 1 tap	-	-
Full 2 taps	-	-

10-bit Tap Configurations

Monochrome 10-bit Tap Configurations

CL2.1 Name	Euresys Name	Compatible products
Lite	LITE_1T10	Base DualBase
Base 1 tap	BASE_1T10	Value Express Base DualBase Full FullXR
Base 2 taps	BASE_2T10	Value Express Base DualBase Full FullXR
Medium 3 taps	MEDIUM_3T10	Full FullXR
Medium 4 taps	MEDIUM_4T10	Full FullXR
Full 5 taps	-	-
Full 6 taps	-	-
80-bit 7 taps	-	-
80-bit 8 taps	DECA_8T10	Full FullXR

RGB 10-bit Tap Configurations

CL2.1 Name	Euresys Name	Compatible products
Medium 1 tap	MEDIUM_1T30	Full FullXR
Full 2 taps	-	-
80-bit 8 taps (3 slots)	DECA_8T30B3	Full FullXR

RGBI 10-bit Tap Configurations

CL2.1 Name	Euresys Name	Compatible products
Medium 1 tap	-	-
Full 2 taps	DECA_2T40	Full FullXR

12-bit Tap Configurations

Monochrome 12-bit Tap Configurations

CL2.1 Name	Euresys Name	Compatible products
Base 1 tap	BASE_1T12	Value Express Base DualBase Full FullXR
Base 2 taps	BASE_2T12	Value Express Base DualBase Full FullXR
Medium 3 taps	MEDIUM_3T12	Full FullXR
Medium 4 taps	MEDIUM_4T12	Full FullXR
Full 5 taps	-	-
Full 6 taps	-	-

RGB 12-bit Tap Configurations

CL2.1 Name	Euresys Name	Compatible products
Medium 1 tap	MEDIUM_1T36	Full FullXR
Full 2 taps	-	-

RGBI 12-bit Tap Configuration

CL2.1 Name	Euresys Name	Compatible products
Medium 1 tap	-	-

14-bit Tap Configurations

Monochrome 14-bit Tap Configurations

CL2.1 Name	Euresys Name	Compatible products
Base 1 tap	BASE_1T14	Value Express Base DualBase Full FullXR
Medium 2 taps	-	-
Medium 3 taps	-	-
Full 4 taps	-	-
72-bit 5 taps	-	-

RGB 14-bit Tap Configuration

CL2.1 Name	Euresys Name	Compatible products
Medium 1 tap	-	-

RGBI 14-bit Tap Configuration

CL2.1 Name	Euresys Name	Compatible products
Full 1 tap	-	-

16-bit Tap Configurations

Monochrome 16-bit Tap Configurations

CL2.1 Name	Euresys Name	Compatible products
Base 1 tap	BASE_1T16	Value Express Base DualBase Full FullXR
Medium 2 taps	-	-
Medium 3 taps	-	-
Full 4 taps	-	-
72-bit 5 taps	-	-

RGB 16-bit Tap Configuration

CL2.1 Name	Euresys Name	Compatible products
Medium 1 tap	-	-

RGBI 16-bit Tap Configuration

CL2.1 Name	Euresys Name	Compatible products
Full 1 tap	-	-

TapGeometry Glossary

Definitions

Adjacent taps

Two taps are adjacent when the extracted pixels are adjacent on the same row or on the same column.

Region

A rectangular area of adjacent pixels that are transferred in a raster-scan order through one or multiple adjacent taps.

Tap

One pixel stream output port of the camera that delivers one pixel every clock cycle.

Tap Geometrical Properties

A tap is characterized by the following properties:

XStart: X-position of the first extracted pixel of a camera readout cycle

XEnd: X-position of the last extracted pixel of a camera readout cycle

YStart: Y-position of the first extracted pixel of a camera readout cycle

YEnd: Y-position of the last extracted pixel of a camera readout cycle

YStep: the difference of Y-position between consecutive rows of pixels; it is positive when Y-position values are increasing (top to bottom); it is negative otherwise.

X-Position: the pixel column number in the (non-flipped) image; column 1 is the leftmost column; column W is the rightmost column of an image having a width of W pixels.

Y-Position: the pixel row number in the (non-flipped) image; row 1 is the topmost row; row H is the bottommost row of an image having a height of H pixels.

TapGeometry Values Syntax

There are two variants of the syntax:

1. For cameras delivering two or more rows of pixels every camera readout cycle:

`<TapGeometryX>_<TapGeometryY>`

2. For cameras delivering only one row of pixels every camera, e.g. single line line-scan cameras:

`<TapGeometryX>`

TapGeometryX Syntax

<TapGeometryX> describes the geometrical organization of the taps along one row of the image. It is built as follows:

<XRegions>X(<XTaps>)(<ExtX>)

- **<XRegions>**: an integer declaring the number of regions encountered across one image row (= the X-direction or the horizontal direction). Possible values are 1, 2, 3, 4, 6, 8, and 10.
- **<XTaps>**: an integer declaring the number of consecutive pixels along one region row that are extracted simultaneously.
Possible values are 1, 2, 3, 4, 8, and 10.
The field is omitted when <XTaps> is 1.
- **<ExtX>**: a letter declaring the relative location of the pixels extractors across one row of the image.
 - This field is omitted when all pixel extractors are at the left of each region.
 - Letter E indicates that pixel extractors are at both ends of the image row.
 - Letter M indicates that pixel extractors are at middle of the image row.
 - Letter R indicates that the pixel extractors are all at the right of each region

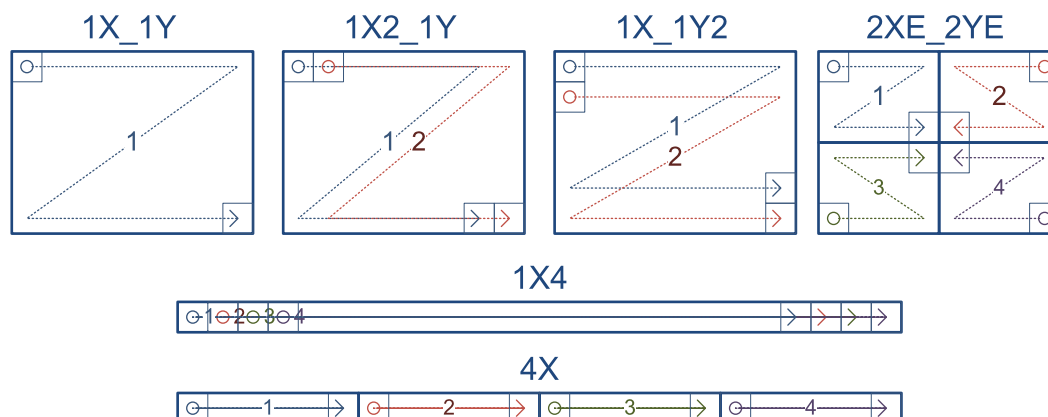
TapGeometryY Syntax

<TapGeometryY> describes the geometrical organization of the taps along one column of the image. It is built as follows:

<YRegions>Y(<YTaps>)(<ExtY>)

- <YRegions>**: an integer declaring the number of regions encountered across vertical direction.
Possible values are 1 and 2.
- <YTaps>**: an integer declaring the number of consecutive pixels along one region column that are extracted simultaneously.
Possible values are 1 and 2. The field is omitted when YTaps is 1.
- <ExtY>**: a letter declaring the relative location of the pixels extractors across one column of the image.
 - This field is omitted when all pixel extractors are at the top of each region.
 - Letter E indicates that pixel extractors are at both ends of the image column.

TapGeometry Values Examples



1X_1Y designates the tap geometry of a single-tap camera having 1 region across the X-direction and 1 region across the Y direction.

The pixels are delivered one at a time on a single tap beginning with the leftmost pixel of the top row, scanning progressively all the rows of the image one by one, and ending with the rightmost pixel of the bottom row.

1X2_1Y designates the tap geometry of a two-tap camera having 1 region across the X-direction and 1 region across the Y direction.

The pixels are delivered two at a time on two taps beginning with the two leftmost pixels of the top row, scanning progressively all the rows of the image one by one, and ending with the two rightmost pixels of the bottom row.

1X_1Y2 designates the tap geometry of a two-tap camera having 1 region across the X-direction and 1 region across the Y direction.

The pixels are delivered two at a time on two taps beginning with the two uppermost pixels of the first column, scanning progressively all the rows of the image two by two, and ending with the two lowermost pixels of the rightmost column.

2XE_2YE designates the tap geometry of a four-tap camera having 2 regions across the X-direction and 2 regions across the Y direction.

The pixels are delivered four at a time on four taps. Each region delivers its pixels on a single-tap using a specific scanning scheme:

The pixels of the upper left quadrant are delivered on tap 1 starting with the upper left pixel and ending with the lower right pixel of the region.

The pixels of the upper right quadrant are delivered on tap 2 starting with the upper rightmost pixel and ending with the lower left pixel of the region.

The pixels of the lower left quadrant are delivered on tap 3 starting with the lower left pixel and ending with the upper right pixel of the region.

The pixels of the lower right quadrant are delivered on tap 4 starting with the lower rightmost pixel and ending with the upper left pixel of the region.

1X4 designates the tap geometry of a four-tap line-scan camera having 1 region across the X-direction.

The pixels are delivered four at a time on four taps beginning with the four leftmost pixels and ending with the four rightmost pixels.

4X designates the tap geometry of a four-tap line-scan camera having 4 regions across the X-direction.

The pixels are delivered four at a time on four taps. Each region delivers its pixels on a single-tap using a common scanning scheme beginning with the leftmost pixel and ending with the rightmost pixel.

Supported Tap Geometries



NOTE

Refer to [TapGeometry](#) in the Parameters Reference for a description of each geometry.

Tap Geometries for Line-scan Cameras

Number of taps	1 X-region	2 X-regions	3 X-regions	4 X-regions	8 X-regions	10 X-regions
1	1X	-	-	-	-	-
2	1X2	2X 2XE 2XM 2XR	-	-	-	-
3	1X3	-	3X	-	-	-
4	1X4	2X2 2X2E 2X2M	-	4X 4XE 4XR	-	-
8	1X8	2X4	-	4X2 4X2E	8X 8XR	-
10	1X10	-	-	-	-	10X

Tap Geometries for Area-scan Cameras having only one tap along the vertical direction

Number of taps	1 X-region	2 X-regions	3 X-regions	4 X-regions	8 X-regions	10 X-regions
1	1X_1Y	-	-	-	-	-
2	1X2_1Y	2X_1Y 2XE_1Y 2XM_1Y 2XR_1Y	-	-	-	-
3	1X3	-	3X	-	-	-
4	1X4_1Y	2X2_1Y 2X2E_1Y 2X2M_1Y	-	4X_1Y 4XE_1Y 4XR_1Y	-	-
8	1X8_1Y	2X4_1Y	-	4X2_1Y 4X2E_1Y	8X_1Y 8XR_1Y	-
10	1X10_1Y	-	-	-	-	10X_1Y

Tap Geometries for Area-scan Cameras having Two taps along the vertical direction

Number of taps	1 X-region	2 X-regions	3 X-regions	4 X-regions	8 X-regions	10 X-regions
2	1X_1Y2 1X_2YE	-	-	-	-	-
4	-	2XE_2YE	-	-	-	-

3.5. Camera Active Area Properties

The Camera Active Area is a rectangular array of pixels containing active video that are delivered by the camera to the frame grabber.



NOTE

For line-scan cameras, the height of the active area is 1 (or 2 for bilinear line-scan).

Hactive_Px Parameter

For all cameras, the MultiCam parameter **Hactive_Px** represents the number of pixels in each line of the Camera Active Area.

The following rules apply:

Rule #1: The width of the Camera Active Area may contain at most 65535 pixels

$$\text{Hactive_Px} \leq 65535$$

Rule #2: Each tap delivers the same amount of pixels every line, **Hactive_Px must be a multiple of X_{Taps}**

$$\frac{\text{Hactive_Px}}{X_{\text{Taps}}} = N$$

- N is an integer number since each tap delivers the same amount of pixels every line:
- X_{Taps} is the number of taps along the X direction.



NOTE

X_{Taps} can be obtained from the value of **TapGeometry** by multiplying together the two numbers surrounding the letter "X". For example, 1X2, 1X2_1Y, 1X2_1Y2, 2X, 2X_1Y have all 2 taps along the X direction.

Rule #3: Each X_{Region} must contain at least $\text{MinBytesPerRegionLine}$ bytes

$$\frac{\text{Hactive_Px} \times \text{BytesPerPixel}}{X_{\text{Regions}}} \geq \text{MinBytesPerRegionLine}$$

- $\text{MinBytesPerRegionLine} = 48$



NOTE

$X_{Regions}$ is the number of geometrical regions in the X direction. This is the number preceding the letter "X" in the **TapGeometry** value: For example: 2X, 2X_1Y, 2X2 have all 2 regions along the X direction.

- BytesPerPixel is the amount of bytes required to store one pixel into the on-board memory:
 - 1 byte for 8-bit monochrome and Bayer CFA cameras
 - 2 bytes for 10-/12-/14- and 16-bit monochrome and Bayer CFA cameras
 - 3 bytes for 24-bit RGB cameras
 - 6 bytes for 30-bit and 36-bit RGB cameras

TapConfiguration	TapGeometry	Min. value	Multiple of	Max. value
BASE_1T8	1X	48	1	65535
BASE_1T10	1X	24	1	65535
BASE_1T24	1X	16	1	65535
MEDIUM_1T30	1X	8	1	65535
BASE_2T8	1X2	48	2	65534
BASE_2T10	1X2	24	2	65534
MEDIUM_2T24	1X2	16	2	65534
BASE_2T8	2X	96	2	65534
BASE_2T10	2X	48	2	65534
MEDIUM_2T24	2X	32	2	65534
BASE_3T8	1X3	48	3	65535
MEDIUM_3T10	1X3	24	3	65535
BASE_3T8	3X	144	3	65535
MEDIUM_3T10	3X	72	3	65535
MEDIUM_4T8	1X4	48	4	65532
MEDIUM_4T10	1X4	24	4	65532
MEDIUM_4T8	4X	192	4	65532
MEDIUM_4T10	4X	96	4	65532
FULL_8T8	1X8	48	8	65528
FULL_8T8	8X	384	8	65528
DECA_10T8	1X10	50	10	65530
DECA_10T8	10X	480	10	65530

Vactive_Ln Parameter

For area-scan cameras only, the MultiCam parameter **Vactive_Ln** represents the number of lines of the Camera Active Area.

The following rules apply:

Rule #1: The camera active window may contain at most 65535 lines.

$$\text{Vactive_Ln} \leq 65535$$

Rule #2: Each tap delivers exactly the same amount of pixels, **Vactive_Ln** must be a multiple of YTaps .

$$\frac{\text{Vactive_Ln}}{\text{YTaps}} = N$$

- N is an integer number
- YTaps is the number of taps along the Y direction. YTaps can be obtained from the value of **TapGeometry** by multiplying together the two numbers surrounding the letter "Y". For example: 1X_1Y2, 1X_2YE have all 2 taps along the Y direction.

3.6. Bayer CFA Color Registration

When **ColorMethod** = **BAYER**, the enumerated parameter **ColorRegistration** specifies the alignment of the color pattern filter over the sensor active area.

Possible values are: **GB**, **BG**, **RG**, **GR**. The two letters indicate respectively the color of the two first pixels of the first line.

Value	Description
GB	The first two pixels are green and blue
BG	The first two pixels are blue and green
RG	The first two pixels are red and green
GR	The first two pixels are green and blue

The information is used by MultiCam to automatically configure the Bayer CFA decoder.

4. Image Processing

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4.1. Overview

Overview of the image data processing on Grablink cards

The acquisition channels of Grablink boards performs the following successive operations on the image data stream:

Image Reconstruction

This operation unscrambles the pixel streams of multi-tap cameras and reconstructs the image exactly like it was captured on the camera sensor

For more information, refer to ["Image Reconstruction" on page 49](#).

Image Cropping

This operation extracts a rectangular area from the Camera Active Area.

For more information, refer to ["Image Cropping" on page 52](#).

Image Flipping

This operation flips the image around an horizontal and/or a vertical axis.

For more information, refer to ["Image Flipping" on page 55](#).

Lookup Table Transformation

Applies to: Base DualBase Full FullXR

This operation performs lookup table processing on individual pixel components.

For more information and configuration instructions, refer to ["Look-up Table Transformation" on page 64](#).

Bayer CFA decoding

Applies to: Base DualBase Full FullXR

This operation transforms the raw Bayer CFA data stream issued by the camera into an RGB color data stream.

For more information and configuration instructions, refer to ["Bayer CFA to RGB Conversion" on page 67](#).

White Balancing

Applies to: Base DualBase Full FullXR

This operation adjusts the gain and the offset of each color channel.

For more information, refer to ["White Balance Operator" on page 71](#).

Pixel Formatting

This stage performs several operations:

- unpacking of 10-bit, 12-bit, and 14-bit pixel components to 8-bit or 16-bit.
- delivery of RGB data in packed or planar formats

For more information and configuration instructions, refer to ["Pixel Formatting" on page 78](#).

Image line build-up

This operation builds concatenates the components data of all pixels of an image line:

- 8-bit pixel components are aligned to byte boundaries
- 16-bit pixel components (possibly expanded by unpacking or lookup table processing) are aligned to word (2-byte) boundaries, the 2 bytes are stored according to the little-endian convention.

Image Transfer

The processed and formatted image data are transferred into a MultiCam Surface over the PCI Express bus using a DMA engine.

For more information, refer to ["Image Transfer" on page 81](#).

Transfer Latencies

Image data are transferred ASAP to the MultiCam Surface, keeping the time latency as short as possible.

For more information, refer to ["Transfer Latency" on page 83](#).

4.2. Image Reconstruction

Grablink boards unscramble the pixel streams of multi-tap cameras and reconstruct the image exactly like it was captured on the camera sensor for most of the tap configurations and geometries.

The following tables list, for each tap configuration, the tap geometries that allows the reconstruction of the image.

- Refer to [TapGeometry](#) in the Parameters Reference for a description of each geometry.

Tap Geometries for Lite Camera Link Configuration

Applies to: Base DualBase

TapConfiguration	TapGeometry (line-scan)	TapGeometry (area-scan, 1 YTap)	TapGeometry (area-scan, 2 YTap)
LITE_1T8 LITE_1T10	1X	1X_1Y	-

Tap Geometries for Base Camera Link Configuration

Applies to: Value Express

TapGeometry (area-scan, 2 YTap)	TapGeometry (area-scan, 2 YTap)	TapGeometry (area-scan, 2 YTap)	TapGeometry (area-scan, 2 YTap)
BASE_1T8 BASE_1T10 BASE_1T12 BASE_1T14 BASE_1T16 BASE_1T24	1X	1X_1Y	-
BASE_2T8 BASE_2T10 BASE_2T12	1X2 2X 2XE 2XM 2XR	1X2_1Y 2X_1Y 2XE_1Y 2XM_1Y 2XR_ 1Y	1X_1Y2 1X_2YE

Applies to:

Base

DualBase

Full

FullXR

TapConfiguration	TapGeometry (line-scan)	TapGeometry (bilinear line-scan)	TapGeometry (area-scan, 1 YTap)	TapGeometry (area-scan, 2 YTap)
BASE_1T8 BASE_1T10 BASE_1T12 BASE_1T14 BASE_1T16 BASE_1T24	1X	-	1X_1Y	-
BASE_2T8 BASE_2T10 BASE_2T12	1X2 2X 2XE 2XM 2XR	1X_1Y2	1X2_1Y 2X_1Y 2XE_1Y 2XM_1Y 2XR_1Y	1X_1Y2 1X_2YE
BASE_3T8	1X3 3X	-	1X3_1Y 3X_1Y	-

Tap Geometries for Medium Camera Link Configuration

Applies to:

Full

FullXR

TapConfiguration	TapGeometry (line-scan)	TapGeometry (bilinear line-scan)	TapGeometry (area-scan, 1 YTap)	TapGeometry (area-scan, 2 YTap)
MEDIUM_1T30 MEDIUM_1T36	1X	-	1X_1Y	-
MEDIUM_2T24	1X2 2XE	-	1X2_1Y 2XE_1Y	-
MEDIUM_3T10 MEDIUM_3T12	1X3 3X	-	1X3_1Y 3X_1Y	-
MEDIUM_4T8 MEDIUM_4T10 MEDIUM_4T12	1X4 2X2 2X2E 2X2M 4X 4XE 4XR	1X2_1Y2 2X_1Y2 2XE_1Y2 2XM_1Y2 2XR_1Y2	1X4_1Y 2X2_1Y 2X2E_1Y 2X2M_1Y 4X_1Y 4XE_1Y 4XR_1Y	1X2_1Y2 2X_1Y2 2XE_1Y2 2XM_1Y2 2XR_1Y2 1X2_2YE 2X_2YE 2XE_2YE 2XM_2YE 2XR_2YE
MEDIUM_6T8	-	1X3_1Y2 3X_1Y2	-	1X3_1Y2 1X3_2YE 3X_1Y2 3X_2YE

Tap Geometries for Full Camera Link Configuration

Applies to: Full FullXR

TapConfiguration	TapGeometry (line-scan)	TapGeometry (bilinear line-scan)	TapGeometry (area-scan, 1 YTap)	TapGeometry (area-scan, 2 YTap)
FULL_8T8	1X8 2x4 4x2 4X2E 8X 8XR	1X4_1Y2 2X2_1Y2 2X2E_1Y2 2X2M_1Y2 4X_1Y2 4XE_1Y2 4XR_1Y2	1X8_1Y 2x4_1Y 4X2_1Y 4X2E_1Y 8X_1Y 8XR_1Y	1X4_1Y2 2X2_1Y2 2X2E_1Y2 2X2M_1Y2 4X_1Y2 4XE_1Y2 4XR_1Y2 1X4_2YE 2X2_2YE 2X2E_2YE 2X2M_2YE 4X_2YE 4XE_2YE 4XR_2YE

Tap Geometries for 72-bit Camera Link Configuration

Applies to: Full FullXR

TapConfiguration	TapGeometry (line-scan)	TapGeometry (trilinear line-scan)	TapGeometry (area-scan, 1 YTap)	TapGeometry (area-scan, 3 YTap)
DECA_3T24	1X3 3X	-	1X3_1Y 3X_1Y	-
DECA_9T8	-	3X_1Y3	-	3X_1Y3

Tap Geometries for 80-bit Camera Link Configuration

Applies to: Full FullXR

TapConfiguration	TapGeometry (line-scan)	TapGeometry (area-scan, 1 YTap)	TapGeometry (area-scan, 2 YTap)
DECA_2T40	-	1X2_1Y	-
DECA_8T10	1X8	1X8_1Y	-
DECA_8T30B3	1X8	1X8_1Y	-
DECA_10T8	1X10 10X	1X10_1Y 10X_1Y	-

4.3. Image Cropping

Grablink boards implement an image cropping operator that selects a subset of the pixels delivered by the camera to build the image delivered to the Host PC.

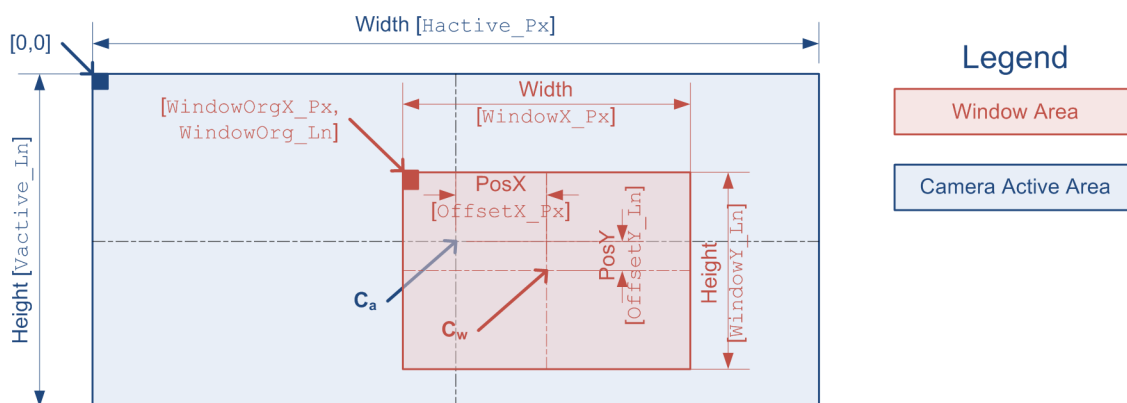
This subset, named *Window Area*, includes:

- For area-scan cameras: a single rectangular region of the 2D image sensor.
- For line-scan cameras: a single segment of the 1D image sensor.

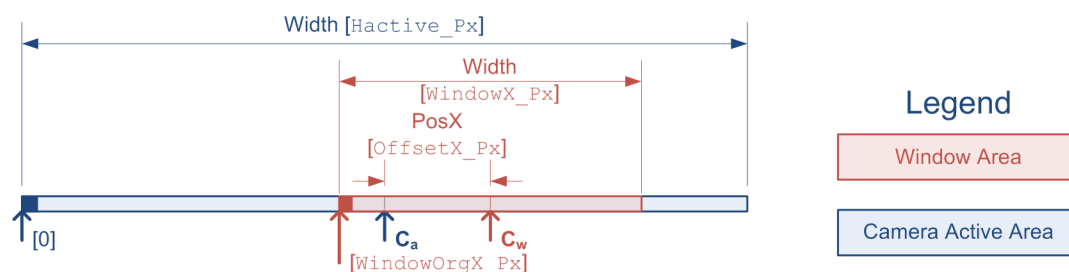
Image Cropping parameters

The operator is controlled through the following Channel Class parameters of the Grabber Timing category:

- **GrabWindow**: the main control parameter.
- **WindowX_Px**, **WindowY_Ln**: integer parameters defining the size of the *Window Area*.
- **OffsetX_Px**, and **OffsetY_Ln**: integer parameters defining the position of the *Window Area* within the Camera Active Area.



Window Area Parameters for Area-Scan cameras



Window Area Parameters for Line-Scan cameras



NOTE

The position of the Window Area within the Camera Active Window is expressed as the difference of coordinates between C_w , the center of the Window Area, and C_a , the center of the Camera Active Area.



NOTE

The range of allowed values of **OffsetX_Px** and **OffsetY_Ln** parameters is automatically adjusted to force the Window Area to stay within the boundaries of the Camera Active Area.

WindowOrgX_Px reports the X-coordinate in the Camera Active Area of the leftmost pixels of the Window Area:

$$\text{WindowOrgX_Px} = (\text{Hactive_Px} - \text{WindowX_Px}) / 2$$

For area-scan cameras only, **WindowOrgY_Ln** reports the Y-coordinate in the Camera Active Area of the topmost pixels of the Window Area:

$$\text{WindowOrgY_Ln} = (\text{Vactive_Ln} - \text{WindowY_Ln}) / 2$$

Configuring Image Cropping

By default, **GrabWindow** is set to **NOBLACK** disabling the ICO: the acquired image includes all active pixels delivered by the camera without any surrounding weak or blind pixels on the image edges.

To enable image cropping, proceed as follows:

- Enable cropping by setting **GrabWindow** to **MAN**.
- Adjust the width of the Window Area using **WindowX_Px**. Any integer value ranging from 8 up to **Hactive_Px** is allowed.
- *For area-scan cameras only:* Adjust the height of the Window Area using **WindowY_Ln**. Any integer value ranging from 1 up to **Vactive_Ln** is allowed.
- Move horizontally the Window Area using **OffsetX_Px**. Increasing the value moves the Window Area towards the right of the Camera Active Area and vice-versa, decreasing the value moves the Window Area towards the left of the Camera Active Area.
- *For area-scan cameras only:* Move vertically the Window Area using **OffsetY_Ln**. Increasing the value moves the Window Area towards the top of the Camera Active Area and vice-versa, decreasing the value moves the Window Area towards the bottom of the Camera Active Area.

Conditions of applicability

Cropping is applicable to the following camera classes:

- Monochrome, RGB color, and Bayer CFA color area-scan cameras: any valid combination of **TapConfiguration** and **TapGeometry** is allowed except when **TapGeometry** = ***_2YE**
- Monochrome, and RGB color line-scan cameras.

4.4. Image Flipping

Grablink boards implement an image flipping operator that performs the mirroring of the image delivered to the Host PC:

- For area-scan cameras, it performs the left/right and the top/bottom mirroring.
- For line-scan camera, it is capable to perform the left/right mirroring only.

The operator controlled through the following Channel Class parameters of the Cluster Category:

- **ImageFlipX** enables the left/right mirroring
- **ImageFlipY** enables the top/bottom mirroring.

By default, both operators are OFF disabling any mirroring.



Image Flipping Operations

Conditions of applicability

IFO is applicable to the following camera classes:

- Monochrome, RGB color, and Bayer CFA color *area-scan cameras*
- Monochrome, and RGB color *line-scan cameras*

4.5. Pixel Data Processing Configurations

Configurations for Monochrome Pixels

Processing elements availability for monochrome pixels vs. products

Capability	Value	Express	Base	DualBase	Full	FullXR
LUT transformation	-	-	OK	OK	OK	OK
Pixel formatting	OK	OK	OK	OK	OK	OK

Valid configurations when LUT transformation is disabled

Applies to: 1625 3304 Value Express Base DualBase Full FullXR

The pixel output format is defined by **ColorFormat**.

Camera PFNC Pixel Format	ColorFormat	Output PFNC Pixel Format
Mono8	Y8	Mono8
Mono10	Y8	Mono8
	Y10	Mono10
	Y16	Mono16
Mono12	Y8	Mono8
	Y12	Mono12
	Y16	Mono16
Mono14	Y8	Mono8
	Y14	Mono14
	Y16	Mono16
Mono16	Y8	Mono8
	Y16	Mono16

Valid configurations when LUT transformation is enabled

Applies to: Base DualBase Full FullXR

The LUT operator operates in the monochrome mode, its input bit depth is the camera pixel bit depth.

The LUT output bit depth and the pixel output format are defined by **ColorFormat**.

Camera PFNC Pixel Format	Lookup Table Configuration	ColorFormat	Output PFNC Pixel Format
Mono8	M_8x8	Y8	Mono8
Mono10	M_10x8	Y8	Mono8
	M_10x10	Y10	Mono10
	M_10x16	Y16	Mono16
Mono12	M_12x8	Y8	Mono8
	M_12x12	Y12	Mono12
	M_12x16	Y16	Mono16

Configurations for Bayer CFA Pixels

Processing elements availability for Bayer CFA pixels vs. products

Elements	Value	Express	Base	DualBase	Full	FullXR
Bayer CFA Decoding	-	-	OK	OK	OK	OK
White Balance	-	-	OK	OK	OK	OK
LUT Transformation	-	-	OK	OK	OK	OK
Pixel Formatting	OK	OK	OK	OK	OK	OK

Valid configurations when Bayer CFA decoding is disabled

Applies to: 1625 3304 Value Express Base DualBase Full FullXR

The pixel output format is defined by **ColorFormat**.

Camera PFNC Pixel Format	ColorFormat	Output PFNC Pixel Format
Bayer**8	BAYER8	Bayer**8
Bayer**10	BAYER8	Bayer**8
	BAYER10	Bayer**10
	BAYER16	Bayer**16
Bayer**12	BAYER8	Bayer**8
	BAYER12	Bayer**12
	BAYER16	Bayer**16
Bayer**14	BAYER8	Bayer**8
	BAYER14	Bayer**14
	BAYER16	Bayer**16
Bayer**16	BAYER16	Bayer**16



NOTE

The white balance and the LUT transformation are not available when Bayer CFA decoding is disabled!

Valid configurations when Bayer CFA decoding is enabled

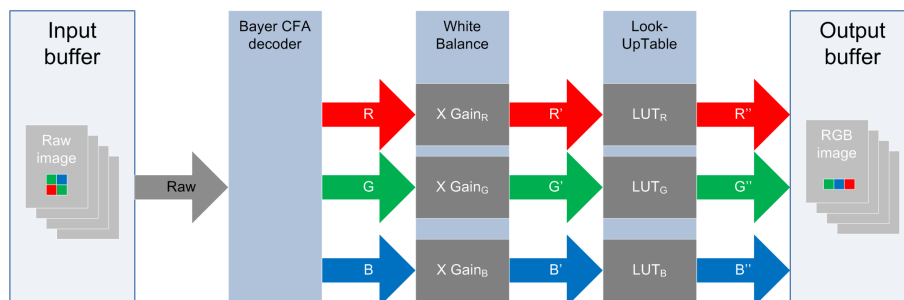
Applies to: Base DualBase Full FullXR

The pixel processing chain uses the following elements:

- Bayer CFA Decoder
- White Balance operator (Optional)
- Look-Up-table operator (Optional)

The Look-Up-table operator is configured for RGB color processing.

The processing chain outputs one RGB pixel for each RAW pixel of the input buffer.



Pixel Processing Chain - Bayer CFA => RGB Configuration

Valid configurations and peak processing rate [Megapixels/sec] for 1624 Grablink Base and 1623 Grablink DualBase

Applies to: Base DualBase

Camera PFNC Pixel Format	WBO	Lookup Table Configuration	ColorFormat	Output PFNC Pixel Format	Max. rate
Bayer**8	Optional	8-bit to 8-bit RGB	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -	125
Bayer**10	Optional	10-bit to 8-bit RGB	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -	125
Bayer**12	Optional	12-bit to 8-bit RGB	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -	125
Bayer14	Optional	Not available	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -	125
Bayer16	Optional	Not available	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -	125

Valid configurations and peak processing rate [Megapixels/sec] for 1622 Grablink Full and 1626 Grablink Full XR

Applies to: Full FullXR

Camera PFNC Pixel Format	WBO	Lookup Table Configuration	ColorFormat	Output PFNC Pixel Format	Max. rate
Bayer8	Optional	8-bit to 8-bit RGB	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -	250
Bayer10	Optional	10-bit to 8-bit RGB	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -	250
Bayer10	Optional	10-bit to 10-bit RGB	RGB30PL	-	125
Bayer10	Optional	10-bit to 16-bit RGB	RGB48PL	-	125
Bayer12	Optional	12-bit to 8-bit RGB	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -	250
Bayer12	Optional	12-bit to 12-bit RGB	RGB36PL	-	125
Bayer12	Optional	12-bit to 16-bit RGB	RGB48PL	-	125
Bayer14	Optional	Not available	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -	250
Bayer14	Optional	Not available	RGB42PL	-	125
Bayer14	Optional	Not available	RGB48PL	-	125
Bayer16	Optional	Not available	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -	250
Bayer16	Optional	Not available	RGB48PL	-	125

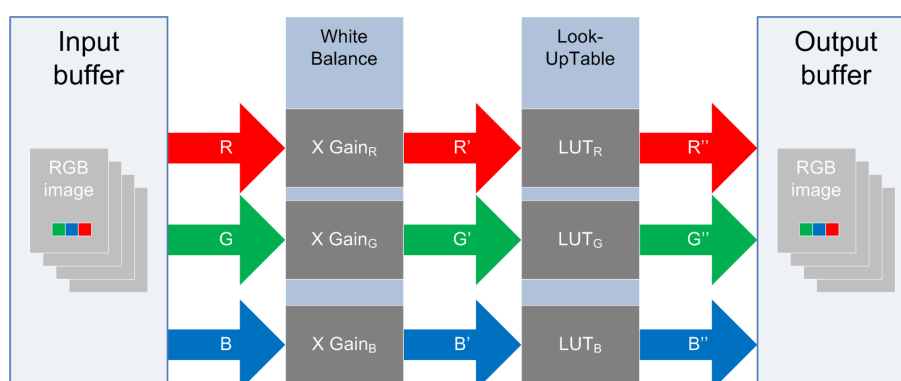
Configurations for RGB Pixels

Processing elements availability for RGB pixels vs. products

Elements	Value	Express	Base	DualBase	Full	FullXR
White Balance	-	-	OK	OK	OK	OK
LUT Transformation	-	-	OK	OK	OK	OK
Pixel Formatting	OK	OK	OK	OK	OK	OK

Valid configurations

The processing chain outputs one RGB pixel for each RGB pixel of the input buffer.



Pixel Processing Chain - RGB => RGB Configuration

Valid configurations for 1624 Grablink Base and 1623 Grablink DualBase

Applies to: Base DualBase

Camera PFNC Pixel Format	WBO	Lookup Table Configuration	ColorFormat	Output PFNC Pixel Format
RGB8	Optional	8-bit to 8-bit RGB	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -

Valid configurations for 1622 Grablink Full and 1626 Grablink Full XR

Applies to: Full FullXR

Camera PFNC Pixel Format	WBO	LUT	ColorFormat	Output PFNC Pixel Format
RGB8	Optional	8-bit to 8-bit RGB	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -
RGB10	Optional	10-bit to 8-bit RGB	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -
RGB10	Optional	10-bit to 10-bit RGB	RGB30PL	-
RGB10	Optional	10-bit to 16-bit RGB	RGB48PL	-
RGB12	Optional	12-bit to 8-bit RGB	RGB24 RGB32 RGB24PL	BGR8 BGRa8 -
RGB12	Optional	12-bit to 10-bit RGB	RGB36PL	-
RGB12	Optional	12-bit to 16-bit RGB	RGB48PL	-

4.6. Look-up Table Transformation

Applies to: Base DualBase Full FullXR

The look-up table operator enables you to process monochrome or RGB color pixel data streams.

Storage for five LUT definitions is available in the main memory. They are indexed from 1 to 4.

Selecting the LUT of index 0 disables the LUT operator and establishes a bypass over the Look-up table operator in the pixel processing stream.

During MultiCam channel activation, the hardware initializes the LUT operator. Therefore it fills the tables of the LUT operator with the selected LUT definition.

Any further modification of the LUT operator configuration that occurs during the acquisition sequence is applied without any further delay. For example, this occurs when:

- Changing the **LUT_UseIndex** parameter.
- Modifying the LUT definition have the same index as the **LUT_UseIndex** value.

Monochrome Operation

When the acquisition channel is configured for acquisition from monochrome cameras, the LUT operator is modeled as a single very high speed RAM inserted into the pixel data stream.

Available configurations and performance of the LUT operator for monochrome cameras

Camera	LUT Input bit depth	LUT output bit depth	Peak pixel rate [Megapixels/s]		Peak pixel rate [Megapixels/s]	
			Base	DualBase	Full	FullXR
8	8	8	500		1000	
10	10	8, 10, 16	250		500	
12	12	8, 12, 16	250		500	
14	12	8, 14, 16	250		500	
16	12	8, 16	250		500	

The input bit depth of the Look-Up-Table is:

- Equal to the camera bit depth for 8-, 10- and 12-bit cameras.
- 12-bit for 14-bit and 16-bit cameras.

**NOTE**

For 14-bit and 16-bit cameras, when the look-up table operator is enabled, only 12 most significant bits of the camera pixel data are effectively considered; the remaining bits are ignored.

The output bit depth of look-up table is equal to the bit depth of the selected output format. The possible bit depths depends on the pixel depth of the camera:

- For 8-bit cameras: 8-bit
- For cameras delivering more than 8-bit: 8-bit, 16-bit and the same bit depth as the camera.

The LUT operator is designed to sustain the highest pixel rate achievable by the board-compatible monochrome cameras.

RGB Color Acquisition

When the board is configured for acquisition from RGB color cameras, the LUT operator is modeled as a triplet of very high speed RAMs inserted into the red, green, and blue pixel components data streams.

The three color components have the same bit depth. Consequently, the three look-up tables have the same input bit depth and the same output bit depth.

Available configurations and performance of the LUT operator for RGB color cameras

Camera	LUT Input bit depth	LUT output bit depth	Peak pixel rate [Megapixels/s]		Peak pixel rate [Megapixels/s]	
			Base	DualBase	Full	FullXR
8	8	8	125		250	
10	10	8, 10, 16	-		125	
12	12	8, 10, 16	-		125	

The input bit depth of each look-up table is equal to the camera bit depth of each color component.

The output bit depth of each look-up table is equal to the bit depth of each color component of the selected output format. The possible color components bit depths depends on the color component pixel bit depth of the camera:

- For 3 x 8-bit RGB cameras: 8-bit.
- **Full** **FullXR** For cameras delivering more than 8-bit per component: 8-bit, 16-bit and the same bit depth as the component bit depth of the camera.

The LUT operator can sustain the highest pixel rate achievable by the board-compatible RGB color cameras.

Bayer Color Acquisition

When the board is configured for acquisition from Bayer color cameras, the LUT operator is modeled as a triplet of very high speed RAM inserted into the red, green, and blue pixel components data streams delivered by the CFA decoder.

Available configurations of the LUT operator for Bayer color cameras when the Bayer decoder is enabled

Camera	LUT Input bit depth	LUT output bit depth	Peak pixel rate [Megapixels/s]		Peak pixel rate [Megapixels/s]	
			Base	DualBase	Full	FullXR
8	8	8	125		250	
10	10	8	-		250	
		10, 16	-		125	
12	12	8	-		250	
		12, 16	-		125	
14	12	8	-		250	
		14, 16	-		125	
16	12	8	-		250	
		16	-		125	

The input bit depth of the look-up table is:

- Equal to the camera bit depth for 8-, 10- and 12-bit cameras.
- 12-bit for 14-bit and 16-bit cameras.



NOTE

For 14-bit and 16-bit cameras, when the look-up table operator is enabled, only 12 most significant bits of each component delivered by the CFA decoder are effectively considered; the remaining bits are ignored.

The output bit depth of each of the 3 look-up tables is equal to the bit depth of each color component of the selected output format. The possible color components bit depths depends on the pixel bit depth of the camera and board type:

- For 8-bit Bayer cameras: 8-bit per component.
- **Full** **FullXR** For cameras delivering more than 8-bit per component; 8-bit, 16-bit and the same bit depth as the component bit depth of the camera.

The performance of the RGB LUT operator matches the performance of the Bayer CFA decoder.

4.7. Bayer CFA to RGB Conversion

Applies to: Base DualBase Full FullXR

The Bayer CFA decoder transforms the raw Bayer CFA data stream issued by the camera into an RGB color data stream.

The missing pixel components are reconstructed using one of the following interpolation methods:

- Legacy interpolation method computes the missing color components using exclusively Mean() functions of the nearest components.
- Advanced interpolation method computes the missing color components using Mean() and Median() functions of the nearest components. The advanced interpolation eliminates the "creneling" effect on the highly contrasted sharp transitions in the image.

The CFA decoder requires up to eight surrounding pixels to compute the missed components of RGB pixels. Surrounding pixels are identified by their geographic location relative to the pixel for computation.

NW	N	NE
W		E
SW	S	SE

Definitions

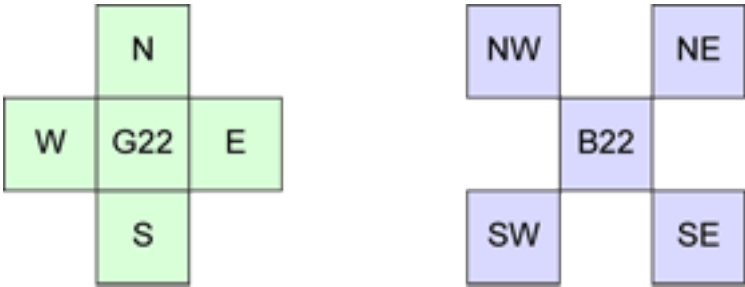
```
Mean2(a,b) = (a+b) / 2
Mean4(a,b,c,d) = (a+b+c+d) / 4
Median2Of4(a,b,c,d) = Mean2( Min [ Max(a,b); Max(c,d) ] ; Max [ Min(a,b); Min(c,d) ] )
```

Computing Missing Components on 4 Positions of a Bayer CFA Array

B11	G21	B31	G41
G12	R22	G32	R42
B13	G23	B33	G43
G14	R24	G34	R44

For red pixel locations (case of R22)

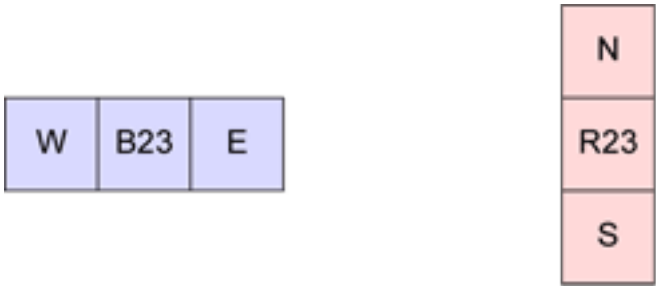
```
G <= Mean4 (GN, GS, GE, GW)
B <= Mean4 (BNE, BSE, BSW, BNW)
```



Case of R22

For green pixel locations in lines with blue (case of G23)

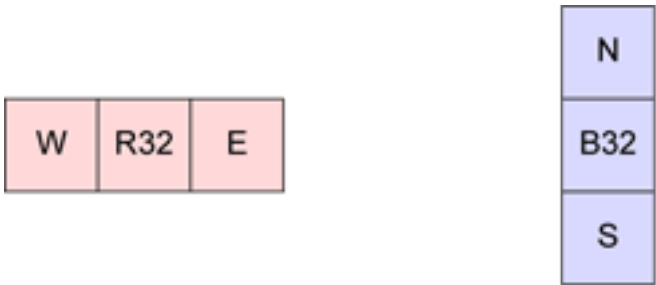
```
R <= Mean4 (RN, RS)
B <= Mean4 (BE, BW)
```



Case of G23

For green pixel locations in lines with red (case of G32)

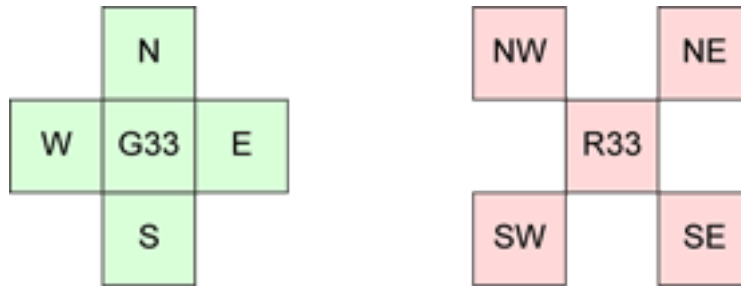
```
R <= Mean2 (RE, RW)
B <= Mean2 (BN, BS)
```



Case of G32

For blue pixel locations (case of B33)

```
G <= Mean4 (GN, GS, GE, GW)
R <= Mean4 (RNE, RSE, RSW, RNW)
```



Case of B33

Enabling the Bayer CFA Decoder

The Bayer CFA decoding function is automatically enabled if all the following conditions are satisfied:

1. The camera is an area-scan camera (**Imaging** = AREA)
2. The camera is a color camera (**Spectrum** = COLOR)
3. The camera delivers raw data from a Bayer Color Filter Array sensor (**ColorMethod** = BAYER)

Registering Bayer CFA

The registration of the BAYER CFA must be correctly set by assigning the appropriate value to the **ColorRegistration** parameter.

There are 4 values: GB, BG, RG, GR corresponding to the colors of the 2 first pixels of the first image line delivered by the camera.

Configuring the Bayer CFA Decoder

The CFA decoder has only one setting to select the interpolation method: the **CFD_Mode** parameter.

The default and recommended setting is **ADVANCED**. The alternate setting is **LEGACY**.

CFA Decoder Performance

The peak pixel processing rate of the CFA decoder of each acquisition channel is 250 megapixels/s.

The performance level of the Bayer CFA decoder is matching approximately the performance of the DMA transfer. However, the peak pixel processing rate is significantly lower than the highest pixel rate achievable by the board-compatible Bayer CFA cameras.

4.8. White Balance

Applies to: Base DualBase Full FullXR

Detailed description of the White Balance Operator

What Is White Balance?

Color image acquisition

A color image acquisition involves the use of three color filters on the camera sensor. Each color filter restricts the light source to a range of wavelengths of the light spectrum, either red (R), green (G), or blue (B).

An ideal capture system renders a white object as a white image. A white stimulation should yield the same signal for R, G and B filters. But practically, there are always unavoidable defects on the signals that introduce a *white imbalance*.

White imbalance factor

Several factors, due to the camera and to the capture conditions, are responsible for the white imbalance:

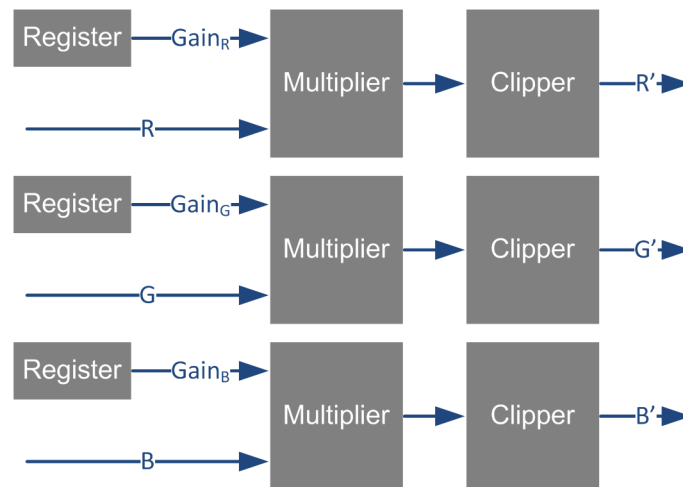
- Object illumination. The color of an object is a combination of its reflectivity and the spectral contents of the illuminating light.
- Camera optical filters response.
- Sensor sensitivity, which is not the same for the three ranges of wavelength.
- Different gain coefficients applied to each color signal before digitization.

White balance correction

MultiCam can correct the white imbalance of the capture system. The operation is called the *white balance*:

- The *white balance operator* applies correcting coefficients (R, G, and B gains) to each color signal, so, for a white object, the combination of the R, G, and B signals renders a white image.
- The *white balance calibration* is the computation of the three R, G, and B gains. It is performed on a representative image area, prior to the image capture. It can be automatic or manual.

White Balance Operator



White Balance Operator - Block Diagram

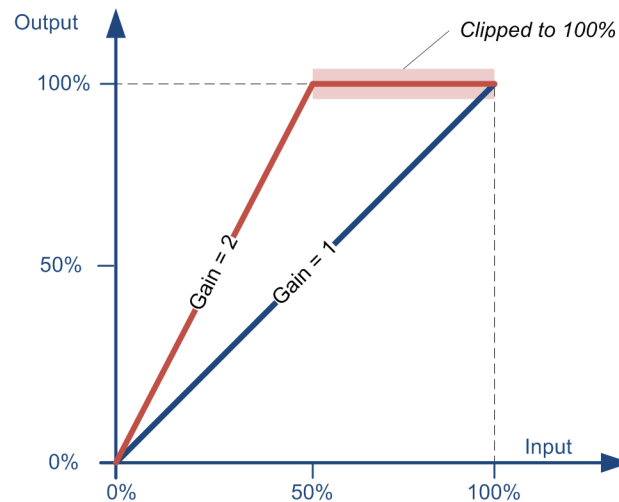
The White Balance Operator is an element of the pixel processing chain. It is composed with 3 identical processing blocks, one for each color component. Each processing block contains 3 elements:

- One register
- One multiplier
- One clipper

The register element holds the gain correction factor. The gain value is registered as a 16-bit unsigned binary value allowing gain correction factors to be accurately defined.

The multiplier computes the product of the gain correction factor and the color component value. It is capable to handle components having 8-bit, 10-bit, 12-bit, 14-bit and 16-bit bit depth.

The multiplier output is clipped to the maximum value of the digital output scale. The digital output scale is in all cases identical to the digital input scale; itself identical to the digital output scale of the camera. For instance, for a camera delivering 10-bit components, the digital scale is [0..1023].



White Balance Operator - Transfer Function

The above drawing shows 2 transfer functions of one component of the White Balance Operator:

- The blue line corresponds to a gain setting of 1.000; i.e., the minimal allowed gain value.
- The red line corresponds to a gain setting of 2.000. The output remains proportional to the input until the 100% full-scale output is reached; for greater input values, the output is clipped to 100% full-scale!

Automatic Calibration Description

The color calibration process takes place during the first acquisition phase of a MultiCam acquisition sequence when the **WBO_Mode** is set to **ONCE**.

The color calibrator analyzes a rectangular area (AWB_AREA) of one uncorrected image and computes a correcting gain factor for each RGB color component.

The correction factor for the color component having the strongest response is always 1; the correction factors for the weakest color components are greater than 1.

Providing that the requirements of the *color source equipment*, the *calibration target* and the *acquisition channel settings* are fulfilled, the calibrator estimates the gain factors with an accuracy better than 1/1000.

Applying the calculated gain correction factors to the White Balance Operator for subsequent image acquisitions allows on-the-fly color balancing of the acquired images.

The calibrator returns a NOT_OK status in the following cases:

- Excessive color imbalance.
- Not enough pixels satisfying the calibration target requirements in the AWB_AREA.

Automatic Calibration Requirements

This topic describes the requirements that must be fulfilled to obtain optimal calibration results.

Image Source Equipment Requirements

The image source equipment including: the camera, the lighting and the optical elements, must exhibit:

- A linear response: The digital value of each color component must be proportional to the light intensity of the corresponding color.
- A moderate color imbalance: The ratio between the response of the strongest color component and the weakest color component must be less than 5.

Calibration Target Requirements

The calibration target is a neutral color object located in the field of view of the camera during the calibration process.

The form of the target can be either:

- Clustered light gray pixels located in a specific area of the camera field of view.
- Non-clustered-light gray pixels located in a specific area of the camera field of view.
- Non-clustered-light gray pixels located anywhere in the camera field of view.

The calibration target can be:

- In the object to inspect.
- A specific object placed in the camera field of view during the calibration phase.

The appearance of the target must be:

- A neutral light gray color.
- The level of the brightest component within 75% to 90% of the full scale.
- The level of the darkest component above 15% of the full scale.

The target must contain at least 256 pixels satisfying the appearance requirements.

Acquisition Channel Settings

The parameter **WBO_Mode** must be set to **ONCE**.

The parameters defining the position and the size of the AWB_AREA must be configured such that:

- It includes at least 256 pixels satisfying the calibration target appearance requirements.

- It contains at least 1 line and 32 columns of pixels.
- It is located entirely within the Camera Active Area.

Specifically on **1624 Grablink Base**, **1623 Grablink DualBase**, **1622 Grablink Full** and **1626 Grablink Full XR**:

- The LUT Operator must be disabled.
- The position and the size of the cropping area must be configured such that it encompasses the AWB_AREA

Automatic Calibration Timing

The color calibration process takes place during the first acquisition phase of a MultiCam acquisition sequence when the **WBO_Mode** is set to **ONCE**.

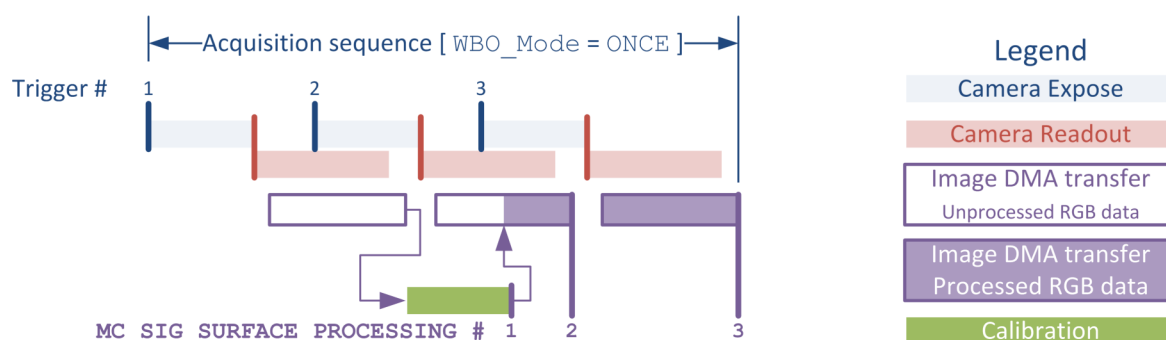
The White Balance Operator is disabled before the sequence starts.

The calibration process begins when the DMA transfer of the first acquisition phase is completed. The first **MC_SIG_SURFACE_PROCESSING** signal of the sequence is delayed until the completion of the calibration process.

- At the completion of a successful calibration process:
 - The value of the parameter **WBO_Status** is set to **OK**.
 - The values of parameters **WBO_GainR**, **WBO_GainG**, and **WBO_GainB** are updated with the calibration results.
 - The White Balance Operator is reconfigured with the new settings.

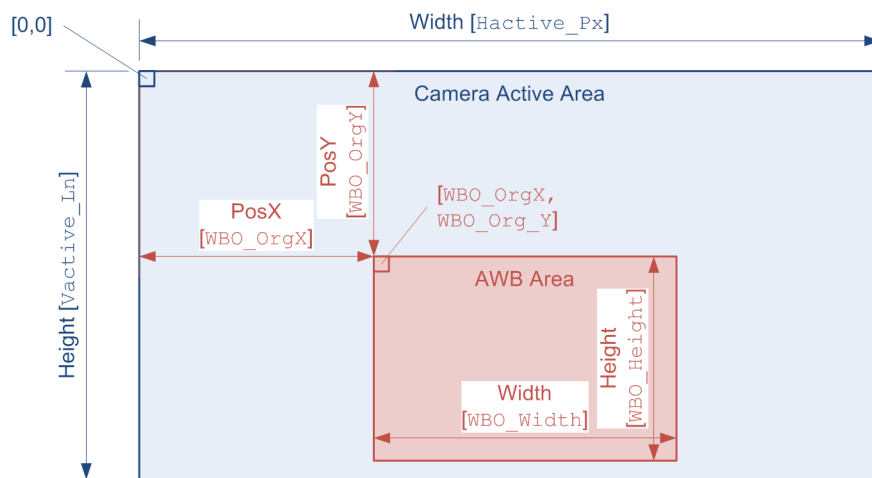
At the completion of an unsuccessful calibration process:

- The value of the parameter **WBO_Status** is set to **NOT_OK**.
- The original values of parameters **WBO_GainR**, **WBO_GainG**, and **WBO_GainB** are restored.
- The White Balance Operator is reconfigured with the original settings.



Calibration Timing Diagram

AWB_AREA Settings Description



AWB_AREA layout

The AWB_AREA is a rectangular area within the Camera Active Window that is analyzed by the color balancing calibrator.

The size and the position of the AWB_AREA within the Camera Active Area is defined by the following parameters: **WBO_Width**, **WBO_Height**, **WBO_OrgX**, and **WBO_OrgY**.

The default size of the AWB_AREA is the whole Camera Active Area.

4.9. Pixel Formatting

Pixel Component Unpacking

Grablink boards unpack 10-bit, 12-bit, and 14-bit pixel component data to 16-bit pixel data.

Two unpacking options are available:

- Unpacking to lsb (Default setting)
- Unpacking to msb

Unpacking to lsb

The significant bits of the pixel component data are aligned to the **least significant bit** of the data container. Padding '0' bits are put as necessary in the **most significant bits** to reach the next 8-bit boundary.

- 10-bit pixels: 0000 00<pp pppp pppp>
- 12-bit pixels: 0000 <pppp pppp pppp>
- 14-bit pixels: 00<pp pppp pppp pppp>



NOTE

Unpacking to lsb doesn't modify the pixel component value.

Unpacking to msb

The significant bits of the pixel component data are aligned to the **most significant bit** of the data container. Padding '0' bits are put as necessary in the **least significant bits** to reach the next 8-bit boundary.

- 10-bit pixels: <pppp pppp pp>00 0000
- 12-bit pixels: <pppp pppp pppp> 0000
- 14-bit pixels: <pppp pppp pppp pp>00



NOTE

Unpacking 10-bit, 12-bit, and 14-bit pixel components to msb multiplies the pixel component value by 64, 16, and 4 respectively.

**NOTE**

Unpacking 8-bit and 16-bit pixel components is a neutral operation:

- The size of the data container is unchanged: One byte for 8-bit pixel components; two bytes for 16-bit pixel components
- The data bits are not modified

**NOTE**

Unpacking 10-bit, 12-bit, and 14-bit pixel components increases the amount of data by 160%, 133%, and 114% respectively.

No unpacking

The packed image data transmitted by the camera through the CoaXPress Link is delivered as is to the output buffer.

Pixel Bit Depth Reduction

Grablink boards are capable to reduce the bit depth of 10-/12-/14- and 16-bit pixel components to 8-bit by truncation of the least significant bits.

Pixel Format Control

Pixel Component Unpacking and Pixel Bit Depth reduction are controlled by the **ColorFormat** parameter.

For instance for monochrome 10-bit pixels:

- Set **ColorFormat** to **Y8** to select the bit depth reduction
- Set **ColorFormat** to **Y16** to select the unpacking to msb
- Keep the default value (or set) **ColorFormat** to **Y10** to select unpacking to lsb

Valid Pixel Format Configurations

Refer to ["Configurations for Monochrome Pixels"](#) on page 57, ["Configurations for Bayer CFA Pixels"](#) on page 59 and ["Configurations for RGB Pixels"](#) on page 62

for an exhaustive list of valid pixel processing configurations.

Refer to *D406EN-MultiCam Storage Formats* PDF document for an exhaustive description of pixel formats.

ColorFormat Properties

The following table describes the properties of each **ColorFormat** values:

ColorFormat	Format class name	Bit depth	Number of components	Number of bytes per pixel	Number of planes
Y8	Monochrome	8	1	1	1
Y10		10		2	
Y12		12			
Y14		14			
Y16		16			
BAYER8	Raw Bayer CFA	8	1	1	1
BAYER10		10		2	
BAYER12		12			
BAYER14		14			
BAYER16		16			
RGB24	Packed (A)RGB Color	8/8/8	3	3	1
RGB32 ARGB32		8/8/8/8	4	4	
RGB24PL	Planar RGB Color	8/8/8	3	3	3
RGB30PL		10/10/10		6	
RGB36PL		12/12/12			
RGB42PL		14/14/14			
RGB48PL		16/16/16			



NOTE

- When the pixel container is larger than the bit depth (namely for 10-, 12-, and 14-bit cases), the significant bits are justified to the LSB and the uppermost unused bits are forced to 0.
- In case of packed (A) RGB formats, the components are stored at successive byte addresses in the following order: B, G, R, (A). B always occupies the LSB position. The A component is forced to 0.
- In case of planar RGB formats, the planes are indexed in the following order: R, G, B; the R plane is the plane of index 0.

4.10. Image Transfer

The processed and formatted image data are transferred into a surface over the PCI Express bus using a DMA engine.

**NOTE**

In the MultiCam driver, a surface is the physical memory space allocated into the host PC memory for the storage of one image.

The transferred image is stored in the destination surface in a progressive-scan order:

- The first pixel (top-left corner) of the transferred image is stored at the first memory location of the surface (at 0).
- The first pixels of the subsequent lines are stored at a byte address that is a multiple of the surface pitch.

Surface Pitch

The MultiCam driver establishes a default surface pitch that corresponds to the amount of bytes required to store one pixel row of the image data. You may increase this value if the application requires it.

Surface Size

The MultiCam driver establishes a default surface size that corresponds to the amount of bytes required to store all the rows of the image data. You may increase this value if its application requires it.

Extended Addressing Capabilities

Applies to: Base DualBase Full FullXR

These Grablink boards are capable of transferring image data anywhere into the physical memory of the system.

Automatic DMA Descriptors Loading

Applies to: Base DualBase Full FullXR

These Grablink boards implement an automatic mechanism for the loading of the DMA descriptors from the host PC memory to the DMA descriptor section of the on-board memory.

The DMA descriptors lists can then be prepared and stored into the host PC memory; the automatic loader pre-fetches the descriptors such that they are available for the DMA engine when it needs them.

Notes

- The transfer of the descriptors is performed with the second DMA engine, it doesn't require any CPU work.
- The transfer of the descriptors does not impact the available bandwidth on the PCI Express link for image transfer since it conveys data on the opposite direction.

Transfer Rate

Applies to: **Base**

This board sustains image data transfer over the PCI Express bus:

- Up to 200 megabytes/s for a PCI Express payload size of 256 bytes
- Up to 180 megabytes/s for a PCI Express payload size of 128 bytes.



WARNING

The effective data rate depends on the performance of the PCI Express link.

Transfer Rate

Applies to: **DualBase** **Full** **FullXR**

This board sustains image data transfer over the PCI Express bus:

- Up to 833 megabytes/s for a PCI Express payload size of 256 bytes and 64-bit addressing.
- Up to 844 megabytes/s for a PCI Express payload size of 256 bytes and 32-bit addressing.
- Up to 754 megabytes/s for a PCI Express payload size of 128 bytes and 64-bit addressing.
- Up to 780 megabytes/s for a PCI Express payload size of 128 bytes and 32-bit addressing.

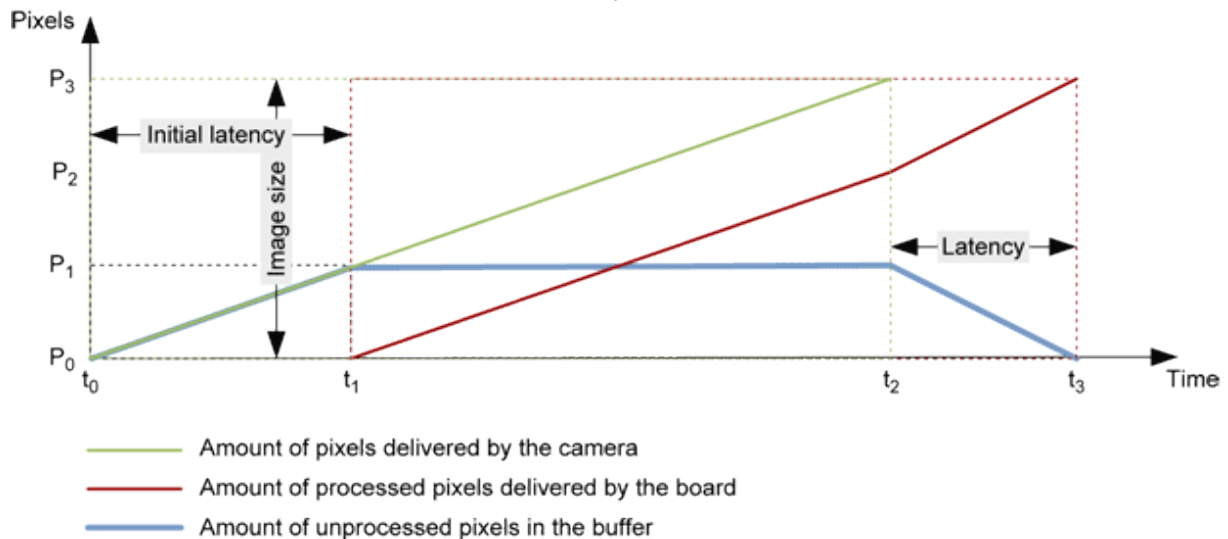


WARNING

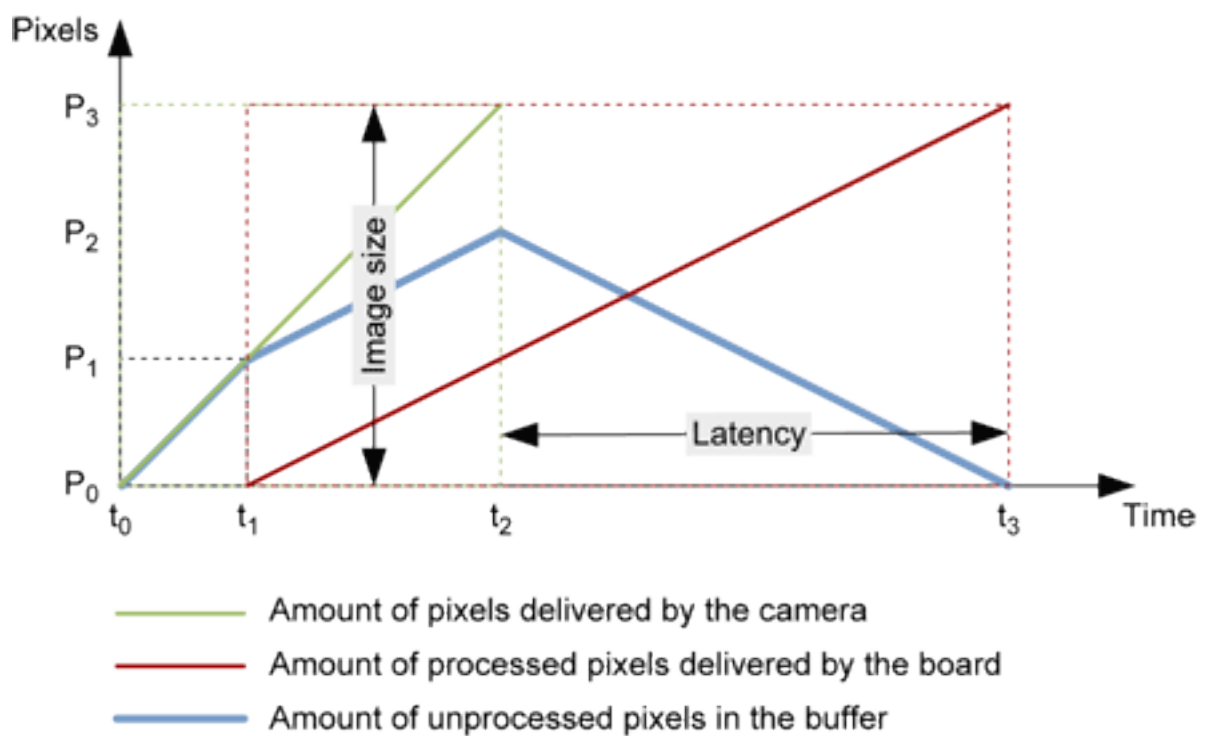
The effective data rate depends on the performance of the PCI Express link.

4.11. Transfer Latency

The transfer latency is the time interval between the time when the last camera pixel of the image enters the frame grabber and the same processed pixel is stored in the host PC memory.



Case #1: Camera pixel rate < max pixel processing or pixel delivery rate



Case #2: Camera pixel rate > max pixel processing or pixel delivery rate*For both cases*

- The camera delivers the image starting at time t_0 and ending at time t_2 . For the simplicity, the line blanking intervals are not shown.
- The on-board pixel processing and DMA transfer to the host PC memory begins at time t_1 and ends at time t_3 .
- The pixel processing requires that a minimum of pixel data (P_1 pixels) is available into the buffer. Consequently, the processing of the first pixel is retarded until the minimum amount of pixel is available. The time interval ($t_1 - t_0$) is the "initial latency"; it depends only on the camera pixel data rate and P_1 .
- The time interval ($t_3 - t_2$) is the "latency".

For case #1

- After t_1 , the effective pixel processing rate is equal to the camera pixel rate; the processing is eventually halted at line boundaries until enough data is available from the camera to resume it.
- When the camera has delivered the last pixel of the image, the processing continues at the max pixel processing (or the maximum delivery rate) until the last pixel of the image. The latency depends only on P_1 and the maximum processing/delivery rate.

For case #2

- After t_1 , the effective pixel processing rate is the maximum pixel processing/delivery rate until the last pixel of the image. The time interval ($t_3 - t_2$) is the "latency"; it depends only on P_1 and on the maximum processing/delivery rate.

For the estimation of the latency

The following board characteristics are be considered:

- P_1 is the amount of pixel that must be received from the camera before initiating the pixel processing. This amount is equal to the number of pixels contain into:
 - Two lines of the image if the Bayer CFA decoder is used.
 - One line of the image if the Bayer CFA decoder is not used.
- The maximum processing/delivery pixel rate is the smallest value of :
 - The pixel processing rate.
 - The pixel delivery rate over the PCIe interface.
- For the pixel processing rate, the following processing functions may limit the processing rate if they are used:
 - For the Bayer CFA decoder performances, refer to [Bayer CFA to RGB Conversion](#) for more details

- For LUT processor performances, refer to [Look-up Table Transformation](#) for more details.
- For the pixel delivery rate, refer to ["Image Transfer" on page 81](#) for an estimation of the data transfer rate and divide by the number of bytes/pixel to obtain an evaluation in pixels/s.

5. Acquisition Control

Introduction

The fundamental acquisition mode is selected by means of the MultiCam **AcquisitionMode** parameter. This board supports the following parameter values:

- for the area-scan cameras: **SNAPSHOT**, and **HFR**
- for the line-scan and TDI line-scan cameras: **WEB**, **PAGE**, and **LONGPAGE**

The user must select the fundamental acquisition mode by assigning the appropriate value to the **AcquisitionMode** before any other acquisition control parameter.

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5.1. SNAPSHOT Acquisition Mode

The **SNAPSHOT** acquisition mode is intended for the acquisition of snapshot images from area-scan cameras. The unique sequence is capable of acquiring **SeqLength_Fr** frames within the channel activity period.

The **SNAPSHOT** acquisition mode is the default mode enforced automatically by MultiCam for all area-scan cameras; it can also be explicitly invoked by assigning value **SNAPSHOT** to **AcquisitionMode**.

For cameras having a frame rate higher than 5000 frames per second, it is mandatory to use the **HFR** mode.

When invoking the **SNAPSHOT** acquisition mode:

- **ActivityLength** is enforced to **1**. The channel goes inactive at the completion of the sequence.
- **PhaseLength_Fr** is enforced to **1**. A single frame is acquired during an acquisition phase.
- **TrigMode** establishes the starting condition of the sequence, and consequently the starting condition of the first phase of the sequence. Possible values are **IMMEDIATE**, **HARD**, **SOFT**, and **COMBINED**. The default MultiCam setting is **IMMEDIATE**.
- **NextTrigMode** establishes the starting condition of the subsequent phases within the sequence. Possible values are **SAME**, **REPEAT**, **HARD**, **SOFT**, and **COMBINED**. The default value is **SAME**.
- **TrigDelay_us** can be used to insert a delay between the hardware trigger signal and the effective trigger event.
- The sequence length is specified by **SeqLength_Fr**. Assigning a value **-1** enforces an indefinite acquisition sequence.
- **EndTrigMode** is enforced to **AUTO**. The sequence terminates automatically after the acquisition of the specified number of frames. An indefinite acquisition sequence stops when the channel is forced to its inactive state.
- In case of a user break event, the **BreakEffect** parameter value is irrelevant. The acquisition terminates ALWAYS at a phase —frame— boundary ensuring the integrity of the frame.
- **Elapsed_Fr** reports the number of acquired frames in the sequence.
- When the sequence length is defined (**SeqLength_Fr** ≠ **1**), **Remaining_Fr** reports the number of remaining frames in the sequence.
- When the sequence contains more than 1 frame (**SeqLength_Fr** > **1**), **PerSecond_Fr** reports the measured average frame rate.

5.2. HFR Acquisition Mode

The **HFR** acquisition mode is intended for the acquisition of snapshot images from high frame rate area-scan cameras. The unique sequence is divided into phases, each phase acquiring **PhaseLength_Fr** frames into a single destination surface.

The **HFR** acquisition mode is applicable to area-scan cameras with a frame rate not exceeding 1,275,000 frames per second.

The **HFR** acquisition mode is explicitly invoked by assigning value **HFR** to **AcquisitionMode**.

When invoking the **HFR** acquisition mode:

- The **ActivityLength** parameter is enforced to **1**. The channel goes inactive at the completion of the sequence.
- The number of frames per acquisition phase is specified by **PhaseLength_Fr**. The minimal applicable value is the camera frame rate divided by 5000.
- The **TrigMode** parameter establishes the starting condition of the sequence and consequently the starting condition of the first slice of the first phase —the first frame— of the sequence. Possible values are **IMMEDIATE**, **HARD**, **SOFT**, and **COMBINED**. The default MultiCam setting is **IMMEDIATE**.
- The **NextTrigMode** parameter establishes the starting condition of the subsequent slices — frames— within the sequence. Possible values are **SAME**, **REPEAT**, **HARD**, **SOFT**, and **COMBINED**. The default value is **SAME**.
- The **TrigDelay_us** parameter can be used to insert a delay between the hardware trigger signal and the effective trigger event.
- The sequence length is specified by **SeqLength_Fr**. Assigning a value **-1** enforces an indefinite acquisition sequence.
- The **EndTrigMode** parameter is enforced to **AUTO**. The sequence terminates automatically after the acquisition of the specified number of frames. An indefinite acquisition sequence stops when the channel is forced to its inactive state.
- In case of a user break event, the **BreakEffect** parameter value is irrelevant. The acquisition terminates ALWAYS at a slice —frame— boundary ensuring the integrity of the frame.
- The **Elapsed_Fr** parameter reports the number of acquired frames in the sequence.
- When the sequence length is defined (**SeqLength_Fr**≠**-1**), **Remaining_Fr** reports the number of remaining frames in the sequence.
- When the sequence contains more than 1 frame (**SeqLength_Fr**>**1**), **PerSecond_Fr** reports the measured average frame rate.

5.3. WEB Acquisition Mode

This mode is intended for image acquisition of a continuous object, like a web, from a line-scan camera.

A single sequence acquiring **SeqLength_Ln** contiguous lines is available within the channel activity period. The sequence is divided in contiguous phases, each phase acquiring **PageLength_Ln** lines.

In the case **SeqLength_Ln** is not a multiple of **PageLength_Ln**, the surface is partially filled during the last phase.

The sequence and the first acquisition phase are initiated according to **TrigMode**. Subsequent acquisition phases are automatically initiated without any line loss.

BreakEffect specifies the behavior in case of a user break. The WEB acquisition mode is applicable to any line-scan camera.

The **WEB** acquisition mode is the default mode enforced automatically by MultiCam for line-scan cameras; it can also be explicitly invoked by assigning value **WEB** to **AcquisitionMode**.

When invoking the **WEB** acquisition mode:

- **ActivityLength** is enforced to **1**. The channel goes inactive at the completion of the sequence.
- **PhaseLength_Pg** is enforced to **1**. A single page is acquired during an acquisition phase.
- **PageLength_Ln** is automatically set to a working value. However, the page length can be enforced by setting **PageLength_Ln**. The minimal applicable value is the camera line rate divided by 500.
- **TrigMode** establishes the starting condition of the sequence and consequently the starting condition of the first phase of the sequence. Possible values are **IMMEDIATE**, **HARD**, **SOFT**, and **COMBINED**. The default MultiCam setting is **IMMEDIATE**.
- **NextTrigMode** is enforced to **REPEAT**. This ensures that no lines are missed between subsequent acquisition phases.
- The sequence length is specified by **SeqLength_Ln**. Assigning a value **-1** enforces an indefinite acquisition sequence.
- **EndTrigMode** is enforced to **AUTO**. The sequence terminates automatically after the acquisition of the specified number of pages. An indefinite acquisition sequence stops when the channel is forced to its inactive state.
- **BreakEffect** establishes the effect of a user break on the channel. When set to **FINISH**, it ensures the integrity of the last acquired phase —page— even when the user break event occurs during its execution; this is the default value. When set to **ABORT**, the effect of the user break is immediate (at line boundary); the current acquisition might be incomplete; the portion of image already acquired is available. Note that when **SeqLength_Ln** is not a multiple of **PageLength_Ln**, the last acquired page is partially filled despite the **FINISH** setting.

- **Elapsed_Ln** reports the number of acquired lines in the sequence.
- When the sequence length is defined (**SeqLength_Ln**≠1), **Remaining_Ln** reports the number of remaining lines in the sequence.

5.4. PAGE Acquisition Mode

This mode is intended for image acquisition of discrete objects from a line-scan camera.

Each page is constituted of contiguous lines; the page length, expressed in lines, is specified by **PageLength_Ln**.

A single sequence is capable to acquire **SeqLength_Pg** pages within the channel activity period. The **PAGE** acquisition mode is applicable to any line-scan camera.

The **PAGE** acquisition mode is explicitly invoked by assigning value **PAGE** to **AcquisitionMode**.

When invoking the **PAGE** acquisition mode:

- The **ActivityLength** parameter is enforced to **1**. The channel goes inactive at the completion of the sequence.
- The **PhaseLength_Pg** parameter is enforced to **1**. A single page is acquired during an acquisition phase.
- The page length is specified by **PageLength_Ln**. The minimal applicable value is the camera line rate divided by 500. Note for longer pages, use the **LONGPAGE** acquisition mode.
- The **TrigMode** parameter establishes the starting condition of the sequence and consequently the starting condition of the first phase—the first **PAGE**—of the sequence. Possible values are **IMMEDIATE**, **HARD**, **SOFT**, and **COMBINED**. The default MultiCam setting is **COMBINED**.
- The **NextTrigMode** parameter establishes the starting condition of the subsequent phases—pages—within the sequence. Possible values are **SAME**, **REPEAT**, **HARD**, **SOFT**, and **COMBINED**. The default value is **SAME**.
- The **PageDelay_Ln** parameter can be used to insert a delay between the hardware trigger signal and the page trigger event.
- The sequence length is specified by **SeqLength_Pg**. Assigning a value **-1** enforces an indefinite acquisition sequence.
- The **EndTrigMode** parameter is enforced to **AUTO**. The sequence terminates automatically after the acquisition of the specified number of pages. An indefinite acquisition sequence stops when the channel is forced to its inactive state.
- The **BreakEffect** parameter establishes the effect of a user break on the channel. When set to **FINISH**, it ensures the integrity of the last acquired phase—page— even when the user break event occurs during its execution; this is the default value. When set to **ABORT**, the effect of the user break is immediate (at line boundary); the current acquisition might be incomplete; the portion of image already acquired is available.
- The parameter **Elapsed_Pg** reports the number of acquired pages in the sequence
- When the sequence length is defined (**SeqLength_Pg**≠**-1**), **Remaining_Pg** reports the number of remaining pages in the sequence.

5.5. LONGPAGE Acquisition Mode

This mode is intended for image acquisition of long or variable size discrete objects from a line-scan camera.

The **ActivityLength** parameter specifies the number of sequences within the channel activity period. Each sequence is capable to acquire **SeqLength_Ln** contiguous lines.

A sequence is divided in phases, each phase acquiring **PageLength_Ln** lines. The **LONGPAGE** acquisition mode is applicable to any line-scan camera.

The **LONGPAGE** acquisition mode is explicitly invoked by assigning value **LONGPAGE** to **AcquisitionMode**.

When invoking the **LONGPAGE** acquisition mode:

- The **ActivityLength** parameter is enforced to **INDETERMINATE**. The channel remains active at the completion of the sequence.
- The **PhaseLength_Pg** parameter is enforced to **1**. A single page is acquired during an acquisition phase.
- The **PageLength_Ln** parameter is automatically set to a working value. However, the page length can be enforced by setting **PageLength_Ln**. The minimal applicable value is the camera line rate divided by 500.
- The sequence length is specified by **SeqLength_Ln**. Assigning a value **-1** enforces an indefinite acquisition sequence.
- The **TrigMode** parameter establishes the starting condition of the sequence and consequently the starting condition of the first slice of the first phase of the sequence. Possible values are **IMMEDIATE**, **HARD**, **SOFT**, and **COMBINED**. The default MultiCam setting is **COMBINED**.
- The **NextTrigMode** parameter is enforced to **REPEAT**. This ensures that no lines are missed between subsequent acquisition phases.
- The **PageDelay_Ln** parameter can be used to insert a delay between the hardware trigger signal and the page trigger event.
- The **EndTrigMode** parameter establishes the conditions of a sequence termination. When **EndTrigMode** = **AUTO**, the sequence terminates automatically after the acquisition of the specified number of frames. When **EndTrigMode** = **HARD**, the sequence terminates upon an external End Trigger event.
- The **BreakEffect** parameter establishes the effect of a user break on the channel. When set to **FINISH**, it ensures the integrity of the last acquired sequence —long page— even when the user break event occurs during its execution; this is the default value. When set to **ABORT**, the effect of the user break is immediate (at line boundary); the current acquisition might be incomplete; the portion of image already acquired is available. Note when **SeqLength_Ln** is not multiple of **PageLength_Ln**, the last acquired page is partially filled despite the **FINISH** setting.
- The **Elapsed_Ln** parameter reports the number of acquired lines in the sequence

- When the sequence length is defined (**SeqLength_Ln**≠-1), **Remaining_Ln** reports the number of remaining lines in the sequence.

6. I/O Operation

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6.1. I/O Ports Overview

I/O Ports

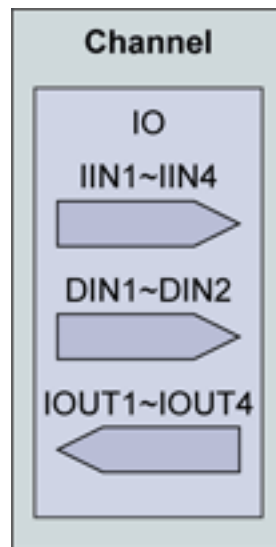
Applies to: Base DualBase Full FullXR

Every acquisition channel owns a dedicated set of 10 system I/O ports.

The composition and the functionality of the system I/O sets are identical for all acquisition channels.

Each system I/O set includes:

- 4 isolated input ports named IIN1, IIN2, IIN3, IIN4
- 2 high-speed differential input ports named DIN1, DIN2
- 4 isolated output ports named IOUT1, IOUT2, IOUT3, IOUT4



System I/O

6.2. Port Usage Selection

OutputConfig and **OutputFunction** are Board class MultiCam parameters belonging to the [Input/Output Control Category](#).

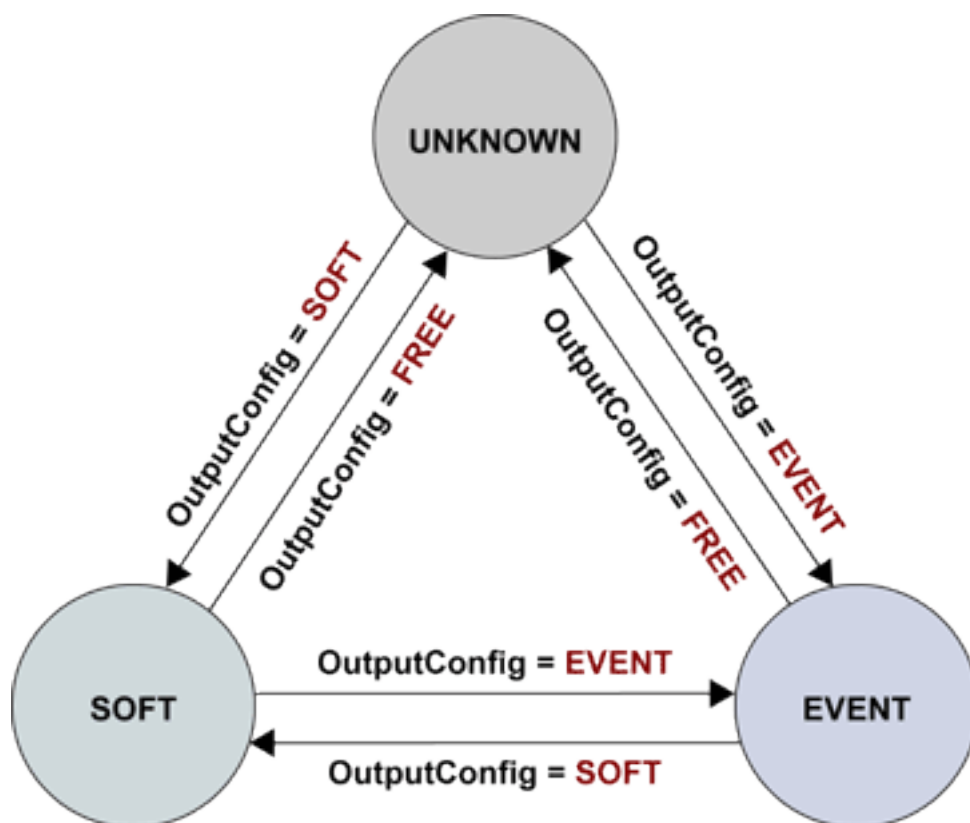
OutputConfig is a set-only collection parameter that must be used by the application software to configure an output port for a particular usage.

OutputFunction is a get-only collection parameter that reports the actual function assigned to the output port designated.

**NOTE**

The ports are individually designated by their I/O index. The I/O indices of the corresponding member can be found in [General-Purpose Output](#).

The following state diagram shows the 3 states of **OutputFunction** and all the possible inter-state transition:



OutputFunction state diagram

The "UNKNOWN" state means that the function of the output port is not known by the MultiCam Board object. The output port is then free to be used by a MultiCam channel for strobe output or SyncBus driver functions. Setting **OutputConfig** to **FREE** forces immediately **OutputFunction** to the value **UNKNOWN**. This is the default state after board startup.

The "SOFT" state means that the output port is directly under control of the application software for general-purpose usage. Setting **OutputConfig** to **SOFT** forces immediately **OutputFunction** to the value **SOFT**.

The "EVENT" state means that the output port is driven by the respective event signal generator for the event signaling usage. Setting **OutputConfig** to **EVENT** forces immediately **OutputFunction** to the value **EVENT**.

6.3. I/O Indices

I/O indices for input lines Value

Index	ConnectorName	InputPinName	InputStyle
17	SYSTEM	TRG	TTL
21	CAMERA	LVAL	CHANNELLINK
22	CAMERA	FVAL	CHANNELLINK
23	CAMERA	DVAL	CHANNELLINK
24	CAMERA	SPARE	CHANNELLINK
27	CAMERA	CK	CHANNELLINK
28	SYSTEM	A1	ITTL
29	SYSTEM	A2	ITTL

I/O indices for output lines Value

Index	ConnectorName	OutputPinName	OutputStyle
17	SYSTEM	STR	TTL
21	CAMERA	CC1	LVDS
22	CAMERA	CC2	LVDS
23	CAMERA	CC3	LVDS
24	CAMERA	CC4	LVDS
28	SYSTEM	A1	ITTL
29	SYSTEM	A2	ITTL

I/O indices for input lines Express

Index	ConnectorName	InputPinName	InputStyle
1	SYSTEM	Enhanced_I01	TTL
2	SYSTEM	Enhanced_I02	TTL
3	SYSTEM	Enhanced_I03	TTL
4	SYSTEM	Enhanced_I04	TTL
28	SYSTEM	ISOA1	ITTL, I12V
29	SYSTEM	ISOA2	TTL, I12V
17	SYSTEM	TRA1	LVDS
18	SYSTEM	TRA2	LVDS
21	CAMERA	LVAL	CHANNELLINK
22	CAMERA	FVAL	CHANNELLINK
23	CAMERA	DVAL	CHANNELLINK
24	CAMERA	SPARE	CHANNELLINK

I/O indices for output lines Express

Index	ConnectorName	OutputPinName	OutputStyle
1	SYSTEM	Enhanced_I01	TTL
2	SYSTEM	Enhanced_I02	TTL
3	SYSTEM	Enhanced_I03	TTL
4	SYSTEM	Enhanced_I04	TTL
28	SYSTEM	ISOA1	ITTL, IOC, IOE
29	SYSTEM	ISOA2	ITTL, IOC, IOE
17	SYSTEM	STA	OPTO
21	CAMERA	CC1	LVDS
22	CAMERA	CC2	LVDS
23	CAMERA	CC3	LVDS
24	CAMERA	CC4	LVDS
51	LED	RED	TTL
52	LED	GREEN	TTL

I/O indices for input lines

Base

Index	ConnectorName	InputPinName	InputStyle
1	IO	IIN1	ISO
2	IO	IIN2	ISO
3	IO	IIN3	ISO
4	IO	IIN4	ISO
5	IO	DIN1	DIFF
6	IO	DIN2	DIFF
7	CAMERA	LVAL	CHANNELLINK
8	CAMERA	FVAL	CHANNELLINK
9	CAMERA	DVAL	CHANNELLINK
10	CAMERA	SPARE	CHANNELLINK
11	CAMERA	CK_PRESENT	CHANNELLINK
23	IO	POWER_5V	POWERSTATE5V
24	IO	POWER_12V	POWERSTATE12V

I/O indices for output lines

Base

Index	ConnectorName	OutputPinName	OutputStyle
1	IO	IOUT1	ISO
2	IO	IOUT2	ISO
3	IO	IOUT3	ISO
4	IO	IOUT4	ISO
7	CAMERA	CC1	CHANNELLINK
8	CAMERA	CC2	CHANNELLINK
9	CAMERA	CC3	CHANNELLINK
10	CAMERA	CC4	CHANNELLINK
25	BRACKET	LED	NA

I/O indices for input lines DualBase

Index	ConnectorName	InputPinName	InputStyle
1	IO_A	IIN1	ISO
2	IO_A	IIN2	ISO
3	IO_A	IIN3	ISO
4	IO_A	IIN4	ISO
5	IO_A	DIN1	DIFF
6	IO_A	DIN2	DIFF
7	CAMERA_A	LVAL	CHANNELLINK
8	CAMERA_A	FVAL	CHANNELLINK
9	CAMERA_A	DVAL	CHANNELLINK
10	CAMERA_A	SPARE	CHANNELLINK
11	CAMERA_A	CK_PRESENT	CHANNELLINK
12	IO_B	IIN1	ISO
13	IO_B	IIN2	ISO
14	IO_B	IIN3	ISO
15	IO_B	IIN4	ISO
16	IO_B	DIN1	DIFF
17	IO_B	DIN2	DIFF
18	CAMERA_B	LVAL	CHANNELLINK
19	CAMERA_B	FVAL	CHANNELLINK
20	CAMERA_B	DVAL	CHANNELLINK
21	CAMERA_B	SPARE	CHANNELLINK
22	CAMERA_B	CK_PRESENT	CHANNELLINK
23	IO_A	POWER_5V	POWERSTATE5V
24	IO_A	POWER_12V	POWERSTATE12V
25	IO_B	POWER_5V	POWERSTATE5V
26	IO_B	POWER_12V	POWERSTATE12V

I/O indices for output lines DualBase

Index	ConnectorName	OutputPinName	OutputStyle
1	IO_A	IOUT1	ISO
2	IO_A	IOUT2	ISO
3	IO_A	IOUT3	ISO
4	IO_A	IOUT4	ISO
7	CAMERA_A	CC1	CHANNELLINK
8	CAMERA_A	CC2	CHANNELLINK
9	CAMERA_A	CC3	CHANNELLINK
10	CAMERA_A	CC4	CHANNELLINK
12	IO_B	IOUT1	ISO
13	IO_B	IOUT2	ISO
14	IO_B	IOUT3	ISO
15	IO_B	IOUT4	ISO
18	CAMERA_B	CC1	CHANNELLINK
19	CAMERA_B	CC2	CHANNELLINK
20	CAMERA_B	CC3	CHANNELLINK
21	CAMERA_B	CC4	CHANNELLINK
22	BRACKET	LED_A	NA
28	BRACKET	LED_B	NA

I/O indices for input lines

Full

FullXR

Index	ConnectorName	InputPinName	InputStyle
1	IO	IIN1	ISO
2	IO	IIN2	ISO
3	IO	IIN3	ISO
4	IO	IIN4	ISO
5	IO	DIN1	DIFF
6	IO	DIN2	DIFF
7	CAMERA	LVAL_X	CHANNELLINK
8	CAMERA	FVAL_X	CHANNELLINK
9	CAMERA	DVAL_X	CHANNELLINK
10	CAMERA	SPARE_X	CHANNELLINK
11	CAMERA	CK_PRESENT_X	CHANNELLINK
12	CAMERA	LVAL_Y	CHANNELLINK
13	CAMERA	FVAL_Y	CHANNELLINK
14	CAMERA	DVAL_Y	CHANNELLINK
15	CAMERA	SPARE_Y	CHANNELLINK
16	CAMERA	CK_PRESENT_Y	CHANNELLINK
17	CAMERA	LVAL_Z	CHANNELLINK
18	CAMERA	FVAL_Z	CHANNELLINK
19	CAMERA	DVAL_Z	CHANNELLINK
20	CAMERA	SPARE_Z	CHANNELLINK
21	CAMERA	CK_PRESENT_Z	CHANNELLINK
23	IO	POWER_5V	POWERSTATE5V
24	IO	POWER_12V	POWERSTATE12V



NOTE

The I/O indices 0 and 22 have no input-related function.

I/O indices for output lines

Full FullXR

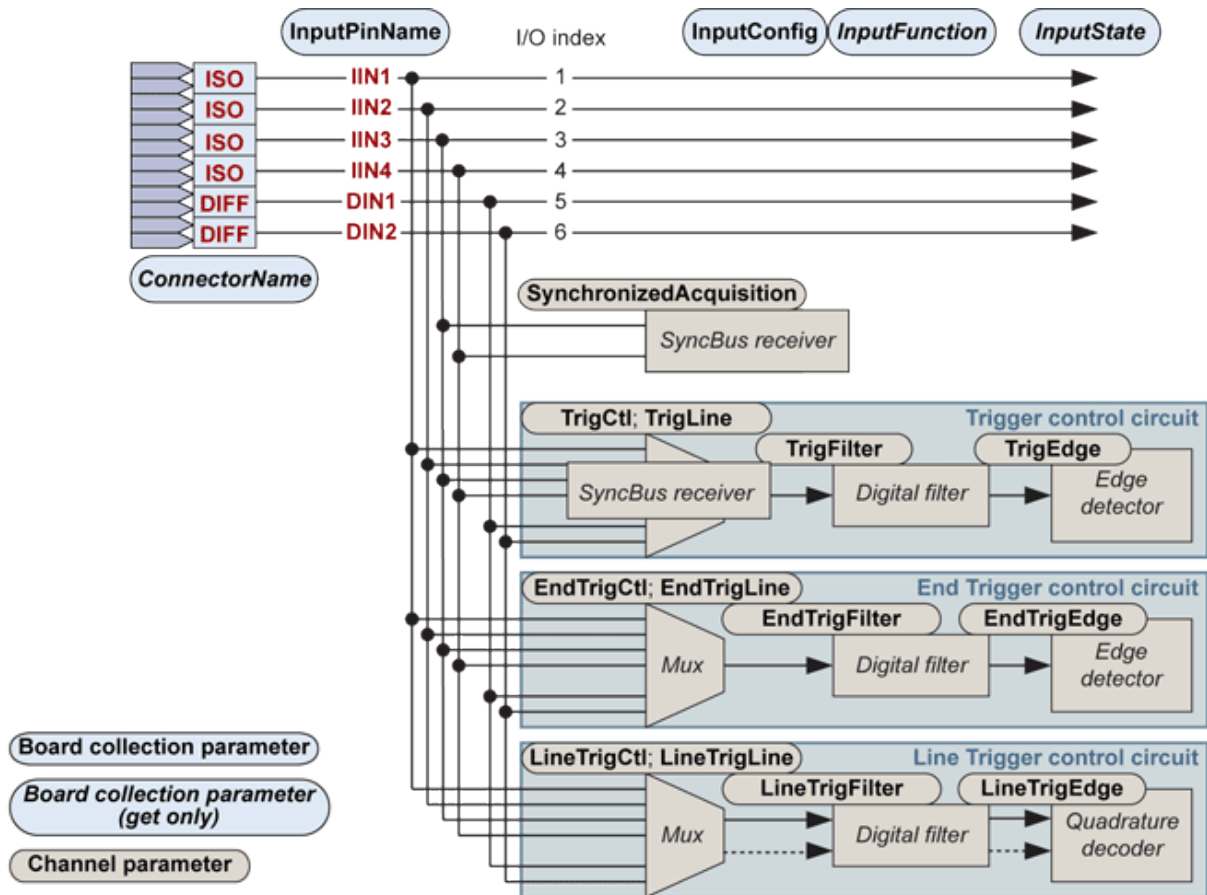
Index	ConnectorName	OutputPinName	OutputStyle
1	IO	IOUT1	ISO
2	IO	IOUT2	ISO
3	IO	IOUT3	ISO
4	IO	IOUT4	ISO
7	CAMERA	CC1	CHANNELLINK
8	CAMERA	CC2	CHANNELLINK
9	CAMERA	CC3	CHANNELLINK
10	CAMERA	CC4	CHANNELLINK
25	BRACKET	LED	NA



NOTE
The I/O indices 0, 5, 6, and {11 24} have no output-related function.

6.4. Input Ports Structure

Applies to: Base DualBase Full FullXR



General structure of the input ports of a set of system I/O

Possible usages of the input ports are:

- General-purpose inputs
- "Isolated I/O SyncBus Receiver" on page 113
- Trigger input
- EndTrigger input
- LineTrigger input

An **output multiplexer** that selects the signal to be issued on the output port. Possible selections are :

- "LOW" to connect any of the 4 output port to the logical level corresponding to the OFF state of the opto-coupler
- "HIGH" to connect any of the 4 output port logical level corresponding to the OFF state of the opto-coupler
- "EVSIGx" to connect any of the four output port to the signal issued by the respective event signal generator
- "STROBE" to connect the IOUT1 port to the signal produced by the acquisition and camera controller
- "SB1" to connect the IOUT3 port to the first of the two signals produced by the SyncBus transmitter of the acquisition and camera controller
- "SB2" to connect the IOUT4 port to the second of the two signals produced by the SyncBus transmitter of the acquisition and camera controller
- The **ISO electrical interface** built with an opto-coupler device.
- A **readback circuit** allowing getting at any time the actual logic state of the output multiplexer.

Possible usages of the output ports

Port usage	IOUT1	IOUT2	IOUT3	IOUT4
General-purpose output	OK	OK	OK	OK
Event signaling	OK	OK	OK	OK
Strobe output	OK	-	-	-
"Isolated I/O SyncBus Driver" on page 112	-	-	OK	OK

The output ports are primarily managed using parameters belonging to the Board **Input/Output Control Category** . However, for the strobe function and the SyncBus configuration, the ports are configured using Channel class parameters.

The following topics provide explanations about the **selection of the port usage** and each output port usage.

6.6. General-Purpose Inputs

All the system I/O input port can be used as a general-purpose digital input port.

For that usage, use the MultiCam parameters belonging to the board [Input/Output Control Category](#).

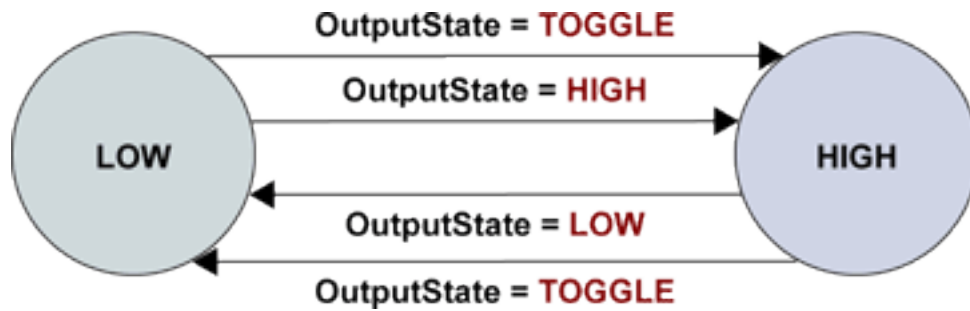
Prior to the first attempt to get the state of an input port, it is mandatory to configure the port for that usage by assigning the value **SOFT** to the corresponding member of the collection parameter **InputConfig**.

When configured for general-purpose input usage, the corresponding member of the get only **InputFunction** parameter reports the value **SOFT**.

The digital state of the input port can then be read at any time by getting the value of the corresponding member of the **InputState** parameter.

6.7. General-Purpose Output

When configured for general-purpose usage, the output multiplexer is restricted to two positions: LOW and HIGH. The following state diagram shows the 2 states and all the possible inter-state transition:



Output multiplexer state diagram (OutputFunction = SOFT)

The position of the output multiplexer is controlled by means of **OutputState**, a Board class MultiCam parameter belonging to the [Input/Output Control Category](#).

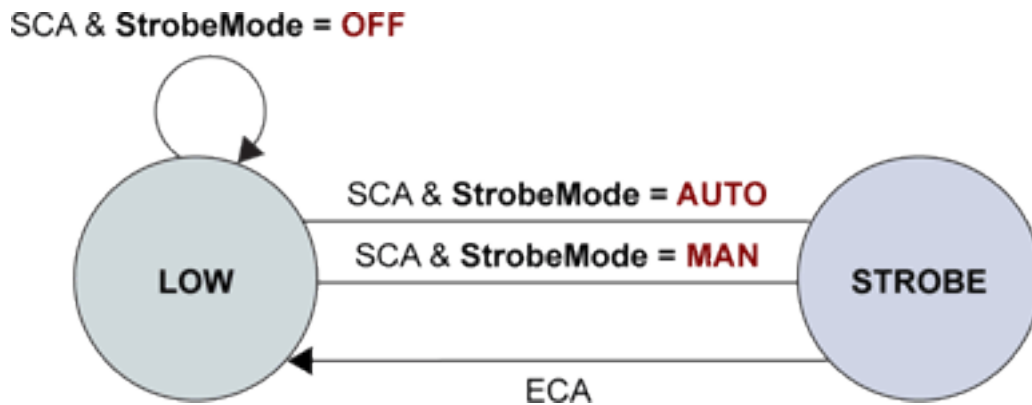
The "LOW" state means that the output multiplexer is in the "LOW" position. Setting **OutputState** to **LOW** forces immediately the output multiplexer to the "LOW state".

The "HIGH" state means that the output multiplexer is in the "HIGH" position. Setting **OutputState** to **HIGH** forces immediately the output multiplexer to the "HIGH state".

Setting **OutputState** to **TOGGLE** forces immediately the output multiplexer to change its position from LOW to HIGH, if it was at the LOW position or vice-versa.

6.8. Strobe Output

When **InputFunction** is **UNKNOWN**, the output multiplexer of IOUT1 is under control of the **StrobeMode** parameter.



Output multiplexer state diagram (OutputFunction = UNKNOWN)

The output multiplexer is reconfigured at channel activation (SCA event) and deactivation (ECA event) according to the value of **StrobeMode**:

- When **StrobeMode** = **AUTO** or **MAN**, the output multiplexer is configured as follows:
 - At channel activation, it is forced to the STROBE position, allowing the STROBE signal to drive the opto-coupler of the output port.
 - At channel deactivation, it is forced to the LOW position, turning off the opto-coupler and preventing any strobe pulses to occur while the channel is deactivated.
- When **StrobeMode** = **OFF**, the output multiplexer is configured at channel activation to the LOW position, turning off the opto-coupler and preventing any strobe pulses to occur during channel activity. The OFF state persists after channel deactivation.

When **StrobeMode** = **NONE**, the output multiplexer is left unchanged.

6.9. Event Signaling

Applies to: Base DualBase Full FullXR

When configured for event signaling usage, the output multiplexer is restricted to one position that selects the output of the event signal generator.

The generator is configured by means of **SetSignal** and **ResetSignal**.

SetSignal is a collection parameter that configures the multiplexer on the set branch of the SR flip-flop.

ResetSignal is a collection parameter that configures the multiplexer on the reset branch of the SR flip-flop.

All the multiplexers of the event signal generator exhibit the same set of events sources.

- Start and end of : channel activity, acquisition phase, acquisition sequence
- Rising and falling edges of Camera Link downstream control signals: FVAL, LVAL, DVAL
- Rising and falling edges of Camera Link upstream control signals: CC1, CC2, CC3, CC4

In addition, the multiplexer can be set to the position NONE ensuring that no more event are considered.



NOTE

When the set and reset condition are identical, the SR flip/flop toggles at every event.

6.10. Isolated I/O SyncBus Driver

Applies to: Base DualBase Full FullXR

When **InputFunction** is **UNKNOWN**, the output multiplexer of IOUT3 and IOUT4 are under control of the **SynchronizedAcquisition** parameter.

The output multiplexer is reconfigured at channel activation (SCA event) according to the value of **SynchronizedAcquisition** :

- When **SynchronizedAcquisition** = **MASTER** or **LOCAL_MASTER**, it is forced to the SYNCBUS position, allowing the respective signal of the SyncBus transmitter to drive the output port.
- For other values of **SynchronizedAcquisition**, the output multiplexer is left unchanged.

At channel deactivation, the output multiplexer remains unchanged.

6.11. Isolated I/O SyncBus Receiver

Applies to: Base DualBase Full FullXR

For applications using the [synchronized acquisition](#) feature, **IIN3** and **IIN4** ports can be used for the SyncBus receiver. For that usage, use the **SynchronizedAcquisition** parameter.

7. Triggers

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7.1. Trigger Input Controls

About Trigger Input

When the Grablink board is configured for area-scan acquisition using the **SNAPSHOT** or the **HFR** acquisition modes, a **(frame) trigger pulse** is an electrical signal sent by the external system to instruct the frame grabber to take control over the camera, including exposure control, and to perform a frame acquisition. This is usually used when an asynchronous capture of a moving object is involved. The trigger pulse is issued by a position sensor indicating when the observed object is adequately located in the field of view.

When the Grablink board is configured for line-scan acquisition using the **PAGE** or the **LONGPAGE** acquisition modes, a **(page) trigger pulse** is an electrical signal sent by the external system to instruct the frame grabber to perform the acquisition of a set of several successive lines. This is usually used when a moving object is about to enter the field of view of the line-scan camera.

Each MultiCam channel elaborates a clean **trigger** event using a dedicated set of hardware resources including: source multiplexer, edge detector, noise filter, delay line and decimation filter.

Function Availability

The hardware trigger input function is available for a restricted set of **AcquisitionMode**, **TrigMode** and **NextTrigMode** acquisition control parameters.

In the following tables, a OK indicates that the hardware trigger input function is effectively used:

SNAPSHOT Acquisition Mode

TrigMode	NextTrigMode				
	HARD	SOFT	COMBINED	SAME	REPEAT
IMMEDIATE	OK	-	OK	-	-
HARD	OK	OK	OK	OK	OK
SOFT	OK	-	OK	-	-
COMBINED	OK	OK	OK	OK	OK

HFR Acquisition Mode

TrigMode	NextTrigMode				
	HARD	SOFT	COMBINED	SAME	REPEAT
IMMEDIATE	OK	-	OK	-	-
HARD	OK	OK	OK	OK	OK
SOFT	OK	-	OK	-	-
COMBINED	OK	OK	OK	OK	OK

PAGE Acquisition Mode

TrigMode	NextTrigMode			
	HARD	SOFT	COMBINED	SAME
IMMEDIATE	OK	-	OK	-
HARD	OK	OK	OK	OK
SOFT	OK	-	OK	-
COMBINED	OK	OK	OK	OK

LONGPAGE Acquisition Mode

TrigMode	NextTrigMode	
	SAME	REPEAT
IMMEDIATE	-	-
HARD	Invalid	OK
SOFT	Invalid	-
COMBINED	Invalid	OK

Source Selection and Electrical Style Control Value

Applies to: Value

The trigger signal can originate from the following type of devices:

1. A TTL compatible detector attached to the **TRG input port** or any of the two **Isolated I/O ports**
2. A 12V CMOS compatible device attached to any of the two **Isolated I/O ports**

Sourcing device type	TrigCtl	Input Port	TrigLine
TTL	TTL	TRG	NOM or TRG
TTL and 12V CMOS	ITTL I12V	A1	ALT1
		A2	NOM or ALT2

To select a port:

1. Set the value of the **TrigCtl** parameter corresponding to the electrical style of the sensor device used as trigger source.
2. Optionally, set the value of the **TrigLine** parameter corresponding to the I/O port used to attach the trigger detector.



NOTE

The default value of **TrigLine** is **NOM**.

The hardware trigger input ports are available on the [System I/O Connector](#).

Source Selection and Electrical Style Control Express

Applies to: Express

The trigger signal can originate from the following type of devices:

1. A TTL compatible detector attached to any of the four **Enhanced I/O ports** or any of the two **Isolated I/O ports**
2. A 12V CMOS compatible device attached to any of the two **Isolated I/O ports**
3. A LVDS or RS-422 compatible detector attached to any of the two **differential input ports**

Sourcing device type	TrigCtl	Input Port	TrigLine
TTL	TTL	Enhanced IO1	NOM or IO1
		Enhanced IO2	IO2
		Enhanced IO3	IO3
		Enhanced IO4	IO4
TTL and 12V CMOS	ITTL I12V	IsoA1	NOM or ISOA1
		IsoA2	ISOA2
LVDS or RS-422	LVDS	TRA1	NOM or TRA1
		TRA2	TRA2

To select a port:

1. Set the value of the **TrigCtl** parameter corresponding to the electrical style of the sensor device used as trigger source.
2. Optionally, set the value of the **TrigLine** parameter corresponding to the I/O port used to attach the trigger detector.



NOTE

The default value of **TrigLine** is **NOM**.

The hardware trigger input ports are available on the (External) **System Connector** and on the **Internal System Connector**

Source Selection and Electrical Style Control

Base

DualBase

Full

FullXR

Applies to:

Base

DualBase

Full

FullXR

The trigger signal can originate from the following type of devices:

1. An RS-422 compatible detector attached to any of the two **high-speed differential input ports** belonging to the channel or ...
2. ... another type of device attached to any of the 4 **isolated current-sense input ports** belonging to the channel.

Sourcing device type	TrigCtl	Input Port	TrigLine
RS-422	DIFF	Diff. Input #1	DIN1
		Diff. Input #2	NOM or DIN2
Other detectors	ISO	Isolated input #1	IIN1
		Isolated input #2	NOM or IIN2
		Isolated input #3	IIN3
		Isolated input #4	IIN4

To select a port:

1. Set the value of the **TrigCtl** parameter corresponding to the electrical style of the sensor device used as trigger source.
2. Optionally, set the value of the **TrigLine** parameter corresponding to the I/O port used to attach the trigger detector.



NOTE

The default value of **TrigLine** is **NOM**.

The hardware trigger input ports are available on Internal IO and External IO connectors:

Product	Camera	Connector(s)	Trigger Input Ports
Base	-	Internal I/O External I/O	DIN1, DIN2 IIN1, IIN2, IIN3, IIN4
DualBase	A	Channel A Internal I/O	DIN1A, DIN2A IIN1A, IIN2A, IIN3A, IIN4A
		External I/O	DIN1A, DIN2A IIN1A, IIN2A
DualBase	B	Channel B Internal I/O	DIN1B, DIN2B IIN1B, IIN2B, IIN3B, IIN4B

Product	Camera	Connector(s)	Trigger Input Ports
		External I/O	DIN1B, DIN2B IIN1B, IIN2B
Full FullXR	-	Internal I/O External I/O	DIN1, DIN2 IIN1, IIN2, IIN3, IIN4

Filter Control

The hardware signal flows through a digital filter that removes any pulse narrower than its time constant.

The filter strength is configurable in 3 steps by means of the **TrigFilter** parameter. Each step corresponds to a specific filter time constant.

TrigFilter	Time Constant
OFF	100 nanoseconds
ON or MEDIUM	500 nanoseconds
STRONG	2.5 microseconds (Default value)



TIP

To avoid unexpected loss of trigger events, check that the selected time constant is shorter than the trigger pulse width sent by the detector!

Product specific notes

Product	Description
Value	The value OFF is not allowed for isolated ports A1 and A2
Express	The value OFF is not allowed for isolated ports IsoA1 and IsoA2
Base DualBase Full FullXR	There is no digital filter for isolated inputs.

Polarity Control

A trigger event is generated on a positive-going or a negative-going transition of the electrical signal.

To select the transition, set accordingly the value of the **TrigEdge** parameter:

TrigEdge	Description
GOHIGH	The trigger event is generated at each positive-going transition of the trigger line
GOLOW	The trigger event is generated at each negative-going transition of the trigger line

The default value for **TrigEdge** is **GOHIGH**.

Delay Control

For area-scan cameras operated with the **SNAPSHOT** or the **HFR** acquisition modes, the hardware frame trigger signal can be delayed by a **user-programmable time delay**.

For (TDI) line-scan cameras operated with the **PAGE** or the **LONGPAGE** acquisition modes, the hardware page trigger signal can be delayed by a **user-programmable number of captured lines**.

The following table shows the respective control parameters, their value range and the default value:

AcquisitionMode	Parameter	Value Range	Default Value
SNAPSHOT or HFR	TrigDelay_us	0 to 2,000,000 (2 seconds)	0
PAGE or LONGPAGE	PageDelay_Ln	0 to 65,534 lines	0



NOTE

The "number of captured lines" is equal to the "number of camera cycles" when the frame grabber is configured to capture all lines. When the frame grabber performs "downweb resampling", the "number of captured lines" might be different than the "number of camera cycles".



NOTE

When the downweb line rate is linked to the web speed through an encoder, the delay expressed as a number of captured lines represents a fixed length on the web!

Decimation Control

Applies to: Base DualBase Full FullXR

The trigger decimation feature applies:

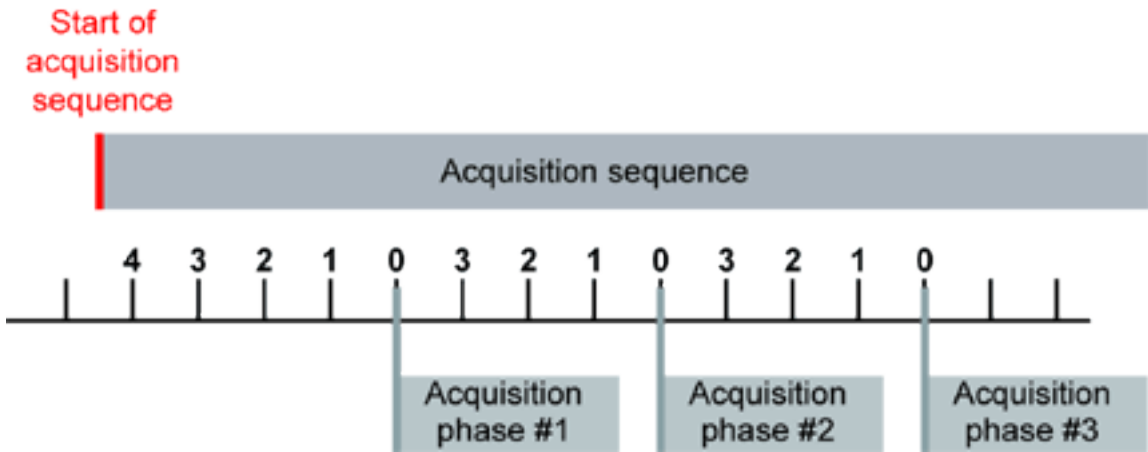
- when a hardware frame trigger source is selected for the area-scan **SNAPSHOT** or **HFR** acquisition modes.
- When a hardware page trigger source is selected for the (TDI) line-scan **WEB**, **PAGE** or **LONGPAGE** acquisition modes.

Trigger decimation discards a configurable number of trigger events **after the start of acquisition sequence** and **after every start of acquisition phase**.

TrigDelay_Pls specifies the number of detected pulses on the hardware trigger line to be skipped after the acquisition sequence begins.

NextTrigDelay_Pls specifies the number of detected pulses on the hardware trigger or page trigger line to be skipped between successive acquisition phases.

Parameter	Value Range	Default Value
TrigDelay_Pls	0 to 65,536 pulses	0
NextTrigDelay_Pls	0 to 65,536 pulses	0



Acquisition rate and trigger decimation, with TrigDelay_Pls = 4 and NextTrigDelay_Pls = 3

**WARNING**
NextTrigDelay_Pls is irrelevant for the LONGPAGE acquisition mode.

7.2. End Trigger Input Controls

About End Trigger Input

When the Grablink board is configured for line-scan acquisition using the **LONGPAGE** acquisition mode, a **end trigger pulse** is an electrical signal sent by the external system to instruct the frame grabber to stop the acquisition of a set of several successive lines.

Each MultiCam channel elaborates a clean **end trigger** event using a dedicated set of hardware resources including: source multiplexer, edge detector, noise filter and delay line.

Function Availability

The hardware end trigger input function is available for a restricted set of **AcquisitionMode**, **TrigMode** and **NextTrigMode** acquisition control parameters.

In the following tables, a OK indicates that the hardware end trigger input function is effectively used:

LONGPAGE Acquisition Mode

TrigMode	NextTrigMode	
	SAME	REPEAT
IMMEDIATE	-	-
HARD	Invalid	OK
SOFT	Invalid	-
COMBINED	Invalid	OK

Source Selection and Electrical Style Control Value

Applies to: Value

The end trigger signal can originate from the following type of devices:

1. A TTL compatible detector attached to the **TRG input port** or to the **ALT2 Isolated I/O port**
2. A 12V CMOS compatible device attached to the **ALT2 Isolated I/O port**

Sourcing device type	End TrigCtl	Input Port	EndTrigLine
TTL	TTL	TRG	NOM
TTL and 12V CMOS	ITTL I12V	A2	NOM

To select a port:

1. Set the value of the **EndTrigCtl** parameter corresponding to the electrical style of the sensor device used as trigger source.



NOTE

The unique and default value of **EndTrigLine** is **NOM**.

The hardware end trigger input ports are available on the [System I/O Connector](#).

Source Selection and Electrical Style Control Express

Applies to: Express

The end trigger signal can originate from the following type of devices:

1. A TTL compatible detector attached to any of the four **Enhanced I/O ports** or any of the two **Isolated I/O ports**
2. A 12V CMOS compatible device attached to any of the two **Isolated I/O ports**
3. A LVDS or RS-422 compatible detector attached to any of the two **differential input ports**

Sourcing device type	EndTrigCtl	Input Port	EndTrigLine
TTL	TTL	Enhanced IO1	NOM or IO1
		Enhanced IO2	IO2
		Enhanced IO3	IO3
		Enhanced IO4	IO4
TTL and 12V CMOS	ITTL I12V	IsoA1	NOM or ISOA1
		IsoA2	ISOA2
LVDS or RS-422	LVDS	TRA1	NOM or TRA1
		TRA2	TRA2

To select a port:

1. Set the value of the **EndTrigCtl** parameter corresponding to the electrical style of the sensor device used as trigger source.
2. Optionally, set the value of the **EndTrigLine** parameter corresponding to the I/O port used to attach the trigger detector.



NOTE

The default value of **EndTrigLine** is **NOM**.

The hardware trigger input ports are available on the (External) **System Connector** and on **the Internal System Connector**

Source Selection and Electrical Style Control

Base

DualBase

Full

FullXR

Applies to:

Base

DualBase

Full

FullXR

The end trigger signal can originate from the following type of devices:

1. An RS-422 compatible detector attached to any of the two **high-speed differential input ports** belonging to the channel or ...
2. ... another type of device attached to any of the 4 **isolated current-sense input ports** belonging to the channel.

Sourcing device type	EndTrigCtl	Input Port	EndTrigLine
RS-422 compatible detector	DIFF	Diff. Input #2	NOM
		Diff. Input #1	DIN1
		Diff. Input #2	DIN2
Other detectors	ISO	Isolated input #2	NOM
		Isolated input #1	IIN1
		Isolated input #2	IIN2
		Isolated input #3	IIN3
		Isolated input #4	IIN4

To select a port:

1. Set the value of the **EndTrigCtl** parameter corresponding to the electrical style of the sensor device used as end trigger source.
2. Optionally, set the value of the **EndTrigLine** parameter corresponding to the I/O port used to attach the end trigger detector.



NOTE

The default value of **EndTrigLine** is **NOM**.

The hardware end trigger input ports are available on Internal IO and External IO connectors:

Product	Camera	Connector(s)	End Trigger Input Ports
Base	-	Internal I/O Connector External I/O Connector	DIN1, DIN2 IIN1, IIN2, IIN3, IIN4
DualBase	A	Channel A Internal I/O Connector	DIN1A, DIN2A IIN1A, IIN2A, IIN3A, IIN4A
		External I/O Connector	DIN1A, DIN2A IIN1A, IIN2A

Product	Camera	Connector(s)	End Trigger Input Ports
DualBase	B	Channel B Internal I/O Connector	DIN1B, DIN2B IIN1B, IIN2B, IIN3B, IIN4B
		External I/O Connector	DIN1B, DIN2B IIN1B, IIN2B
Full FullXR	-	Internal I/O Connector External I/O Connector	DIN1, DIN2 IIN1, IIN2, IIN3, IIN4

Filter Control

Applies to: Base DualBase Full FullXR

The hardware signal issued from **differential inputs only** flows through a digital filter that removes any pulse narrower than its time constant.

The filter strength is configurable in 3 steps by means of the **EndTrigFilter** parameter. Each step corresponds to a specific filter time constant.

EndTrigFilter	Time Constant
OFF	100 nanoseconds
ON or MEDIUM	500 nanoseconds
STRONG	2.5 microseconds (Default value)



TIP

To avoid unexpected loss of end trigger events, check that the selected time constant is shorter than the end trigger pulse width sent by the detector!

Product specific notes

Product	Description
Value	The value OFF is not allowed for isolated ports A1 and A2
Express	The value OFF is not allowed for isolated ports IsoA1 and IsoA2
Base DualBase Full FullXR	There is no digital filter for isolated inputs.

Polarity Control

A end trigger event is generated on a positive-going or a negative-going transition of the electrical signal.

To select the transition, set accordingly the value of the **EndTrigEdge** parameter:

EndTrigEdge	Description
GOHIGH	The end trigger event is generated at each positive-going transition of the trigger line
GOLOW	The end trigger event is generated at each negative-going transition of the trigger line

The default value for **EndTrigEdge** is **GOLOW**.

Delay Control

For (TDI) line-scan cameras operated with the **LONGPAGE** acquisition mode, the hardware page end trigger signal can be delayed by a **user-programmable number of captured lines**.

The following table shows the control parameter, its value range and its default value:

AcquisitionMode	Parameter	Value Range	Default Value
LONGPAGE	EndPageDelay_Ln	0 to 65,534 lines	0



NOTE

The "number of captured lines" is equal to the "number of camera cycles" when the frame grabber is configured to capture all lines. When the frame grabber performs "downweb resampling", the "number of captured lines" might be different than the "number of camera cycles".



NOTE

When the downweb line rate is linked to the web speed through an encoder, the delay expressed as a number of captured lines represents a fixed length on the web.!

8. Line Trigger Controls

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8.1. Line Trigger Input

Applies to: Base DualBase Full FullXR

For line-scan applications using a hardware acquisition line trigger, anyone or some pairs of the system I/O input ports can be selected as the source(s) for the line trigger control circuit of the acquisition channel.

For that usage, use the parameters belonging to the [Encoder Control Category](#).

The line trigger signal can be elaborated from one or two external signals provided by one of the following type of devices:

- Through **one pair of high-speed differential input ports** driven by an RS-422 compatible dual output phase quadrature incremental motion encoder.
- Through **one pair of isolated current-sense input ports** driven by a dual output phase quadrature incremental motion encoder that is not RS-422 compliant.
- Through **a single high-speed differential input port** driven by an RS-422 compatible single output incremental motion encoder or a similar device. This can be a dual output type of incremental motion encoder for which only one output signal is connected.
- Through **a single isolated current-sense input port** driven by an incremental motion encoder or a similar device that is not RS-422 compliant.

Possible ports assignments for each case

Sourcing device type	Default port(s) assignment	Alternate port assignment(s)
RS-422 compatible dual output phase quadrature incremental motion encoder	High-speed diff. inputs #1 and #2	-
Other dual output phase quadrature incremental motion encoder	Isolated input #1 and #2	Isolated input #3 and #4
RS-422 compatible single output incremental motion encoder or similar device	High-speed diff. input #1	High-speed diff. input #2
Single output incremental motion encoder or similar device	Isolated input #1	Isolated input #2 Isolated input #3 Isolated input #4



NOTE

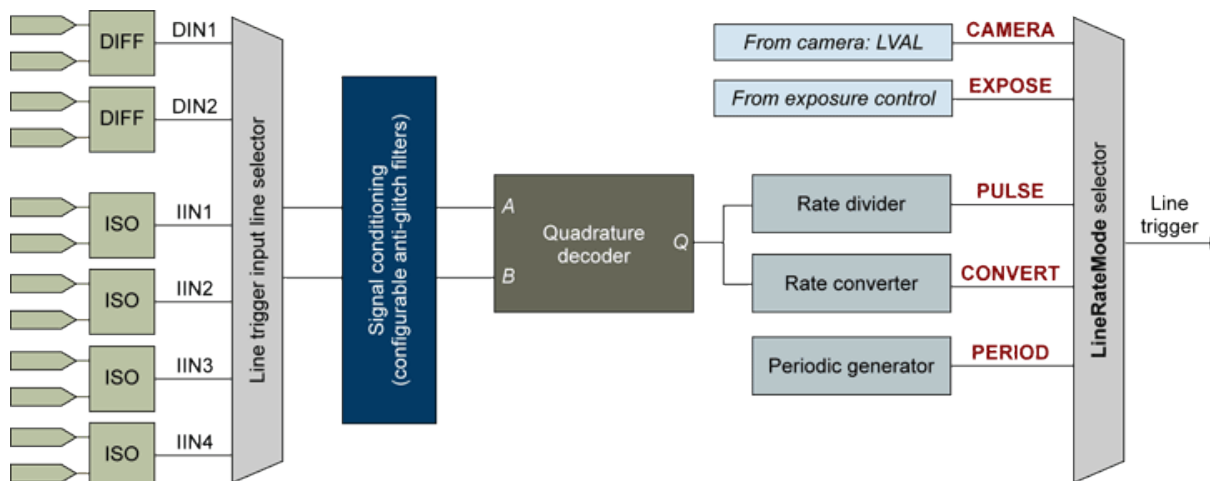
Any input port and hence any electrical style can be specified for that function.

**NOTE**

The default port assignment for both single signal electrical styles is different of the one of the trigger.

8.2. Line Trigger Overview

The "line trigger" signal triggers the acquisition of one line of the image.



Block diagram



NOTE

The line trigger is available only for the line-scan acquisition modes of MultiCam, namely **WEB**, **PAGE**, and **LONGPAGE**. It is inoperative for the area-scan acquisition mode.



NOTE

The line trigger is effective within a MultiCam acquisition sequence. Any line trigger applied outside a MultiCam acquisition sequence is lost.

8.3. Source Path Selector

The line trigger has five possible source paths. The path is selectable by means of the MultiCam parameter **LineRateMode**:

- When **LineRateMode** is set to **PULSE**, the line trigger signal is generated by the quadrature decoder circuit through a 1/N rate divider.
- When **LineRateMode** is set to **CONVERT**, the line trigger signal is generated by the quadrature decoder circuit through a P/Q rate converter.
- When **LineRateMode** is set to **PERIOD**, the line trigger signal is generated by a periodic pulse generator.
- When **LineRateMode** is set to **CAMERA**, the line trigger is generated by the camera through the LVAL signal. The time interval between line triggers is actually driven by the camera.
- When **LineRateMode** is set to **EXPOSE**, the line trigger is generated by the exposure control circuit. The time interval between line triggers is actually driven by the exposure time.

8.4. Rate Divider

The rate divider circuit generates a line trigger signal at a frequency that is an integer fraction $1/N$ of the frequency of the pulses delivered by the quadrature decoder circuit.

For N consecutive incoming pulses issued by the quadrature decoder circuit, the $1/N$ rate divider:

- generates one output pulse (one line trigger)
- skips $N-1$ input pulse

The rate divider is initialized at the beginning of every MultiCam acquisition sequence. The first output pulse is produced from the first clock input pulse occurring after the sequence trigger event.



Beginning of an acquisition sequence for a value N of 4

The division factor N is user-programmable. Possible values of N include all integer numbers ranging from 1 up to 512, the default value is 1.

N is configured by the MultiCam parameter **RateDivisionFactor**.

Notes

- The output frequency is lower than ($N > 1$) or equal to ($N = 1$) the input frequency. It cannot be higher.
- The output pulse is generated immediately after a non-skipped input pulse. The line trigger pulses are phase-locked to the quadrature decoder output.
- The rate divider settings may not be modified while acquisition is in progress.

8.5. Rate Converter

The rate converter circuit generates a line trigger signal at a frequency that is proportional to the frequency of the pulses delivered by the quadrature decoder circuit.

Rate conversion ratio

The rate converter is capable to multiply or divide the input rate of encoder ticks by any rational number ranging from 1/1000 up to 1000/1

The rate conversion ratio RCR is defined as the ratio between the output rate and the input rate:

$$\text{RCR} = \text{EncoderPitch} / \text{LinePitch}$$

The possible values of RCR range from 0.001 up to 1000.0. The recommended values range from 0.01 up to 1.

Operating range

The rate converter operates within a limited range of frequencies:

- The upper limit of the output frequency range is user configurable with parameter **MaxSpeed**. By default, it is set to the maximum line rate sustainable by the camera defined by **LineRate_Hz**.
- The lower limit of the output frequency takes is automatically set by the driver and reported to the application via the **MinSpeed** parameter. The **MaxSpeed/MinSpeed** ratio is typically greater than 100.

When the input rate drops below **MinSpeed**, the rate converter behaves according to the **OnMinSpeed** setting:

- When set to **MUTING**, it stops delivering line trigger ticks.
- When set to **IDLING**, it continues delivering line trigger ticks at a constant frequency.



WARNING

To enlarge the usable speed range, it is mandatory to set **MaxSpeed** at a value slightly above the actual max camera line rate.

Notices

- The line trigger pulses are NOT phase-locked to the quadrature decoder output.
- The rate converter settings may not be modified while acquisition is in progress.

8.6. Periodic Generator

The periodic generator circuit generates a line trigger signal at a constant frequency.

The time interval T between two consecutive pulses is user programmable. Values of T range from 1 μs up to 5 seconds.

The period T is configured by **Period_us** and **PeriodTrim**.

8.7. Line Trigger Line Selection

When **LineRateMode** is **PULSE** or **CONVERT**, the line trigger signal originates from an external motion encoder device. In that case line trigger input port(s) must be selected.

The selection is performed in two steps:

- Determine the electrical style of the input port(s) and set **LineTrigCtl** accordingly. MultiCam selects the default input port(s) for the selected style.
- Optionally select one alternate choice for the input ports by setting the **LineTrigLine** parameter.

LineTrigCtl	Description	Targeted type of motion encoder device
DIFF	Single differential high-speed input port compatible with EIA/TIA-422 signaling	Single output RS-422 compatible incremental motion encoder
DIFF_PAISED	Pair of differential high-speed input ports compatible with EIA/TIA-422 signaling. Default value.	Dual-output RS-422 compatible phase quadrature incremental motion encoder
ISO	Single isolated current loop input port compatible with TTL, +12V, + 24V signaling	Single output incremental motion encoder
ISO_PAISED	Pair of isolated current loop input ports compatible with TTL, +12V, + 24V signaling	Dual-output phase quadrature incremental motion encoder

LineTrigCtl	Default port(s) assignment	Alternate port assignment(s)
DIFF	DIN1	DIN2
DIFF_PAISED	DIN1 (A) + DIN2 (B)	-
ISO	IIN1	IIN2 IIN3 IIN4
ISO_PAISED	IIN1 (A) + IIN2 (B)	IIN3 (A) + IIN4 (B)

Notes

- Dual output devices where only one output is connected are assimilated to single output devices.
- Any input port and hence any electrical style can be specified for that function.
- The default port assignment for both single signal electrical styles is different of the one of the trigger.

8.8. Signal Conditioning

Applies to: Base DualBase Full FullXR

The hardware Trigger and EndTrigger signals issued from differential inputs are conditioned before being applied to the acquisition controller.

Each signal is sampled at a constant frequency of 50 MHz and flows through a digital filter.

With such a filter:

- All the pulses having a duration larger than $T_{HIGH}[ns]$ are transmitted to the output.
- All the pulses having a duration smaller than $T_{LOW}[ns]$ are blocked.
- Pulses having duration in between the above mentioned limits can be transmitted or blocked.

The following table shows for each possible value of the respective **TrigFilter** and **EndTrigFilter** parameters:

- $T_{LOW}[ns]$: the lower limit of the time constant
- $T_{HIGH}[ns]$: the upper limit of the time constant

TrigFilter EndTrigFilter	$T_{LOW}[ns]$	$T_{HIGH}[ns]$	Note
OFF	96	112	
ON	496	512	
MEDIUM	496	512	
STRONG	2496	2512	Default value

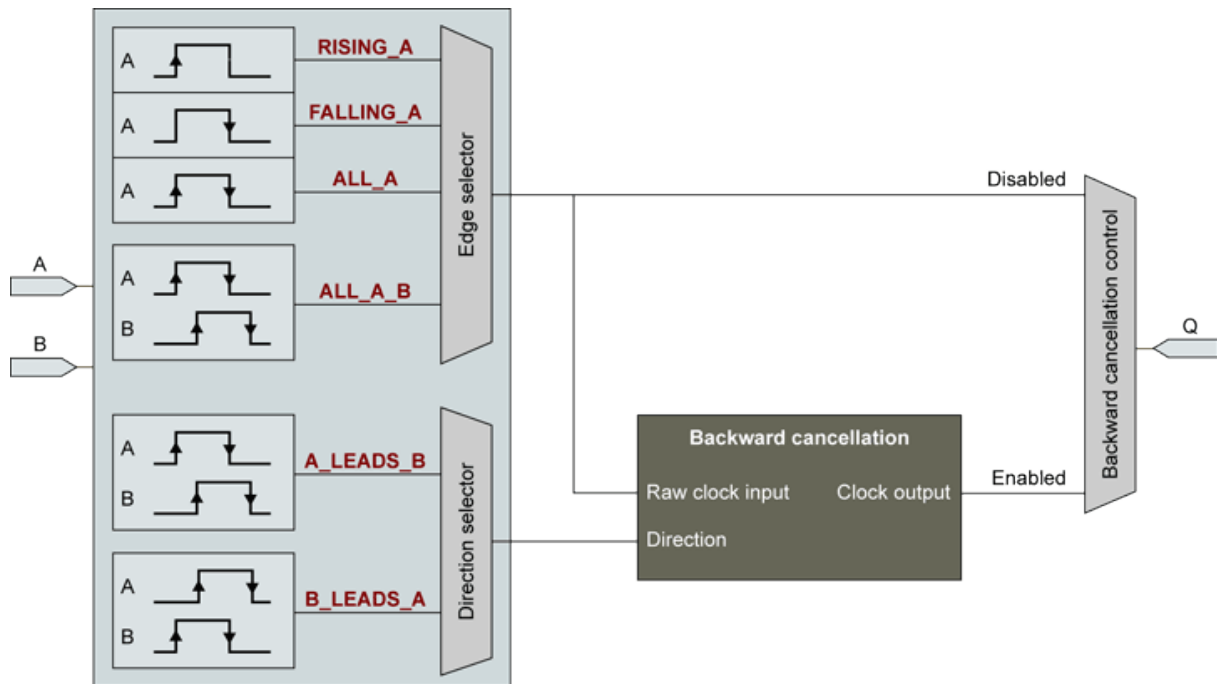


NOTE

TrigFilter and **EndTrigFilter** are not relevant for isolated inputs.

8.9. Quadrature Decoder

The quadrature decoder circuit interfaces directly with dual output phase quadrature incremental motion encoders. It decodes the encoder A and B signals and generates pulses on the Q output.



Quadrature decoder block diagram

Edge Selection

The rate of the output pulse can be 1, 2, or 4 times the frequency of the A signal according to the position of the edge selector.

- When the edge selector is in the rising A position, an output pulse is generated for every rising edge of the A signal. The falling edge on the A signal and both edges on the B-signal are ignored.
- When the edge selector is in the falling A position, an output pulse is generated for every falling edge of the A signal. The rising edge on the A signal and both edges on the B-signal are ignored.
- When the selector is in the All A position, an output pulse is generated for every rising and falling edges of the A signal. The B-signal is ignored.
- When the selector is in the All A B position, an output pulse is generated for every rising and falling edges of the A and B signals.

The edge selector is controlled by the enumerated MultiCam parameter **LineTrigEdge**.

Direction Selector

The motion direction is determined by the phase relationship of the A and B signals.

By construction, the dual-output phase quadrature incremental motion encoder maintains a phase relationship of about 90 degrees between the two signals. For motion in one direction, the A signal leads the B signal by about 90 degrees. For a motion in the other direction, the B signal leads the A signal by about 90 degrees.

The direction selector provides the capability to define which one of the phase relationships is considered as the forward direction for the application.

The default assignment identifies the case "A Leads B" as the forward direction.

Backward Motion Cancellation

The backward cancellation circuit stops sending line trigger pulses as soon as a backward motion is detected. If such an event occurs, the acquisition is stopped.

When the backward cancellation control is configured in the filter mode (F-Mode), the line acquisition resumes when the motion changes to the forward direction. Therefore, the cancellation circuit filters out all the pulses corresponding to the backward direction.

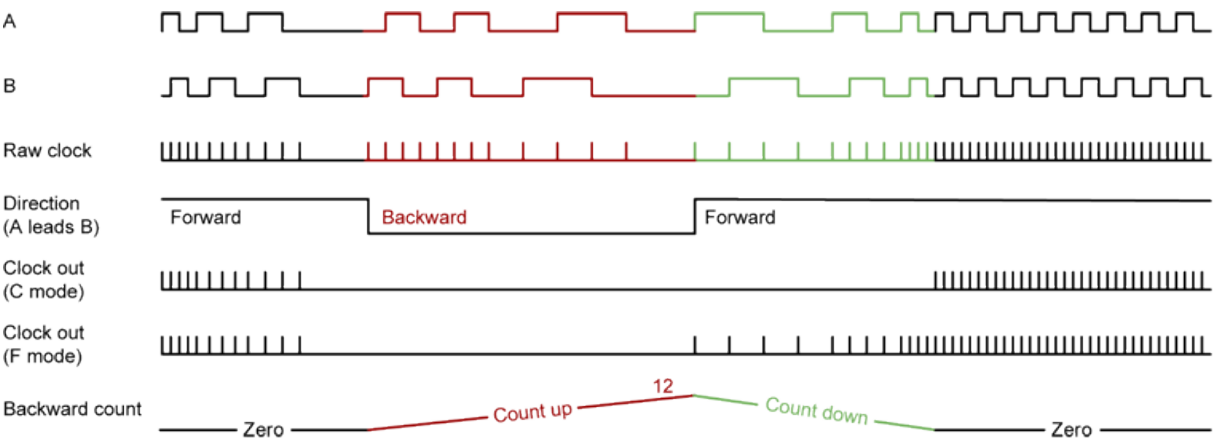
When the backward cancellation control is configured in the compensation mode (C-Mode), the line acquisition resumes when the motion changes to the the forward direction at the place it was interrupted. Therefore, the cancellation circuit filters out not only the pulses corresponding to the backward direction, but a number of forward pulses equal to the number of skipped backward pulses.

In C-Mode, the cancellation circuit uses a "backward pulse counter" that:

- Increments by 1 every clock in the backward direction
- Decrements by 1 every clock in the forward direction until it reaches 0
- Resets at the beginning of each MultiCam acquisition sequence. More precisely, at the first trigger event of the sequence. This trigger is considered as the reference for the position along the web for the whole acquisition sequence.

In C-Mode, all pulses occurring when the counter value is different of zero are blocked.

The counter has a 16-bit span, backward displacement up to 65535 pulses can be compensated.



The backward cancellation circuit operates exclusively with quadrature motion encoders when **LineRateMode = PULSE**.

The backward cancellation mode is configured by **BackwardMotionCancellationMode**.

9. Line-Scan Synchronization

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9.1. Line Capture Modes

This board provides two line capturing methods. The line capture mode is user-selectable by means of **LineCaptureMode**:

LineCaptureMode	Description
ALL	<i>"Take-All-Line" line capture mode</i> The board acquires all the lines delivered by the camera providing that the acquisition channel is active and the trigger conditions are satisfied. If the downweb motion speed is varying, the line-scanning process of the camera should be rate-controlled accordingly. This is the default line capture mode.
PICK	<i>"Pick-A-Line" line capture mode</i> Each pulse occurring at the downweb line rate determines the acquisition of the next line delivered by the camera providing that the acquisition channel is active and the trigger conditions are satisfied. This downweb resampling method allows the camera to be operated at a constant line rate while acquiring lines at a variable downweb line rate.

The relevant and the applicable values of **LineCaptureMode** depend on the two following prerequisites settings: **Imaging**, **CamConfig**.

Imaging	CamConfig	LineCaptureMode
AREA	-	Irrelevant
LINE	LxxxxSP LxxxxSC LxxxxRP	ALL (*) PICK
LINE	LxxxxRC LxxxxRG LxxxxRG2	ALL (*)
TDI	LxxxxSP LxxxxRP	ALL (*) PICK

(*) Default value.

9.2. Line Rate Modes

Line Rate Mode expresses how the *Downweb Line Rate* is determined in a line-scan acquisition system.

The user specifies the *Line Rate Mode* by means of MultiCam parameter **LineRateMode**. Five *Line Rate Modes* are identified in MultiCam:

LineRateMode	Description
CAMERA	Camera – The <i>Downweb Line Rate</i> is originated by the camera.
PULSE	Trigger Pulse – The <i>Downweb Line Rate</i> originates from a train of pulses applied on the line trigger input belonging to the grabber.
CONVERT	Rate Converter – The <i>Downweb Line Rate</i> originates from a train of pulses applied on the line trigger input and processed by a rate converter belonging to the grabber.
PERIOD	Periodic – The <i>Downweb Line Rate</i> originates from an internal periodic generator belonging to the grabber
EXPOSE	Exposure Time – The <i>Downweb Line Rate</i> is identical to the camera line rate and established by the exposure time settings

LineRateMode = CAMERA

This mode is applicable exclusively for free-run permanent exposure – **LxxxxSP** – class of line scan cameras when **LineCaptureMode** = **ALL**. The grabber does not perform any sampling in the downweb direction; the *Downweb Line Rate* is equal to the camera line rate. The camera line rate is entirely under control of the camera. Notice that most of the line scan cameras provide an internal line rate adjustment.

LineRateMode = PULSE

When the speed of motion is varying, the *Downweb Line Rate* should be slaved to this motion. To achieve this, a motion encoder is a good solution.

The motion encoder delivers an electrical pulse each time the moving web advances by a determined amount of length. The continuous motion results in a train of pulses the frequency of which is proportional to the web speed.

There exists another way to take knowledge of the web speed. In some applications, the motion is caused by a stepping motor controlled by pulses. The controlling train of pulses is also a measure of relative motion.

In both cases, the pulses are called line trigger pulses, and their repetition rate is the Line Trigger Rate. The line trigger pulses are applied to the frame grabber to determine the *Downweb Line Rate*.

Each line trigger pulse may result into the generation of one line in the acquired image. This means that the *Downweb Line Rate* is equal to the Trigger Rate.

LineRateMode = CONVERT

Alternatively to the "PULSE" mode, for more flexibility, the Line Trigger Rate may be scaled up or down to match the required *Downweb Line Rate*. The proportion between the two rates is freely programmable to any value lower or greater than unity, with high accuracy. This makes possible to accommodate a variety of mechanical setups, and still maintain a full control over the downweb resolution. The hardware device responsible for this rate conversion is called the rate converter. This device is a unique characteristic of Euresys line-scan frame grabbers.

LineRateMode = PERIOD

Other circumstances necessitate the *Downweb Line Rate* to be hardware-generated by a programmable timer, called the "periodic generator".

LineRateMode = EXPOSE

Applies to: Express Base DualBase Full FullXR

This mode is applicable exclusively for line rate controlled permanent exposure – **LxxxxRP** – class of line scan cameras when **LineCaptureMode = ALL**. The grabber does not perform any sampling in the downweb direction; the *Downweb Line Rate* is equal to the camera line rate. The camera line rate is entirely under control of the grabber through the exposure time settings.

This mode is the default and recommended mode for **LxxxxRP** class of cameras on **1621 Grablink Express**.



NOTE

This mode is not available on **1191 Grablink Value**. Instead, use **LineRateMode = PERIOD**, specify the desired exposure time through **Period_us** and assign a value to **Expose_us** that is smaller than the desired exposure time.

9.3. Operating Limits

When **LineCaptureMode** is set to **PICK** or **ALL**, the downweb line rate may not exceed the maximum camera line rate.

$$\text{Maximum downweb line rate} = \text{Maximum effective camera line rate}$$

The maximum effective camera line rate is the highest line rate that the line-scan camera can achieve in the operating conditions. It can be evaluated using following formula:

$$\text{Maximum effective camera line rate} = \text{Minimum} \left(\frac{1}{\text{Exposure time}} ; \frac{1}{\text{Readout time}} \right)$$

- For free-running operation modes: **LxxxxSP** and **LxxxxSC**, it is mandatory to set the **LineRate_Hz** parameter to a value equal to (or smaller than) the actual camera line rate.
- For grabber-controlled rate operation modes: **LxxxxRP**, **LxxxxRC**, **LxxxxRG**, and **LxxxxRG2**, it is mandatory to set the **LineRate_Hz** parameter to a value equal to (or smaller than) the maximum rate allowed by the camera.
- For grabber-controlled exposure operation modes: **LxxxxRP**, **LxxxxRG**, and **LxxxxRG2**, the exposure time is defined by means of exposure control parameters.
- For grabber-controlled rate and camera-controlled exposure operation mode: **LxxxxRC**, it is mandatory to set **TrueExp_us** to a value equal to (or higher than) the actual exposure time.

10. Strobe Control

Each MultiCam acquisition channel embeds one strobe controller capable of delivering one strobe signal.

Function Availability

The strobe function is only available for these [camera operation modes](#):

Area-scan cameras (*Imaging = AREA*)

Short name	Camera operation class
PxxRC	Asynchronous progressive scan, camera-controlled exposure
PxxRG	Asynchronous progressive scan, grabber-controlled exposure

Line-scan cameras (*Imaging = LINE*)

Short name	Camera operation class
LxxxxRC	Grabber-controlled line scanning, camera controlled exposure
LxxxxRG	Grabber-controlled line-scanning, grabber controlled exposure
LxxxxRG2	Grabber-controlled line-scanning, grabber controlled exposure, dual signal
LxxxxRP	Grabber-controlled line-scanning, permanent exposure

TDI line-scan cameras (*Imaging = TDI*)

Short name	Camera operation class
LxxxxRP	Grabber-controlled line-scanning, permanent exposure



NOTE

The strobe function is inoperative for PxxSC, LxxxxSC, and LxxxxSP camera operation modes; therefore, the **StrobeMode** parameter is filtered in MultiCam Studio for these classes of cameras.

Mode Control

The main control parameter is **StrobeMode**:

StrobeMode	Description
NONE	The strobe function is disabled. No strobe line is allocated to the channel.
MAN	The strobe function is enabled with a manual timing control feature.
AUTO	The strobe function is enabled with an automatic timing control feature.
OFF	The designated strobe line is set to the inactive level; no more strobe pulses are issued.

The default and allowable values of are depending on the camera [camera operation mode](#):

Camera operation modes	StrobeMode			
	NONE	MAN	AUTO	OFF
Grabber-controlled rate and exposurePxxRG, LxxxxRG, LxxxxRG2	OK	-	Default	OK
Grabber-controlled rate, permanent exposure LxxxxRP	OK	-	Default	OK
Grabber-controlled rate, camera controlled exposure PxxRC, LxxxxRC	OK	Default	-	OK

Duration and Position Controls

Parameter	Value Range	Default Value
StrobeDur	1 to 100	50
StrobePos	0 to 100	50

The **strobe duration** is configured as a **percentage of the exposure time** using the **StrobeDur** parameter.

The effective strobe pulsation is:

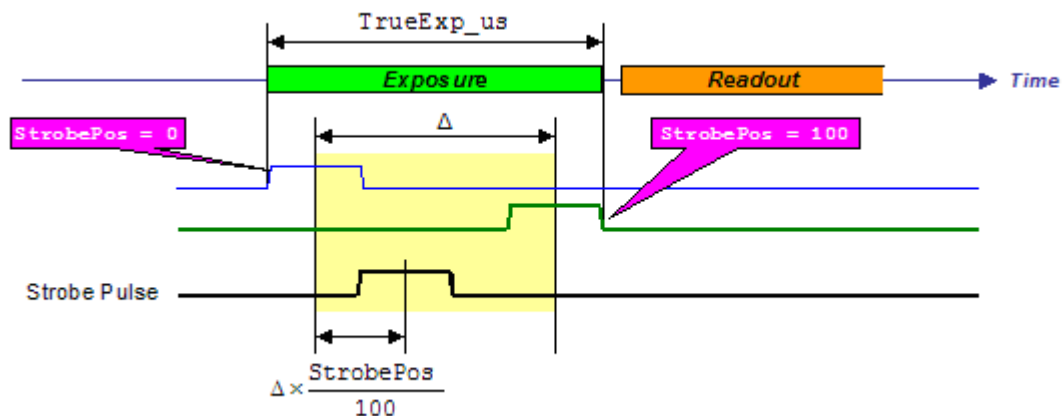
- $\text{Expose_us} \times \text{StrobeDur}$ when the exposure time is controlled by the frame grabber (RG and RP modes)
- $\text{TrueExp_us} \times \text{StrobeDur}$ when the exposure time is controlled by the camera (RC modes)

The strobe position is adjustable in 100 steps within the exposure time using the **StrobePos** parameter.

A value of 0 establishes the earliest position. The leading edge of the strobe pulse is simultaneous with the beginning of the exposure.

A value of 100 establishes the latest position. The trailing edge of the strobe pulse is simultaneous with the end of exposure.

A value of 50 % means that the strobe pulse is located in the middle of the exposure period.



Output Selection, Polarity and Electrical Style Controls

Applies to: Value

Output Line Type	StrobeCtl	StrobeLevel	Input Port	StrobeLine
TTL	TTL	PLSLOW or PLSHIGH	STR	NOM or STR
Isolated TTL	ITTL		ALT1	NOM or ALT1
			ALT2	ALT2
Isolated open-collector	IOC		ALT1	NOM or ALT1
			ALT2	ALT2
Isolated open-emitter	IOE		ALT1	NOM or ALT1
			ALT2	ALT2

To select a strobe output port:

1. Set the value of the **StrobeCtl** parameter corresponding to the electrical style of the strobe light device.
2. Set the value of the **StrobeLevel** parameter corresponding to the desired signal polarity:
3. Optionally, set the value of the **StrobeLine** parameter corresponding to the I/O port used to attach the strobe light device.



NOTE

The default value of **StrobeLine** is **NOM**.

The strobe output ports are available on the [System I/O Connector](#).

Output Selection, Polarity and Electrical Style Controls

Applies to: **Express**

Output Line Type	StrobeCtl	StrobeLevel	Input Port	StrobeLine
Opto-isolated	OPTO	PLSLOW	STA	NOM or STA
TTL	TTL	PLSLOW or PLSHIGH	Enhanced IO1	IO1
			Enhanced IO2	NOM or IO2
			Enhanced IO3	IO3
			Enhanced IO4	IO4
Isolated TTL	ITTL		IsoA1	ISOA1
			IsoA2	NOM or ISOA2
Isolated open-collector	IOC		IsoA1	ISOA1
			IsoA2	NOM or ISOA2
Isolated open-emitter	IOE		IsoA1	ISOA1
			IsoA2	NOM or ISOA2

To select a strobe output port:

1. Set the value of the **StrobeCtl** parameter corresponding to the electrical style of the strobe light device.
2. Set the value of the **StrobeLevel** parameter corresponding to the desired signal polarity:
3. Optionally, set the value of the **StrobeLine** parameter corresponding to the I/O port used to attach the strobe light device.



NOTE

The default value of **StrobeLine** is **NOM**.

The strobe output ports are available on the [System I/O Connector](#).

Output Selection and Electrical Style Control

Applies to: Base DualBase Full FullXR

These Grablink boards only have one dedicated strobe output line per camera.

The strobe output line is available on both the Internal IO and External IO connectors:

Product	Camera	Connector(s)	Pins name
Base	-	Internal I/O Connector External I/O Connector	IOUT1+/IOUT1-
DualBase	A	Channel A Internal I/O Connector External I/O Connector	IOUT1A+/IOUT1A-
DualBase	B	Channel B Internal I/O Connector External I/O Connector	IOUT1B+/IOUT1B-
Full FullXR	-	Internal I/O Connector External I/O Connector	IOUT1+/IOUT1-

The strobe line drives an optically-isolated pair of pins. The + pin is the collector and the - pin is the emitter of an uncommitted photo-transistor driven by LED-emitted light. The photo-transistor remains OFF during the board initialization.

There is no line selection control: the **Strobeline** parameter is not applicable.

There is no electrical style control. The **StrobeCtl** parameter is not applicable.



NOTE

The strobe being always enabled by default, the IOUT1 output port is configured by default as a strobe output and driven by the strobe controller. To disconnect the IOUT1 output port from the strobe generator, set **StrobeMode** to **OFF**.

11. Exposure Control

Grabber-Controlled Exposure

This board supports the following grabber-controlled exposure class of cameras:

Imaging	CamConfig	Camera operation class
AREA	PxxRG	Asynchronous progressive scan, grabber-controlled exposure
LINE	LxxxxRG	Grabber-controlled line-scanning, grabber controlled exposure
LINE	LxxxxRG2	Grabber-controlled line-scanning, grabber controlled exposure, dual signal
LINE	LxxxxRP	Grabber-controlled line-scanning, permanent exposure
TDI	LxxxxRP	Grabber-controlled line-scanning, permanent exposure

If the camera belongs to the grabber-controlled exposure class, you can specify the desired exposure time through parameters **Expose_us** and optionally **ExposeTrim**. This board provides an exposure time range of 1 μ s up to 5 s.

The lower limit of the exposure time range is defined by the camera parameter **ExposeMin_us**. The upper limit of usable exposure time range is defined by the camera parameter **ExposeMax_us**.



NOTE

Both parameters prevent using the camera outside the permitted range.

The grabber controls the exposure time through the reset line and optionally—in case of **LxxxxRG2** operation mode—through the auxiliary reset line.

Any of the four Camera Link upstream control line can be used for these purposes with independent polarity control. MultiCam selects automatically the exposure control line(s) according to the value of camera parameters **CC1Usage**, **CC2Usage**, **CC3Usage** and **CC4Usage**; it configures automatically the polarity according to camera parameters **ResetEdge** and **AuxResetEdge**.

Uncontrolled Exposure

This board supports the following grabber-controlled exposure class of cameras:

Imaging	CamConfig	Camera operation class
AREA	PxxSC	Synchronous progressive scan, camera-controlled exposure
	PxxRC	Asynchronous progressive scan, camera-controlled exposure
LINE	LxxxxSP	Free-running, permanent exposure
	LxxxxSC	Free-running, camera-controlled exposure
	LxxxxRC	Grabber-controlled line scanning, camera controlled exposure

If the camera belongs to the camera-controlled exposure class, you can specify the actual exposure time through parameter **TrueExp_us**. This board provides an exposure time range of 1 microsecond up to 20 s.

12. Advanced Features

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12.1. Synchronized Line-scan Acquisition

Applies to: Base DualBase Full FullXR

Introduction

The "Synchronized Line-scan Acquisition" feature synchronizes accurately two or more line-scan cameras using a **SyncBus**.

The SyncBus interconnects one **Master** MultiCam channel to one or more **Slave** MultiCam channel(s).

The SyncBus conveys two synchronization signals:

- a **line trigger** signal, that, typically, originates from the motion encoder attached to the **Master** MultiCam channel
- a **start (and optionally end) of acquisition** signal, that originates from the acquisition controller of the **Master** MultiCam channel.

The synchronization signals are propagated to all the camera and acquisition controllers of all the participating channels, including the master one.

The SyncBus interconnection is made using one of the following wiring schemes:

- ☐ Internal SyncBus wiring scheme
- ☐ C2C SyncBus wiring scheme
- ☐ Isolated I/O SyncBus wiring scheme

Internal SyncBus wiring

Applies to: DualBase

In the specific case of a SyncBus linking only the 2 channels of a **1623 Grablink DualBase**, the SyncBus doesn't require any physical interconnect wiring; instead, it is routed locally inside the FPGA.

C2C SyncBus wiring

Applies to: Full FullXR

The SyncBus wiring interconnects the **C2C/JTAG connector** of the **Master** channel to the **C2C/JTAG connector(s)** of all the participating channels using a flat cable.

**NOTE**

This wiring scheme interconnects FPGA I/O's allowing very accurate synchronization with the fastest line-scan cameras.

Isolated I/O SyncBus wiring

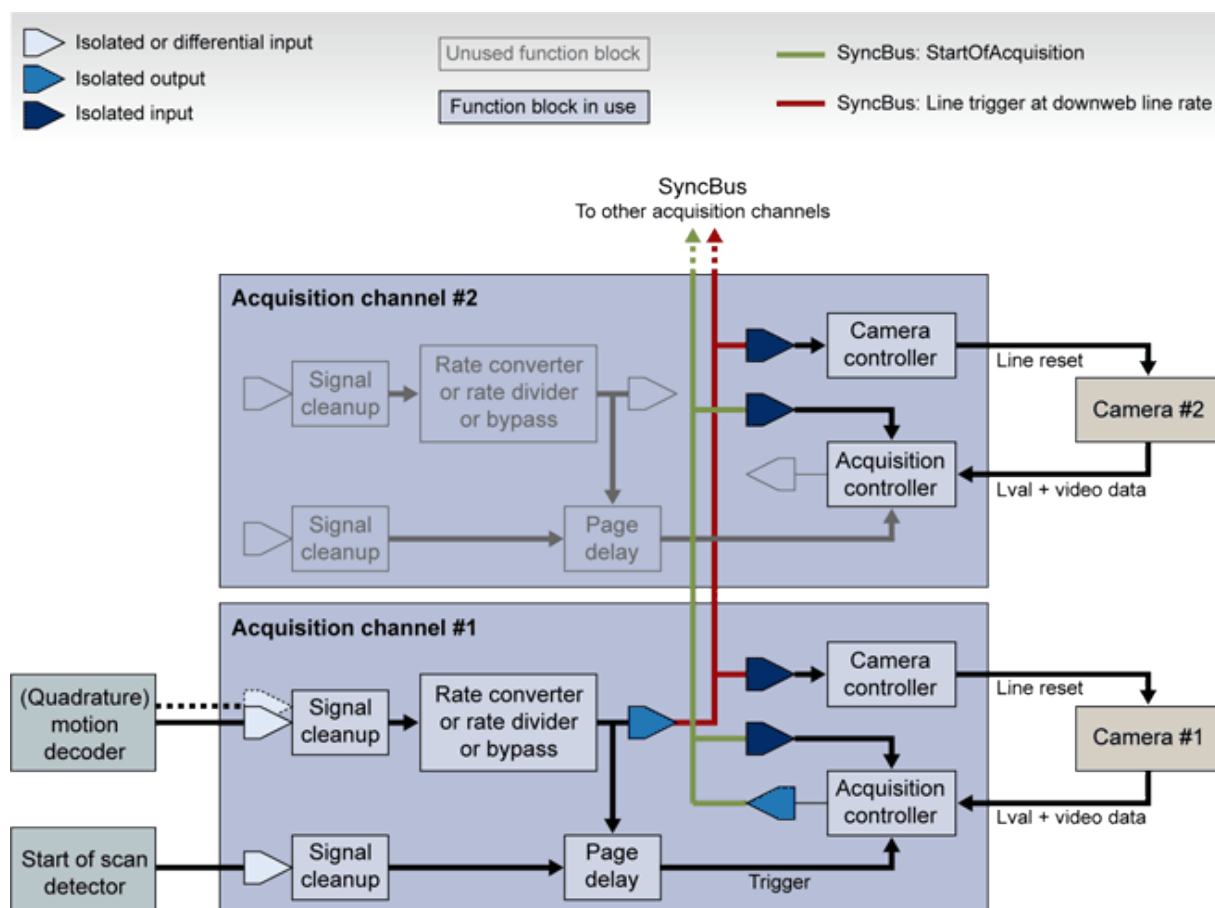
Applies to: **Base** **DualBase** **Full** **FullXR**

The SyncBus wiring interconnects the **IOUT3/IOUT4** output ports of the **Master** channel to the **IIN3/IIN4** input ports of all the participating channels.



NOTE

This wiring scheme restricts the camera line rate range to 40 kHz.



Synchronized acquisition using Isolated I/O SyncBus wiring

Requirements

For adequate synchronization, the line rate of the cameras must be controlled by the frame grabber. Synchronization acquisition is allowed exclusively for the following values of **CamConfig**: **LxxxxRC**, **LxxxxRG**, **LxxxxRG2**, and **LxxxxRP**.

The exposure settings of all cameras must be identical:

- For cameras having the exposure time controlled by the frame grabber, the value of **Expose_us** and **ExposeTrim** must be identical on all channels.
- When the exposure time is controlled by the camera, all cameras must be configured identically, and the value of **TrueExp_us** must be identical on all channels.

The line capture controller must be configured to take all lines: **LineCaptureMode** = **ALL**.



NOTE

The Pick-A-Line mode **LineCaptureMode** = **PICK** is not allowed.

Control Parameters

The **SynchronizedAcquisition** Channel parameter is the main control:

- when set to **OFF** (default value), the synchronized acquisition feature is disabled.
- when set to **MASTER**, **LOCAL_MASTER**, **SLAVE** or **LOCAL_SLAVE**, the synchronized acquisition feature is enabled.

To configure the synchronized acquisition and the participating channels, proceed as follows:

The **SynchronizedAcquisitionBus** Channel parameter provides an additional option to select the SyncBus wiring:

- when set to **ISO** (default value), the SyncBus uses the Isolated I/O SyncBus wiring (or the internal SyncBus wiring of **1623 Grablink DualBase**).
- when set to **C2C**, the SyncBus uses the C2C SyncBus wiring available only on **1622 Grablink Full** and **1626 Grablink Full XR**.

Master channel setup procedure

1. Select a line-scan acquisition mode by setting the value **WEB**, **PAGE** or **LONGPAGE** to **AcquisitionMode**.
2. Set the value **MASTER** (or **LOCAL_MASTER** if local wiring can be used) to **SynchronizedAcquisition**.
3. Set the value **C2C** to **SynchronizedAcquisitionBus** if C2C SyncBus wiring is required or leave the **ISO** value
4. Configure remaining acquisition control parameters as for a stand-alone channel.
5. Configure the trigger control, and the encoder control parameters as for a stand-alone channel.
6. Configure the exposure control parameters.

Slave channel(s) setup procedure

1. Assign to **AcquisitionMode** the same value as on the master channel.
2. Set the value **SLAVE** (or **LOCAL_SLAVE** if internal SyncBus wiring can be used) to **SynchronizedAcquisition**.
3. Set the value **C2C** to **SynchronizedAcquisitionBus** if C2C SyncBus wiring is required or leave the **ISO** value
4. Configure the remaining acquisition control parameters as follows:
 - a. **TrigMode** and **NextTrigMode** are automatically set by MultiCam to the appropriate values, they may not be modified.
 - b. **EndTrigMode** must be set to **SLAVE** if it is configured to **HARD** on the master channel. Otherwise, it remains at its default value **AUTO**.
 - c. **BreakEffect**, **PageLength_Ln**, **SeqLength_Pg** and **SeqLength_Ln**, when applicable, must be configured to the same values as on the master channel.
5. All the trigger control parameters are irrelevant and don't need to be configured.
6. The **LineRateMode** parameter is automatically set to **SLAVE**; the other parameters of the encoder control category are irrelevant.
7. Apply to exposure control parameters the same settings as the master channel.

12.2. Metadata Insertion

Applies to: Base DualBase Full FullXR

Insertion of real-time metadata on the Camera Link data stream

Introduction

The metadata insertion feature inserts metadata in the Camera Link pixel data stream.

Requirements

Metadata insertion is available only with **1624 Grablink Base**, **1623 Grablink DualBase**, **1622 Grablink Full** and **1626 Grablink Full XR** for a subset of cameras.

Refer to ["Camera Requirements" on page 164](#) for an extensive list of camera requirements.

The metadata insertion feature is NOT compatible with the following frame grabber pixel processing options:

- Look-up table transformation
- On-board Bayer CFA to RGB color conversion
- Pixel bit depth reduction

Capabilities

One, two or three fields of metadata are inserted in the Camera Link data stream according to the camera configuration. Refer to ["Metadata Content" on page 165](#) for a detailed description.

Refer to ["Metadata Fields" on page 167](#) for an extensive description of each field.

Refer to ["Memory Layouts" on page 170](#) for an extensive description of the metadata layout in the captured images.

Refer to ["Metadata Controls" on the next page](#) for an exhaustive list of control parameters.

Metadata Controls

Main control

The activation of the Metadata Insertion feature is controlled through the MultiCam Channel parameter **MetadataInsertion**.

By default, **MetadataInsertion** is set to **DISABLE**.

To activate the feature, set the parameter to **ENABLE** at channel creation. The setting takes effect at the next channel activation.

**NOTE**

An error is reported when setting to **ENABLE** if the camera interface configuration is not compatible.

Metadata location control

There are two distinct methods to insert metadata in the Camera Link pixel data stream according to the value of the **MetadataLocation** parameter: **LVALRISE** or **TAP10**.

- When set to **LVALRISE**, the image data conveyed during the first Camera Link time slot after an LVAL are replaced by metadata.
- When set to **TAP10**, the image data conveyed on the Camera Link port J (Tap 10) is replaced by metadata during the first 10 Camera Link Clock cycles of every line after the LVAL rising edge. For the other clock cycles, the frame grabber inserts bytes of '0'.

By default, **MetadataLocation** is set to **LVALRISE**. This value is always available.

The **TAP10** option is applicable only to 80-bit (10-tap 8-bit) line-scan and TDI line-scan cameras characterized by **TapConfiguration** = **DECA_10T8** and **TapGeometry** = **1X10**.

General Purpose Pulse Counter (GPPC) controls

Applies to: Full FullXR

The GPPC, a 32-bit binary counter, is available only on **1622 Grablink Full** and **1626 Grablink Full XR** when `MetadataContent` = `THREE_FIELD`.

The GPPC value is sampled at each rising edge of LVAL is reported in the GPPC metadata field.

Main control

The `MetadataGPPCInputLine` channel parameter is the main control of the GPPC:

- when set to `NONE` (default value), the GPPC is disabled,
- when set to `IIN1`, the GPPC counts the rising edge events of the electrical signal applied to the IIN1 isolated input line.

GPPC field location control

The `MetadataGPPCLocation` parameter controls the location of the GPPC field in the metadata:

- when set to `NONE` (default value), the GPPC metadata is not inserted into the Camera Link data stream.
- when set to `INSTEAD_LVALCNT`, the GPPC metadata replaces the LVAL Count metadata in the Camera Link data stream,
- when set to `INSTEAD_QCNT`, the GPPC metadata replaces the Q Count metadata in the Camera Link data stream,

GPPC reset control

The `MetadataGPPCResetLine` parameter controls the reset feature of the GPPC:

- when set to `NONE` (default value), the GPPC has no reset input line; the IIN4 input remains available for another purpose.
- when set to `IIN4`, the GPPC resets when a high-level is applied to the IIN4 isolated input line.

Camera Requirements

Camera Characteristic	Description
Image sensor geometry Imaging	All the image sensor geometries are compatible: • AREA: Area-scan cameras • LINE and TDI: Line-scan cameras including bi-linear and TDI
Tap configurations TapConfiguration	• For area-scan cameras: all tap configurations are compatible. • For line-scan and TDI line-scan cameras, only the following configurations are compatible: MEDIUM_6T8, MEDIUM_2T24, FULL_8T8, DECA_10T8, DECA_8T10, DECA_2T40 and DECA_8T30B3.
Tap geometries TapGeometry	Only the tap geometries having a single X region 1X* are compatible.

Metadata Content

The get-only MultiCam Channel parameter **MetadataContent** reports the number of distinct **metadata fields** in the metadata content.

MetadataContent	Description
NONE	There is no metadata content.
ONE_FIELD	The metadata content includes one single field: I/O State .
TWO_FIELD	The metadata content includes two fields: I/O State + LVAL Count
THREE_FIELD	The metadata content includes three fields.

When **MetadataContent** = **THREE_FIELD**, the composition of the metadata depends on the **MetadataGPPCLocation** control value:

MetadataGPPCLocation	Field 1	Field 2	Field 3
NONE	I/O State	LVAL Count	Q Count
INSTEAD_LVALCNT	I/O State	GPPC Count	Q Count
INSTEAD_QCNT	I/O State	LVAL Count	GPPC Count



NOTE

Refer to "[Memory Layouts](#)" on page 170 for the bit assignment and memory layout applicable to each configuration

Camera configurations providing 3 fields

Imaging	TapConfiguration	TapGeometry	Memory Layout
LINE or TDI	DECA_10T8	1X10	"3-field 8-bit " on page 186
	DECA_8T10	1X8	"3-field 10-bit Packed" on page 190
	DECA_8T30B3	1X8	"3-field 30-bit Packed" on page 192
	DECA_2T40	1X2	"3-field 40-bit Packed" on page 194

Camera configurations providing 2 fields

Imaging	TapConfiguration	TapGeometry	Memory Layout
LINE or TDI	MEDIUM_6T8	1X3_1Y2	"2-field 8-bit " on page 183
	FULL_8T8	1X8	
	MEDIUM_2T24	1X2	"2-field 24-bit " on page 184

Camera configurations providing 1 field

All the other [compatible camera configurations](#) provide 1 field of metadata.

Metadata Fields

I/O State Field

The I/O State field is a 6-bit field reporting the logical state of all System I/O input lines belonging to the Channel.

The reported state is the logical state measured right at the input stage of the Grablink card.

**WARNING**

As the measurement takes place before any glitch removal filters, spurious state transitions may occur!

The state of System I/O input lines and the values of the counters are sampled at each rising edge of the Camera Link LVAL signal. The sampling time is not adjustable.

LVAL Count Field

The LVAL Count field is a 32-bit field reporting the current value of the LVAL pulse counter

LVAL pulse counter

The LVAL pulse counter is a 32-bit binary counter that counts the Camera Link LVAL pulses.

The counter is not resettable, it is set to 0 at driver initialization. As soon as the Camera Link de-serializers are initialized, it is incremented by 1 at every LVAL cycle whatever the acquisition conditions, i.e. whether the corresponding line data is acquired or not. It wraps around to 0 when it reaches the maximum count 4,294,967,295 ($=2^{32} - 1$).

**NOTE**

The counter is incremented before its value is inserted as metadata: the first line cycle is marked 1.

GPPC Count Field

The GPPC Count field is a 32-bit field reporting the current value of the General Purpose Pulse Counter - GPPC.

General purpose pulse counter

The GGPC is a 32-bit binary counter.

The counter is set to 0 at driver initialization and is incremented by 1 at every rising edge event detected on its input (IIN1). It wraps around to 0 when it reaches the maximum count 4,294,967,295 ($=2^{32} - 1$).

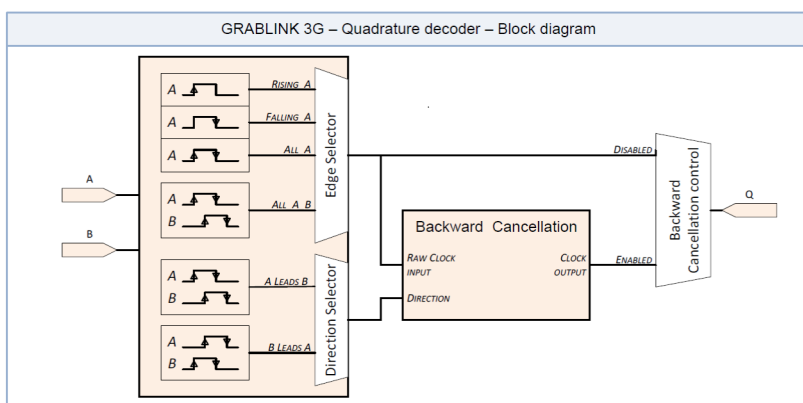
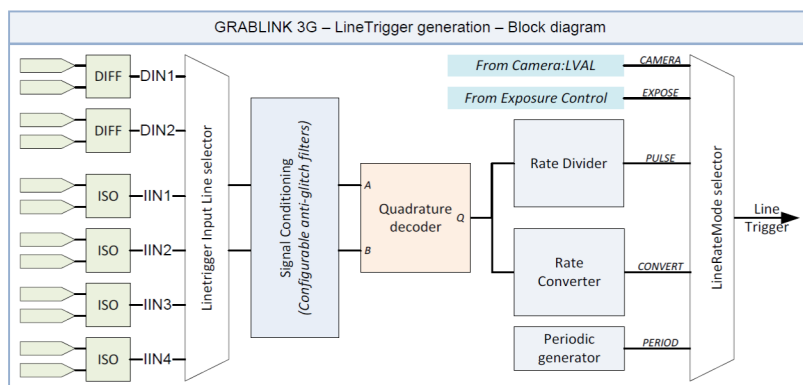
The counter has an optional reset input (IIN4). Applying a high level to the reset input, resets the count value to 0.

**NOTE**

The counter is incremented before its value is inserted as metadata: the first line cycle is marked 1.

Q Count Field

The Q Count Field is a 32-bit field reporting the current value of the Motion Encoder Q counter. This 32-bit binary counter counts the pulses at the Q output of the Quadrature Decoder:



NOTE

Depending on the Quadrature Decoder settings, the counter increments by 0, 1, 2, or 4 units every encoder cycle.

The counter is not resettable:

- It is set to 0 at driver initialization.
- It is incremented by 1 at every Q cycle.
- It wraps around to 0 when it reaches the maximum count $4,294,967,295 (=2^{32} - 1)$.

Memory Layouts

Memory layouts of the metadata as delivered in the MultiCam surface

Introduction

This section provides one topic for each combination of number of metadata fields and pixel bit count.

Each topic provides memory layouts of metadata for applicable combinations of **ColorFormat**, **MetadataLocation** and **ImageFlipX**.

All image lines contain metadata. However, there are exceptions:

1. When **TapGeometry** is set to any ***_1Y2** value, only one line out of two contains metadata.
2. When **TapGeometry** is set to any ***_2YE** value, only the lines belonging to the upper (when **ImageFlipY** = **OFF**) or the lower (when **ImageFlipY** = **ON**) half region contain metadata.

1-field 8-bit

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y8 or BAYER8; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y8 or BAYER8; ImageFlipX = ON

Byte #	7	6	5	4	3	2	1	0
last	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1



NOTE

last = address offset of the last byte of the rightmost active pixel of the line

1-field 10-bit

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y10 or BAYER10; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1	0	0	0	0	0	0	0	0

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y16 or BAYER16; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	IIN2	IIN1	0	0	0	0	0	0
1	0	0	0	0	DIN2	DIN1	IIN4	IIN3

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y10 or BAYER10; ImageFlipX = ON

Byte #	7	6	5	4	3	2	1	0
last - 1	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
last	0	0	0	0	0	0	0	0

**NOTE**

last = address offset of the last byte of the rightmost active pixel of the line

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y16 or BAYER16; ImageFlipX = ON

Byte #	7	6	5	4	3	2	1	0
last - 1	IIN2	IIN1	0	0	0	0	0	0
last	0	0	0	0	DIN2	DIN1	IIN4	IIN3

**NOTE**

last = address offset of the last byte of the rightmost active pixel of the line

1-field 12-bit

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y12 or BAYER12; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1	0	0	0	0	0	0	0	0

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y16 or BAYER16; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	IIN4	IIN3	IIN2	IIN1	0	0	0	0
1	0	0	0	0	0	0	DIN2	DIN1

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y12 or BAYER12; ImageFlipX = ON

Byte #	7	6	5	4	3	2	1	0
last - 1	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
last	0	0	0	0	0	0	0	0

**NOTE**

last = address offset of the last byte of the rightmost active pixel of the line

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y16 or BAYER16; ImageFlipX = ON

Byte #	7	6	5	4	3	2	1	0
last - 1	IIN4	IIN3	IIN2	IIN1	0	0	0	0
last	0	0	0	0	0	0	DIN2	DIN1

**NOTE**

last = address offset of the last byte of the rightmost active pixel of the line

1-field 14-bit

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y14 or BAYER14; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1	0	0	0	0	0	0	0	0

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y16 or BAYER16; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1	0	0
1	0	0	0	0	0	0	0	0

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y14 or BAYER14; ImageFlipX = ON

Byte #	7	6	5	4	3	2	1	0
last - 1	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
last	0	0	0	0	0	0	0	0

**NOTE**

last = address offset of the last byte of the rightmost active pixel of the line

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y16 or BAYER16; ImageFlipX = ON

Byte #	7	6	5	4	3	2	1	0
last - 1	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1	0	0
last	0	0	0	0	0	0	0	0

**NOTE**

last = address offset of the last byte of the rightmost active pixel of the line

1-field 16-bit

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y16 or BAYER16; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1	0	0	0	0	0	0	0	0

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = Y16 or BAYER16; ImageFlipX = ON

Byte #	7	6	5	4	3	2	1	0
last - 1	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
last	0	0	0	0	0	0	0	0



NOTE

last = address offset of the last byte of the rightmost active pixel of the line

1-field 24-bit

MetadataLocation = LVALRISE; MetadataContent = TWO_FIELD

ColorFormat = RGB24; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0

TapConfiguration = MEDIUM_2T24; TapGeometry = 1X2

MetadataLocation = LVALRISE; MetadataContent = TWO_FIELD

ColorFormat = RGB24PL; ImageFlipX = OFF

Plane Byte #	7	6	5	4	3	2	1	0
B 0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
G 0	0	0	0	0	0	0	0	0
R 0	0	0	0	0	0	0	0	0

MetadataLocation = LVALRISE; MetadataContent = TWO_FIELD

ColorFormat = RGB24; ImageFlipX = ON

Byte #	7	6	5	4	3	2	1	0
last - 2	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
last - 1	0	0	0	0	0	0	0	0
last	0	0	0	0	0	0	0	0



NOTE

last = address offset of the last byte of the rightmost active pixel of the line

TapConfiguration = MEDIUM_2T24; TapGeometry = 1X2

MetadataLocation = LVALRISE; MetadataContent = TWO_FIELD

ColorFormat = RGB24PL; ImageFlipX = ON

Plane Byte #	7	6	5	4	3	2	1	0
B last	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
G last	0	0	0	0	0	0	0	0
R last	0	0	0	0	0	0	0	0



NOTE

last = address offset of the last byte of the rightmost active pixel of the line

1-field 30-bit

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = RGB30PL; ImageFlipX = OFF

Plane Byte #	7	6	5	4	3	2	1	0
B 0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
B 1	0	0	0	0	0	0	0	0
G 0	0	0	0	0	0	0	0	0
G 1	0	0	0	0	0	0	0	0
R 0	0	0	0	0	0	0	0	0
R 1	0	0	0	0	0	0	0	0

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = RGB48PL; ImageFlipX = OFF

Plane Byte #	7	6	5	4	3	2	1	0
B 0	IIN2	IIN1	0	0	0	0	0	0
B 1	0	0	0	0	DIN2	DIN1	IIN4	IIN3
G 0	0	0	0	0	0	0	0	0
G 1	0	0	0	0	0	0	0	0
R 0	0	0	0	0	0	0	0	0
R 1	0	0	0	0	0	0	0	0

1-field 36-bit

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = RGB36PL; ImageFlipX = OFF

Plane Byte #	7	6	5	4	3	2	1	0
B 0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
B 1	0	0	0	0	0	0	0	0
G 0	0	0	0	0	0	0	0	0
G 1	0	0	0	0	0	0	0	0
R 0	0	0	0	0	0	0	0	0
R 1	0	0	0	0	0	0	0	0

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = RGB48PL; ImageFlipX = OFF

Plane Byte #	7	6	5	4	3	2	1	0
B 0	IIN4	IIN3	IIN2	IIN1	0	0	0	0
B 1	0	0	0	0	0	0	DIN2	DIN1
G 0	0	0	0	0	0	0	0	0
G 1	0	0	0	0	0	0	0	0
R 0	0	0	0	0	0	0	0	0
R 1	0	0	0	0	0	0	0	0

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = RGB36PL; ImageFlipX = ON

Plane Byte #	7	6	5	4	3	2	1	0
B last - 1	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
B last	0	0	0	0	0	0	0	0
G last - 1	0	0	0	0	0	0	0	0
G last	0	0	0	0	0	0	0	0
R last - 1	0	0	0	0	0	0	0	0
R last	0	0	0	0	0	0	0	0



NOTE

last = address offset of the last byte of the rightmost active pixel of the line

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = RGB48PL; ImageFlipX = ON

Plane Byte #	7	6	5	4	3	2	1	0
B last - 1	IIN4	IIN3	IIN2	IIN1	0	0	0	0
B last	0	0	0	0	0	0	DIN2	DIN1
G last - 1	0	0	0	0	0	0	0	0
G last	0	0	0	0	0	0	0	0
R last - 1	0	0	0	0	0	0	0	0
R last	0	0	0	0	0	0	0	0



NOTE

last = address offset of the last byte of the rightmost active pixel of the line

1-field 42-bit

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = RGB42PL; ImageFlipX = OFF

Plane Byte #	7	6	5	4	3	2	1	0
B 0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
B 1	0	0	0	0	0	0	0	0
G 0	0	0	0	0	0	0	0	0
G 1	0	0	0	0	0	0	0	0
R 0	0	0	0	0	0	0	0	0
R 1	0	0	0	0	0	0	0	0

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = RGB48PL; ImageFlipX = OFF

Plane Byte #	7	6	5	4	3	2	1	0
B 0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1	0	0
B 1	0	0	0	0	0	0	0	0
G 0	0	0	0	0	0	0	0	0
G 1	0	0	0	0	0	0	0	0
R 0	0	0	0	0	0	0	0	0
R 1	0	0	0	0	0	0	0	0

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = RGB42PL; ImageFlipX = ON

Plane Byte #	7	6	5	4	3	2	1	0
B last - 1	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
B last	0	0	0	0	0	0	0	0
G last - 1	0	0	0	0	0	0	0	0
G last	0	0	0	0	0	0	0	0
R last - 1	0	0	0	0	0	0	0	0
R last	0	0	0	0	0	0	0	0



NOTE

last = address offset of the last byte of the rightmost active pixel of the line

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = RGB48PL; ImageFlipX = ON

Plane Byte #	7	6	5	4	3	2	1	0
B last - 1	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1	0	0
B last	0	0	0	0	0	0	0	0
G last - 1	0	0	0	0	0	0	0	0
G last	0	0	0	0	0	0	0	0
R last - 1	0	0	0	0	0	0	0	0
R last	0	0	0	0	0	0	0	0



NOTE

last = address offset of the last byte of the rightmost active pixel of the line

1-field 48-bit

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = RGB48PL; ImageFlipX = OFF

Plane Byte #	7	6	5	4	3	2	1	0
B 0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
B 1	0	0	0	0	0	0	0	0
G 0	0	0	0	0	0	0	0	0
G 1	0	0	0	0	0	0	0	0
R 0	0	0	0	0	0	0	0	0
R 1	0	0	0	0	0	0	0	0

MetadataLocation = LVALRISE; MetadataContent = ONE_FIELD

ColorFormat = RGB48PL; ImageFlipX = ON

Plane Byte #	7	6	5	4	3	2	1	0
B last - 1	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
B last	0	0	0	0	0	0	0	0
G last - 1	0	0	0	0	0	0	0	0
G last	0	0	0	0	0	0	0	0
R last - 1	0	0	0	0	0	0	0	0
R last	0	0	0	0	0	0	0	0



NOTE

last = address offset of the last byte of the rightmost active pixel of the line

2-field 8-bit

TapConfiguration = MEDIUM_6T8; TapGeometry = 1X3_1Y2

TapConfiguration = FULL_8T8; TapGeometry = 1X8

MetadataLocation = LVALRISE; MetadataContent = TWO_FIELD

ColorFormat = Y8 or BAYER8; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1	0	0	0	0	0	0	0	0
2	LV7	LV6	LV5	LV4	LV3	LV2	LV1	LV0
3	LV15	LV14	LV13	LV12	LV11	LV10	LV9	LV8
4	LV23	LV22	LV21	LV20	LV19	LV18	LV17	LV16
5	LV31	LV30	LV29	LV28	LV27	LV26	LV25	LV24

TapConfiguration = MEDIUM_6T8; TapGeometry = 1X3_1Y2

TapConfiguration = FULL_8T8; TapGeometry = 1X8

MetadataLocation = LVALRISE; MetadataContent = TWO_FIELD

ColorFormat = Y8 or BAYER8; ImageFlipX = ON

Byte #	7	6	5	4	3	2	1	0
last - 5	LV31	LV30	LV29	LV28	LV27	LV26	LV25	LV24
last - 4	LV23	LV22	LV21	LV20	LV19	LV18	LV17	LV16
last - 3	LV15	LV14	LV13	LV12	LV11	LV10	LV9	LV8
last - 2	LV7	LV6	LV5	LV4	LV3	LV2	LV1	LV0
last - 1	0	0	0	0	0	0	0	0
last	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1



NOTE

last = address offset of the last byte of the rightmost active pixel of the line

2-field 24-bit

TapConfiguration = MEDIUM_2T24; TapGeometry = 1X2

MetadataLocation = LVALRISE; MetadataContent = TWO_FIELD

ColorFormat = RGB24; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1	0	0	0	0	0	0	0	0
2	LV7	LV6	LV5	LV4	LV3	LV2	LV1	LV0
3	LV15	LV14	LV13	LV12	LV11	LV10	LV9	LV8
4	LV23	LV22	LV21	LV20	LV19	LV18	LV17	LV16
5	LV31	LV30	LV29	LV28	LV27	LV26	LV25	LV24

TapConfiguration = MEDIUM_2T24; TapGeometry = 1X2

MetadataLocation = LVALRISE; MetadataContent = TWO_FIELD

ColorFormat = RGB24PL; ImageFlipX = OFF

Plane Byte #	7	6	5	4	3	2	1	0
B 0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
G 0	0	0	0	0	0	0	0	0
R 0	LV7	LV6	LV5	LV4	LV3	LV2	LV1	LV0
B 1	LV15	LV14	LV13	LV12	LV11	LV10	LV9	LV8
G 1	LV23	LV22	LV21	LV20	LV19	LV18	LV17	LV16
R 1	LV31	LV30	LV29	LV28	LV27	LV26	LV25	LV24

TapConfiguration = MEDIUM_2T24; TapGeometry = 1X2

MetadataLocation = LVALRISE; MetadataContent = TWO_FIELD

ColorFormat = RGB24; ImageFlipX = ON

Byte #	7	6	5	4	3	2	1	0
last - 5	LV15	LV14	LV13	LV12	LV11	LV10	LV9	LV8
last - 4	LV23	LV22	LV21	LV20	LV19	LV18	LV17	LV16
last - 3	LV31	LV30	LV29	LV28	LV27	LV26	LV25	LV24
last - 2	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
last - 1	0	0	0	0	0	0	0	0
last	LV7	LV6	LV5	LV4	LV3	LV2	LV1	LV0



NOTE

last = address offset of the last byte of the rightmost active pixel of the line

TapConfiguration = MEDIUM_2T24; TapGeometry = 1X2

MetadataLocation = LVALRISE; MetadataContent = TWO_FIELD

ColorFormat = RGB24PL; ImageFlipX = ON

	Plane Byte #	7	6	5	4	3	2	1	0
3	B last - 1	LV15	LV14	LV13	LV12	LV11	LV10	LV9	LV8
4	G last - 1	LV23	LV22	LV21	LV20	LV19	LV18	LV17	LV16
5	R last - 1	LV31	LV30	LV29	LV28	LV27	LV26	LV25	LV24
6	B last	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
7	G last	0	0	0	0	0	0	0	0
8	R last	LV7	LV6	LV5	LV4	LV3	LV2	LV1	LV0



NOTE

last = address offset of the last byte of the rightmost active pixel of the line

3-field 8-bit

Case #1

TapConfiguration = DECA_10T8; TapGeometry = 1X10

MetadataLocation = LVALRISE; MetadataGPPCLocation = NONE

MetadataContent = THREE_FIELD

ColorFormat = Y8 or BAYER8; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1	0	0	0	0	0	0	0	0
2	QC7	QC6	QC5	QC4	QC3	QC2	QC1	QC0
3	QC15	QC14	QC13	QC12	QC11	QC10	QC9	QC8
4	QC23	QC22	QC21	QC20	QC19	QC18	QC17	QC16
5	QC31	QC30	QC29	QC28	QC27	QC26	QC25	QC24
6	LV7	LV6	LV5	LV4	LV3	LV2	LV1	LV0
7	LV15	LV14	LV13	LV12	LV11	LV10	LV9	LV8
8	LV23	LV22	LV21	LV20	LV19	LV18	LV17	LV16
9	LV31	LV30	LV29	LV28	LV27	LV26	LV25	LV24



NOTE

- When MetadataGPPCLocation = INSTEAD_LVALCNT replace LV* by GPPC*
- When MetadataGPPCLocation = INSTEAD_QCNT replace QC* by GPPC*

Case #2

TapConfiguration = DECA_10T8; TapGeometry = 1X10

MetadataLocation = TAP10; MetadataGPPCLocation = NONE

MetadataContent = THREE_FIELD

ColorFormat = Y8 or BAYER8; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
9	0	0	0	0	0	0	0	0
19	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
29	QC7	QC6	QC5	QC4	QC3	QC2	QC1	QC0
39	QC15	QC14	QC13	QC12	QC11	QC10	QC9	QC8
49	QC23	QC22	QC21	QC20	QC19	QC18	QC17	QC16
59	QC31	QC30	QC29	QC28	QC27	QC26	QC25	QC24
69	LV7	LV6	LV5	LV4	LV3	LV2	LV1	LV0
79	LV15	LV14	LV13	LV12	LV11	LV10	LV9	LV8
89	LV23	LV22	LV21	LV20	LV19	LV18	LV17	LV16
99	LV31	LV30	LV29	LV28	LV27	LV26	LV25	LV24



NOTE

- When MetadataGPPCLocation = INSTEAD_LVALCNT replace LV* by GPPC*
- When MetadataGPPCLocation = INSTEAD_QCNT replace QC* by GPPC*

Case #3

TapConfiguration = DECA_10T8; TapGeometry = 1X10

MetadataLocation = LVALRISE; MetadataGPPCLocation = NONE

MetadataContent = THREE_FIELD

ColorFormat = Y8 or BAYER8; ImageFlipX = ON

Byte #	7	6	5	4	3	2	1	0
last - 9	LV31	LV30	LV29	LV28	LV27	LV26	LV25	LV24
last - 8	LV23	LV22	LV21	LV20	LV19	LV18	LV17	LV16
last - 7	LV15	LV14	LV13	LV12	LV11	LV10	LV9	LV8
last - 6	LV7	LV6	LV5	LV4	LV3	LV2	LV1	LV0
last - 5	QC31	QC30	QC29	QC28	QC27	QC26	QC25	QC24
last - 4	QC23	QC22	QC21	QC20	QC19	QC18	QC17	QC16
last - 3	QC15	QC14	QC13	QC12	QC11	QC10	QC9	QC8
last - 2	QC7	QC6	QC5	QC4	QC3	QC2	QC1	QC0
last - 1	0	0	0	0	0	0	0	0
last - 0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1



NOTE

last = address offset of the last byte of the rightmost active pixel of the line



NOTE

- When MetadataGPPCLocation = INSTEAD_LVALCNT replace LV* by GPPC*
- When MetadataGPPCLocation = INSTEAD_QCNT replace QC* by GPPC*

Case #4

TapConfiguration = DECA_10T8; TapGeometry = 1X10

MetadataLocation = TAP10; MetadataGPPCLocation = NONE

MetadataContent = THREE_FIELD

ColorFormat = Y8 or BAYER8; ImageFlipX = ON

Byte #	7	6	5	4	3	2	1	0
last - 99	LV31	LV30	LV29	LV28	LV27	LV26	LV25	LV24
last - 89	LV23	LV22	LV21	LV20	LV19	LV18	LV17	LV16
last - 79	LV15	LV14	LV13	LV12	LV11	LV10	LV9	LV8
last - 69	LV7	LV6	LV5	LV4	LV3	LV2	LV1	LV0
last - 59	QC31	QC30	QC29	QC28	QC27	QC26	QC25	QC24
last - 49	QC23	QC22	QC21	QC20	QC19	QC18	QC17	QC16
last - 39	QC15	QC14	QC13	QC12	QC11	QC10	QC9	QC8
last - 29	QC7	QC6	QC5	QC4	QC3	QC2	QC1	QC0
last - 19	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
last - 9	0	0	0	0	0	0	0	0



NOTE

last = address offset of the last byte of the rightmost active pixel of the line



NOTE

- When MetadataGPPCLocation = INSTEAD_LVALCNT replace LV* by GPPC*
- When MetadataGPPCLocation = INSTEAD_QCNT replace QC* by GPPC*

3-field 10-bit Packed

TapConfiguration = DECA_8T10; TapGeometry = 1X8

MetadataLocation = LVALRISE; MetadataGPPCLocation = NONE

MetadataContent = THREE_FIELD

ColorFormat = Y10P; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1	QC7	QC6	QC5	QC4	QC3	QC2	QC1	QC0
2	QC15	QC14	QC13	QC12	QC11	QC10	QC9	QC8
3	QC23	QC22	QC21	QC20	QC19	QC18	QC17	QC16
4	QC31	QC30	QC29	QC28	QC27	QC26	QC25	QC24
5	0	0	0	0	0	0	0	0
6	LV7	LV6	LV5	LV4	LV3	LV2	LV1	LV0
7	LV15	LV14	LV13	LV12	LV11	LV10	LV9	LV8
8	LV23	LV22	LV21	LV20	LV19	LV18	LV17	LV16
9	LV31	LV30	LV29	LV28	LV27	LV26	LV25	LV24



NOTE

ImageFlipX = ON is not available when ColorFormat = Y10P



NOTE

- When MetadataGPPCLocation = INSTEAD_LVALCNT replace LV* by GPPC*
- When MetadataGPPCLocation = INSTEAD_QCNT replace QC* by GPPC*

After unpacking to 16-bit with justification to lsb

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1							QC1	QC0
2	QC9	QC8	QC7	QC6	QC5	QC4	QC3	QC2
3							QC11	QC10
4	QC19	QC18	QC17	QC16	QC15	QC14	QC13	QC12
5							QC21	QC20
6	QC29	QC28	QC27	QC26	QC25	QC24	QC23	QC22
7							QC31	QC30
8	0	0	0	0	0	0	0	0
9							LV1	LV0
10	LV9	LV8	LV7	LV6	LV5	LV4	LV3	LV2
11							LV11	LV10
12	LV19	LV18	LV17	LV16	LV15	LV14	LV13	LV12
13							LV21	LV20
14	LV29	LV28	LV27	LV26	LV25	LV24	LV23	LV22
15							LV31	LV30

3-field 30-bit Packed

TapConfiguration = DECA_8T30B3; TapGeometry = 1X8

MetadataLocation = LVALRISE; MetadataGPPCLocation = NONE

MetadataContent = THREE_FIELD

ColorFormat = RGB30P; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1	QC7	QC6	QC5	QC4	QC3	QC2	QC1	QC0
2	QC15	QC14	QC13	QC12	QC11	QC10	QC9	QC8
3	QC23	QC22	QC21	QC20	QC19	QC18	QC17	QC16
4	QC31	QC30	QC29	QC28	QC27	QC26	QC25	QC24
5	0	0	0	0	0	0	0	0
6	LV7	LV6	LV5	LV4	LV3	LV2	LV1	LV0
7	LV15	LV14	LV13	LV12	LV11	LV10	LV9	LV8
8	LV23	LV22	LV21	LV20	LV19	LV18	LV17	LV16
9	LV31	LV30	LV29	LV28	LV27	LV26	LV25	LV24



NOTE

ImageFlipX = ON is not available when ColorFormat = RGB30P



NOTE

- When MetadataGPPCLocation = INSTEAD_LVALCNT replace LV* by GPPC*
- When MetadataGPPCLocation = INSTEAD_QCNT replace QC* by GPPC*

After unpacking to 16-bit with justification to lsb

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1							QC1	QC0
2	QC9	QC8	QC7	QC6	QC5	QC4	QC3	QC2
3							QC11	QC10
4	QC19	QC18	QC17	QC16	QC15	QC14	QC13	QC12
5							QC21	QC20
6	QC29	QC28	QC27	QC26	QC25	QC24	QC23	QC22
7							QC31	QC30
8	0	0	0	0	0	0	0	0
9							LV1	LV0
10	LV9	LV8	LV7	LV6	LV5	LV4	LV3	LV2
11							LV11	LV10
12	LV19	LV18	LV17	LV16	LV15	LV14	LV13	LV12
13							LV21	LV20
14	LV29	LV28	LV27	LV26	LV25	LV24	LV23	LV22
15							LV31	LV30

3-field 40-bit Packed

TapConfiguration = DECA_2T40; TapGeometry = 1X2

MetadataLocation = LVALRISE; MetadataGPPCLocation = NONE

MetadataContent = THREE_FIELD

ColorFormat = RGBI40P; ImageFlipX = OFF

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1	QC7	QC6	QC5	QC4	QC3	QC2	QC1	QC0
2	QC15	QC14	QC13	QC12	QC11	QC10	QC9	QC8
3	QC23	QC22	QC21	QC20	QC19	QC18	QC17	QC16
4	QC31	QC30	QC29	QC28	QC27	QC26	QC25	QC24
5	0	0	0	0	0	0	0	0
6	LV7	LV6	LV5	LV4	LV3	LV2	LV1	LV0
7	LV15	LV14	LV13	LV12	LV11	LV10	LV9	LV8
8	LV23	LV22	LV21	LV20	LV19	LV18	LV17	LV16
9	LV31	LV30	LV29	LV28	LV27	LV26	LV25	LV24



NOTE

ImageFlipX = ON is not available when ColorFormat = RGBI40P



NOTE

- When MetadataGPPCLocation = INSTEAD_LVALCNT replace LV* by GPPC*
- When MetadataGPPCLocation = INSTEAD_QCNT replace QC* by GPPC*

After unpacking to 16-bit with justification to lsb

Byte #	7	6	5	4	3	2	1	0
0	0	0	DIN2	DIN1	IIN4	IIN3	IIN2	IIN1
1							QC1	QC0
2	QC9	QC8	QC7	QC6	QC5	QC4	QC3	QC2
3							QC11	QC10
4	QC19	QC18	QC17	QC16	QC15	QC14	QC13	QC12
5							QC21	QC20
6	QC29	QC28	QC27	QC26	QC25	QC24	QC23	QC22
7							QC31	QC30
8	0	0	0	0	0	0	0	0
9							LV1	LV0
10	LV9	LV8	LV7	LV6	LV5	LV4	LV3	LV2
11							LV11	LV10
12	LV19	LV18	LV17	LV16	LV15	LV14	LV13	LV12
13							LV21	LV20
14	LV29	LV28	LV27	LV26	LV25	LV24	LV23	LV22
15							LV31	LV30

12.3. Interleaved Acquisition

Applies to: Base DualBase Full FullXR

Image acquisition from grabber-controlled exposure asynchronous reset cameras driven alternatively by two different camera cycle programs

Introduction

When Interleaved Acquisition is enabled, the Camera and Illumination Controller is configured with two different programs named P1 and P2.

Each program defines entirely a *camera and illumination cycle* including:

- One Reset pulse controlling the start-of-exposure and the end-of-exposure of the camera.
- One Strobe pulse on any of the 2 strobe outputs.

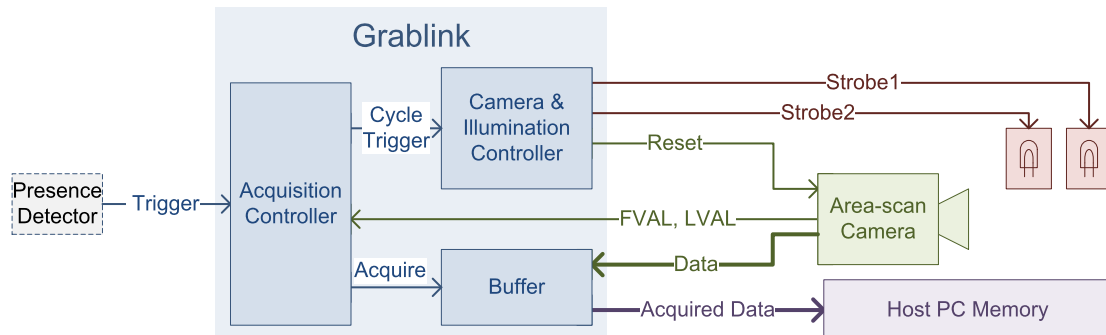
The programs are executed alternatively, starting with P1.

This feature is available for *line-scan cameras* since MultiCam 6.9.7. For more information, refer to ["Interleaved Line-scan Acquisition Principles" on page 199](#).

This feature is available for *area-scan cameras* since MultiCam 6.13. For more information, refer to ["Interleaved Area-scan Acquisition Principles" on the next page](#).

Interleaved Area-scan Acquisition Principles

System Description



Main elements of an area-scan acquisition system configured for Interleaved Area-scan Acquisition

The system is composed of:

- One asynchronous reset grabber-controlled exposure area-scan camera.
- One acquisition channel of a compatible Grablink frame grabber configured for Interleaved Area-scan acquisition.
- Two illumination devices, each being controlled by a specific strobe output of the frame grabber.

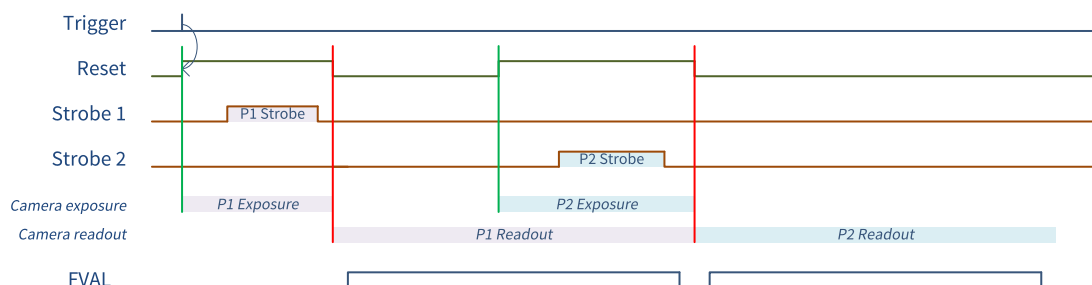
Usually, the Trigger event is delivered by a presence detector.

SNAPSHOT and **HFR** area-scan acquisition modes are compatible with Interleaved Area-scan Acquisition.

Operation

The Interleaved Acquisition feature allows to capture, with a time-optimized sequence of two camera cycles, two images with different exposure time, strobe duration, strobe delay and strobe output settings.

The first cycle of the sequence uses the settings defined by P1, the second cycle uses the settings defined by P2.



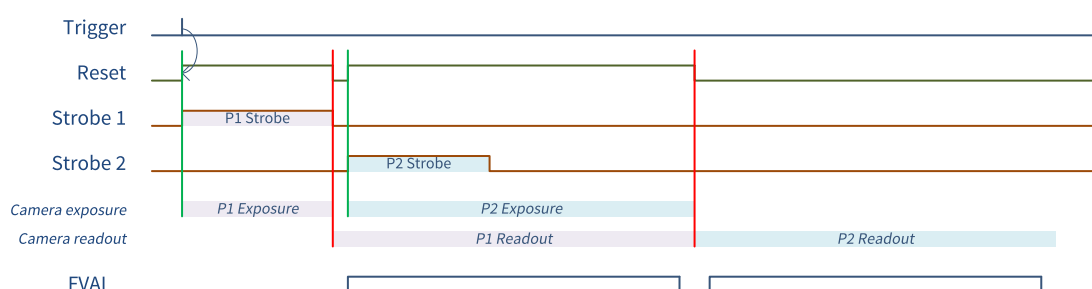
Acquisition sequence of two overlapping cycles (Exposure < Readout for P1 & P2)

The above sequence is time-optimized. Assuming that the exposure time is smaller than the readout time for both cycles:

- The exposure of the second cycle overlaps the readout of the first cycle.
- The exposure of the second cycle terminates exactly when the readout of the first cycle terminates.

Double Exposure Mode Emulation

The following drawing shows a particular Interleaved Area-scan Acquisition sequence of two overlapping cycles where the second exposure time matches the readout time:



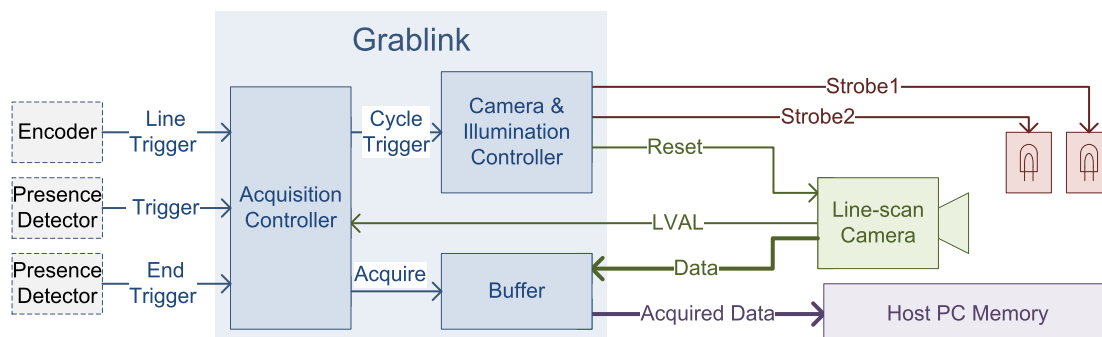
Acquisition sequence of two overlapping cycles (Exposure < Readout for P1; Exposure = Readout for P2)

The exposure time of the second cycle is increased to become equal to the readout time. This allows the second strobe to be issued immediately after the first strobe.

This emulates the *double exposure mode*.

Interleaved Line-scan Acquisition Principles

System Description



Main elements of a line-scan acquisition system configured for Interleaved Area-scan Acquisition

The system is composed of:

- One asynchronous reset grabber-controlled exposure line-scan camera.
- One acquisition channel of a compatible frame grabber configured for Interleaved Line-scan acquisition.
- Two illumination devices, each being controlled by a specific strobe output of the frame grabber.

Usually, the Line Trigger event is obtained by processing signals delivered by a motion encoder. As for any line-scan imaging systems, it can be processed by the rate converter or the rate divider.

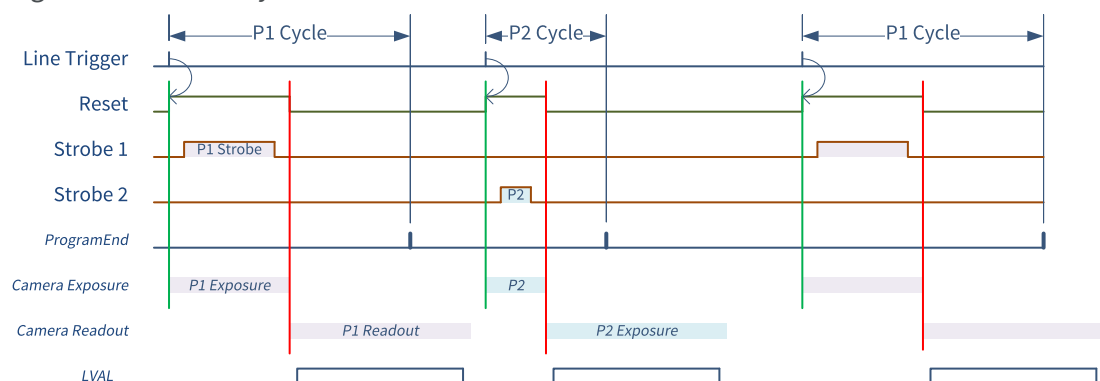
Usually, the Trigger and the EndTrigger events are delivered by a presence detector.

WEB, PAGE and **LONGPAGE** line-scan acquisition modes are compatible with Interleaved Line Acquisition.

Operation

The Interleaved Acquisition feature allows to capture, in a single scanning operation, a composite image where the odd and even lines are captured with different exposure time, strobe duration, strobe delay and strobe output settings.

As shown on the following diagram, the Camera and Illumination Controller executes both programs alternatively: P1 then P2 then P1 ... :



Acquisition sequence showing the alternating P1 and P2 cycles



NOTE

In this example, the line trigger interval is larger than the minimum allowed.

The toggling program sequence is reset at every start-of-scan to ensure that the first captured image line of a scanned object is always built using P1.

In **WEB** acquisition mode, a reset occurs only once at the beginning of the acquisition sequence.

In **PAGE** acquisition mode, a reset occurs at the beginning of every acquisition phase.

In **LONGPAGE** acquisition mode, a reset occurs at the beginning of the first acquisition phase of every acquisition sequence. No reset occurs at the beginning of the subsequent phases of the same sequence.

Reset and Strobe Signals Routing

The reset pulses of both programs are merged into a common Reset signal; the Reset signal can be sent to one or more of the 4 Camera Link Control lines CC1 ... CC4.

The strobe pulse of each program can be routed individually to IOUT1, to IOUT2 or left unused.

When routed to the same output line, the two pulses are merged!

Interleaved Camera and Illumination Control

This section describes the operation of the Camera and Illumination Controller – CIC – when Interleaved Acquisition is enabled.



NOTE

This section applies to line-scan and area-scan interleaved acquisition.

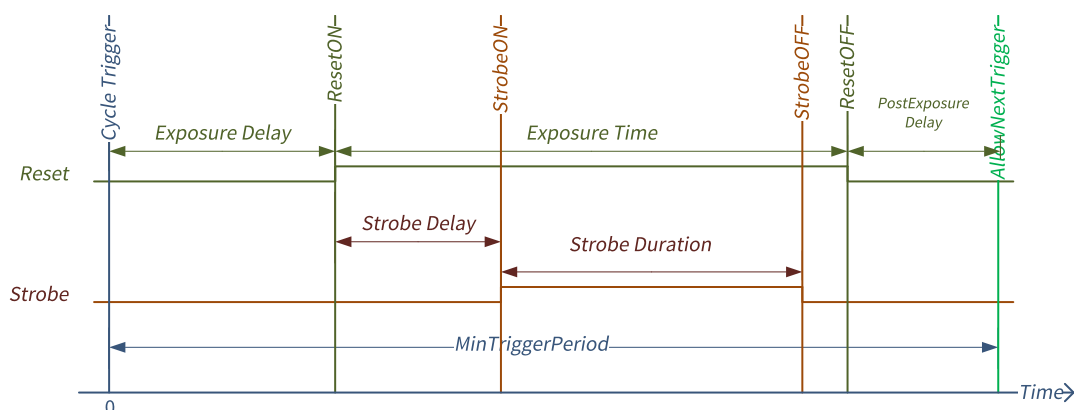
Cycle Trigger designates the event that initiates a CIC cycle: a Line Trigger event in case of line-scan cameras or a Trigger event in case of area-scan cameras.

CIC Cycle Programs

When Interleaved Acquisition is enabled, the Camera and Illumination Controller is configured with two, usually different, camera and illumination cycle programs. These programs are named P1 and P2 respectively.

Each camera and illumination cycle program defines five events on a timeline beginning with a Cycle Trigger event.

1. **ResetON**: turn ON time of the Reset pulse and Start of Exposure
2. **ResetOFF**: turn OFF time of the Reset pulse and End of Exposure
3. **StrobeON**: turn ON time of the Strobe pulse and Start Of Illumination
4. **StrobeOFF**: turn OFF time of the Strobe pulse and End Of Illumination
5. **AllowNextTrigger**: the last event of a program indicating that a new cycle may be initiated.



CIC Program Cycle events and timing definitions

Each program defines two pulses: one Reset pulse and one Strobe pulse. Their timing is user configurable:

- **Exposure Time** is the time interval between the **ResetON** and the **ResetOFF** events.

- **Strobe Duration** is the time interval between the **StrobeON** and the **StrobeOFF** events.
- **Exposure Delay** is the time interval between the **Cycle Trigger** and **ResetON** events.
- **Strobe Delay** is the time interval between the **ResetON** and **StrobeON** events. This value can be positive, null, or negative allowing the Strobe pulse to be positioned anywhere relatively to the start of exposure.

The following restrictions apply on the position order of the events on the timeline:

$$0 \leq \text{ResetON} < \text{ResetOFF} \leq \text{AllowNextTrigger}$$

$$0 \leq \text{StrobeON} < \text{StrobeOFF} \leq \text{AllowNextTrigger}$$

MultiCam Camera Trigger Overrun Protection Principle

At acquisition channel configuration time:

- MultiCam checks if the exposure time user setting can be achieved by the camera. If the user setting of the exposure time is out of bounds, MultiCam corrects its value. The effective exposure time will be set to the nearest boundary.
- MultiCam calculates the position on the timeline of the **AllowNextTrigger** event of P1 and P2 programs. This calculation takes into account the camera operating limits and the user-defined exposure and strobe timing settings for P1 and P2 programs.

At acquisition channel run time, MultiCam reports a "trigger violation" error if a **Cycle Trigger** event is issued before the **AllowNextTrigger** event during the execution of a program.

Camera Operating Limits

The following camera operating limits are considered:

- Exposure time range
- Minimum time interval between two consecutive Exposure
- Maximum line rate

In MultiCam, the following parameters describe the operating limits of a camera:

- **ExposeMin_us**: declares the minimum exposure time, expressed in microseconds (i.e. the minimum duration of a Reset pulse).
- **ExposeMax_us**: declares the maximum exposure time, expressed in microseconds (i.e. the maximum duration of a Reset pulse).
- **ResetDur**: for line-scan cameras only, declares the minimum time interval between two consecutive Reset pulses.
- **ExposeRecovery_us**: for area-scan cameras only, declares the minimum time interval between two consecutive Reset pulses.

- **LineRate_Hz**: for line-scan cameras only, declares the highest line rate supported by the camera (i.e. the reciprocal of the readout time)
- **FrameRate_mHz**: for area-scan cameras only, declares the highest frame rate supported by the camera (i.e. the reciprocal of the readout time)
- **ExposeOverlap**: declares that the camera allows or forbids the next exposure to begin before the completion of the current readout.

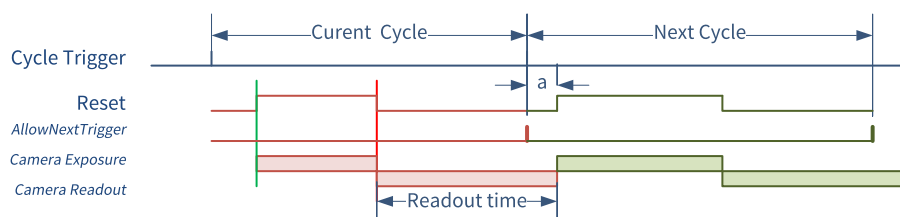
Allow Next Trigger Rules

MultiCam applies the following rules when it calculates the position of the *AllowNextTrigger* event.

RULE 1a - Readout time limitation (Expose Overlapping forbidden)

This rule applies only when **ExposeOverlap** = **FORBID**.

The **start of exposure** of the next cycle may not occur before the end of the current camera readout.



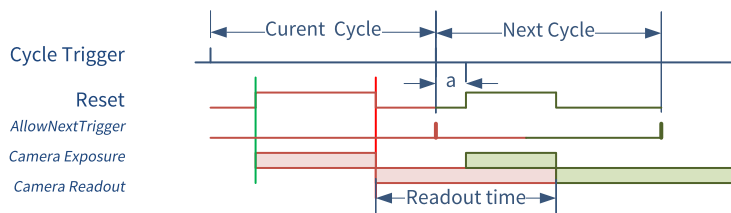
NOTE

If there is any exposure delay (a) in the next cycle, the **AllowNextTrigger** event may be generated earlier.

RULE 1b - Readout time limitation (Expose Overlapping allowed)

This rule applies only when **ExposeOverlap** = **ALLOW**.

The **end of exposure** of the next cycle must not occur before the end of the current camera readout.

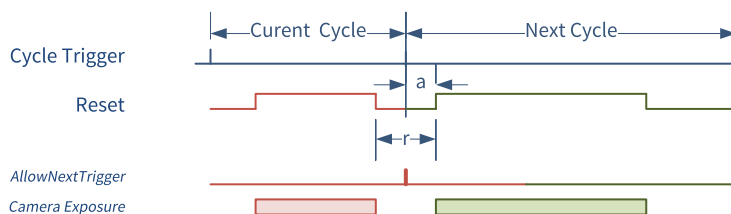


NOTE

If there is any exposure delay (a) in the next cycle, the **AllowNextTrigger** event may be generated earlier.

RULE 2 - Reset interval limitation

The time interval (r) between consecutive Reset pulses may not be shorter than the value specified by **ResetDur**.



NOTE

If there is any exposure delay (a) in the next cycle, the **AllowNextTrigger** event may be generated earlier.

RULE 3 – Next Cycle

**NOTE**

This rule applies only to line-scan interleaved acquisition!

The next cycle after P1 is undetermined, it can be either P2 or P1. The next cycle after P2 is always P1.

For the calculation of the position of the **AllowNextTrigger** event of P1, MultiCam evaluates both hypothesis (P1 and P2) and keeps the worst case.

For the calculation of the position of the **AllowNextTrigger** event of P2, MultiCam assumes that the next cycle is P1.

Exposure Delay

By default, MultiCam configures P1 and P2 with the smallest possible value:

- 0 when $\text{StrobeDelay_P<1,2>_us} \geq 0$
- $(-\text{StrobeDelay_P<1,2>_us})$ when $\text{StrobeDelay_P<1,2>_us} < 0$

If required, the exposure delay can be configured using any of the following methods:

Exposure Delay - Manual Method

This method is invoked when **ExposureDelayControl** is set to **MANUAL**. This is the default method.

With this method, the user may specify the exposure delay for P1 and P2 with:

ExposureDelay_MAN_P1_us and **ExposureDelay_MAN_P2_us**.

By default, these parameters are set to 0.

MultiCam calculates the smallest possible value for each program separately as follows:

- $\text{ExposureDelay_MAN_P<1,2>_us}$ when $\text{StrobeDelay_P<1,2>_us} \geq (-\text{ExposureDelay_MAN_P<1,2>_us})$
- $(-\text{StrobeDelay_P<1,2>_us})$ when $\text{StrobeDelay_P<1,2>_us} < (-\text{ExposureDelay_MAN_P<1,2>_us})$

Exposure Delay - Automatic method 1 (Same Start of Exposure)

Select this method by setting **ExposureDelayControl** to **SAME_START_EXPOSURE**. MultiCam calculates, the smallest value ensuring that the start of exposure occurs at the same position on the program timeline.

Exposure Delay - Automatic method 2 (Same End of Exposure)

Select this method by setting **ExposureDelayControl** to **SAME_END_EXPOSURE**.

MultiCam calculates the smallest values ensuring that the end of exposure occurs at the same position on the program timeline.

Effective Exposure Time

The effective exposure time values are reported by `ExposureTime_P1_Effective_us` and `ExposureTime_P2_Effective_us`.



NOTE

In general, the effective values are very close to the user settings, the slight differences can be explained by rounding issues to the nearest timer tick period.



NOTE

Large differences can be observed in the case of an out-of-bound user setting.

Effective Strobe Duration and Strobe Delay

The effective strobe duration and strobe delay values are reported by `StrobeDuration_P1_Effective_us`, `StrobeDuration_P2_Effective_us`, `StrobeDelay_P1_Effective_us` and, `StrobeDelay_P2_Effective_us`.



NOTE

The effective values are, in any case, very close to the user settings, the slight differences can be explained by rounding issues to the nearest timer tick period.

Effective Exposure Delay

The effective exposure delay values are reported by `ExposureDelay_P1_Effective_us` and `ExposureDelay_P2_Effective_us`.



NOTE

When `ExposureDelayControl` = `MANUAL`, the effective values are very close to the user settings, the slight differences can be explained by rounding issues to the nearest timer tick period.



NOTE

Large differences can be observed in the case of negative strobe delay values.

Effective Minimum Trigger Period

The run time of each program is reported by `MinTriggerPeriod_P1_us` and `MinTriggerPeriod_P2_us`.



NOTE

The values can be different. Considering that programs are executed alternatively, the user should only consider the larger value as the minimum time interval between line triggers.

Interleaved Area-scan Acquisition Channel Setup

PxxRG_IA CAM File Template

```
; *****
*****
; Camera Manufacturer: Templates
; Camera Model: MyCameraLink
; Camera Configuration: Interleaved Area-Scan Acquisition, Asynchronous Reset,
Grabber-Controlled Exposure
; Board: Grablink

; *****
*****
; This CAM file template is suitable for the following camera configuration:
;   - Progressive area-scan camera
;   - Asynchronous Reset
;   - Pulse-Width grabber-controlled exposure
; This CAM file template is suitable for the following system configuration:
;   - SNAPSHOT and HFR Acquisition Modes
;   - Interleaved Acquisition
;
; *****
; ** CAUTION: **
; ** This file is a template, it can be further customized! **
; ** The lines that can be edited are marked with an arrow followed by the most **
; ** popular alternate values for that parameter. **
; ** For a complete list of possible values; refer to MultiCam Studio and/or to **
; ** the MultiCam Reference documentation. **
; *****
;

; *****
*****
; ==Begin of "Camera properties Section"==
;
; -Camera Specification category-
;   Camera =           MyCameraLink;
;   CamConfig =        PxxRG;
;   Imaging =          AREA;
;   Spectrum =         BW;                <== BW COLOR ...
;
; -Camera Features category-
;   TapConfiguration =  BASE_1T8;          <== BASE_1T8 BASE_1T10 BASE_1T24 ...
;   TapGeometry =       1X_1Y;            <== 1X_1Y 1X2_1Y 2X_1Y ...
;   Expose =            WIDTH;
;   Readout =           INTCTL;
;   ColorMethod =       NONE;             <== NONE PRISM BAYER RGB
;   ColorRegistration = BG;               <== GB BG RG GR (when ColorMethod=BAYER)
;   ExposeOverlap =     FORBID;           <== FORBID ALLOW
;
; --Downstream signals--
;   FvalMode =          FA;
;   LvalMode =          LA;
;   DvalMode =          DN;                <== DN DG
;
; --Upstream signals--
;   ResetCtl =          DIFF;
;   ResetEdge =         GOHIGH;           <== GOHIGH GOLOW
;   CC1Usage =          RESET;            <== LOW HIGH RESET SOFT
;   CC2Usage =          LOW;              <== LOW HIGH RESET SOFT
;   CC3Usage =          LOW;              <== LOW HIGH RESET SOFT
```

```

    CC4Usage =          LOW;                <== LOW HIGH RESET SOFT
;
; -Camera Timing category-
    Hactive_Px =        640;                <==
    Vactive_Ln =        480;                <==
    HSyncAft_Tk =        0;                 <==
    VSyncAft_Ln =        0;                 <==
    FrameRate_mHz =     30000;              <==
    ExposeRecovery_us =  10;                 <==
    ReadoutRecovery_us = 10;                 <==
    ExposeMin_us =       10;                 <==
    ExposeMax_us =      1000000;            <==
;
; ==End of "Camera properties Section"==

;*****
;*****
; ==Begin of "System properties Section"==
;
; -Acquisition Control category-
    AcquisitionMode =    SNAPSHOT;           <== SNAPSHOT HFR
    TrigMode =          IMMEDIATE;          <== IMMEDIATE HARD SOFT COMBINED
    NextTrigMode =      SAME;               <== SAME HARD SOFT COMBINED REPEAT
    ActivityLength =     1;                 <== 1
    SeqLength_Fr =       2;                 <== -1 1..65534
    PhaseLength_Fr =     1;                 <== 1 (when AcquisitionMode = SNAPSHOT)
;                                         <== 1..255 (when AcquisitionMode = HFR)
;
;
; -Trigger Control category-
;   The following 5 parameters are relevant only when EndTrigMode = HARD:
    TrigCtl =           ISO;                <== ISO DIFF
    TrigEdge =          GOHIGH;             <== GOHIGH GOLOW
    TrigFilter =         MEDIUM;            <== OFF ON MEDIUM STRONG
    TrigDelay_us =       0;                 <==
    TrigLine =          NOM;                <== NOM ...
;
;   The following 2 parameters are controlling the Trigger Decimation circuit.
;   Remove the leading semi-column on both parameters if the function is required!
;   TrigDelay_Pls =      0;                 <== 0..65536
;   NextTrigDelay_Pls = 0;                 <== 0..65536
;
; -Interleaved Acquisition category-
    InterleavedAcquisition = ON;            <== Enable interleaved acquisition
;   Define the exposure time for P1 and P2 (= RESET signal pulse width)
    ExposureTime_P1_us = 7000.0;            <== Float (0.16 up to 5000000)
    ExposureTime_P2_us = 35000.0;           <== Float (0.16 up to 5000000)
;   Define the strobe duration for P1 and P2 (= STROBE1 and STROBE2 signals pulse
width)
    StrobeDuration_P1_us = 7000.0;          <== Float (0.16 up to 5000000)
    StrobeDuration_P2_us = 10000.0;         <== Float (0.16 up to 5000000)
;   Define the strobe delay for P1 and P2 (relative time offset from RESET going
ON to STROBEx going ON)
;   The time offset can be positive, null or negative
    StrobeDelay_P1_us = 0.0;                <== Float (-10000 up to 5000000)
    StrobeDelay_P2_us = 0.0;                <== Float (-10000 up to 5000000)
;   Select the Exposure delay control method
    ExposureDelayControl = MANUAL;          <== MANUAL SAME_END_EXPOSURE SAME_START_
EXPOSURE
;   When ExposureDelayControl is MANUAL, select the minimum delay from the trigger
;   to the start of exposure (RESET signal going on)
    ExposureDelay_MAN_P1_us = 0;            <== Float (0 up to 5000000)
    ExposureDelay_MAN_P2_us = 0;            <== Float (0 up to 5000000)
    StrobeLine_P1 =      IOUT1;             <== IOUT1 IOUT2 NONE
    StrobeLine_P2 =      IOUT2;             <== IOUT1 IOUT2 NONE
    StrobeOutput_P1 =     ENABLE;           <== ENABLE DISABLE
    StrobeOutput_P2 =     ENABLE;           <== ENABLE DISABLE

```

```

;
; ==End of "System properties Section"==

;*****
;*****
; ==Begin of "Grabber properties Section"==
;
; -Grabber Configuration, Timing & Conditioning categories-
    GrabWindow =          NOBLACK;                <== NOBLACK MAN ...
;    The following 4 parameters are relevant only when GrabWindow = MAN:
    WindowX_Px =          640;                    <==
    WindowY_Ln =          480;                    <==
    OffsetX_Px =           0;                      <==
    OffsetY_Ln =           0;                      <==
;    The following parameter configures the Bayer CFA Decoder.
;    If the default value is not adequate for your application, it is then
necessary
;    to edit the value and to remove the leading semi-column of the following line:
;    CFD_Mode =           ADVANCED;                <== ADVANCED, LEGACY
;
; -Look-Up Tables category-
;    LUT configuration parameters can be inserted here if required by the application
;
; -Cluster category-
    ColorFormat =          Y8;                    <== Y8 Y10 RGB24 RGB24PL ...
    ImageFlipX =           OFF;                    <== OFF ON
    ImageFlipY =           OFF;                    <== OFF ON
;
; End of "Grabber properties Section"

;*****
;*****
; End of File
;=====

```

Customizing Camera Parameters

The following camera parameters must be set according to the selected camera model:

Spectrum, **TapConfiguration**, **TapGeometry**, **ColorMethod**, **DvalMode**, **ResetEdge**, **CC1Usage**, **CC2Usage**, **CC3Usage**, **CC4Usage**, **Hactive_Px**, **Vactive_Ln**, **HSyncAft_Tk** and, **VSynAft_Ln**.

For correct operation of the camera trigger overrun protection mechanism it is mandatory to carefully set the following parameters: **FrameRate_mHz**, **ExposeMin_us**, **ExposeMax_us** and, **ExposeRecovery_us**.

Customizing Acquisition Control Parameters

AcquisitionMode can optionally be set to **HFR**.

In that case **PhaseLength_Fr** can be set to any value in **1 ... 255** range.

The other parameters are not customizable.

Customizing Trigger Control Parameters

The following trigger parameters must be set according to the application needs: **TrigCtl**, **TrigEdge**, **TrigFilter**, **TrigLine**.

The trigger decimation circuit can optionally be activated using **TrigDelay_Pls** and **NextTrigDelay_Pls**.

Customizing Interleaved Acquisition parameters

Enable Interleaved Acquisition by assigning the value **ON** to **InterleavedAcquisition**.

Customizing Interleaved Acquisition – Exposure and Strobe Timing Parameters

When Interleaved Acquisition is enabled, the following exposure and strobe parameters are irrelevant:

Expose_us, **ExposeTrim**, **StrobeMode**, **StrobeDur** and, **PreStrobe_us**.

Instead, the exposure and strobe timings must be defined for P1 and P2 using the following parameter set:

ExposureTime_P1_us, **ExposureTime_P2_us**, **StrobeDuration_P1_us**, **StrobeDuration_P2_us**, **StrobeDelay_P1_us** and, **StrobeDelay_P2_us**.

Customizing Interleaved Acquisition – Exposure Delay Parameters

By default, MultiCam configures P1 and P2 with the smallest possible Exposure Delay value. This setting is satisfactory for the use cases where the exposure time is shorter than the readout time.

Optionally, keeping **ExposureDelayControl** set to **MANUAL**, you may manually change the minimum exposure delay value of P1 and/or P2 using the **ExposureDelay_MAN_P1_us** and **ExposureDelay_MAN_P2_us** parameters.

Alternatively, you may also change **ExposureDelayControl** to one of the automatic control methods: **SAME_START_EXPOSURE** or **SAME_END_EXPOSURE**.

With **SAME_START_EXPOSURE**, the start of exposure is delayed by the same amount of time for both programs: both exposure delay values are equal.

With **SAME_END_EXPOSURE** the end of exposure is delayed by the same amount of time for both programs.

In case of asymmetric exposure times, when at least one exposure time is greater than the readout time, the minimal line trigger period can be achieved when:

- Assigning the longest exposure time to P2
- Inserting an exposure delay prior to the lowest one

Customizing Interleaved Acquisition – Strobe Control Parameters

The **StrobeLine_P1** and **StrobeLine_P2** parameters designate the I/O lines used as strobe outputs for P1 and P2 respectively. The default values are **IOUT1** for P1 and **IOUT2** for P2.

Setting **StrobeLine_P2** to **IOUT1** or **NONE** disconnects the IOUT2 output from the P2 Strobe and makes it available for another usage (Software controlled I/O).

Setting **StrobeLine_P1** and **StrobeLine_P2** to the same output **IOUT1** merges the two strobe pulses .

The **StrobeOutput_P1** and **StrobeOutput_P2** parameters control the delivery of the strobe pulse for P1 and P2 respectively. The delivery is enabled by default. Assigning the **DISABLE** value, inhibits the delivery of the strobe pulse.

Customizing Grabber Timing Parameters

As for any are-scan application, the following grabber configuration, timing and conditioning parameters must be set according to the application needs: **GrabWindow**, **WindowX_Px**, **WindowY_Ln**, **OffsetX_Px** and **OffsetY_Ln**.

Customizing Cluster Parameters

As for any area-scan application, the following cluster parameters must be set according to the application needs: **ColorFormat**, **ImageFlipX** and, **ImageFlipY**.

Interleaved Line-scan Acquisition Channel Setup

LxxxxRG_IA CAM File Template

```
; *****
*****
; Camera Manufacturer: Templates
; Camera Model: MyCameraLink
; Camera Configuration: Interleaved Line-Scan Acquisition, Grabber-Controlled Rate and
Exposure
; Board: Grablink

; *****
*****
; This CAM file template is suitable for the following camera configuration:
;   - Line-scan camera
;   - Grabber-controlled rate
;   - Pulse-Width grabber-controlled exposure
; This CAM file template is suitable for the following system configuration:
;   - WEB, PAGE, or LONGPAGE Acquisition Modes
;   - Take all lines
;   - Interleaved Acquisition
;
; *****
; ** CAUTION: **
; ** This file is a template, it can be further customized! **
; ** The lines that can be edited are marked with an arrow followed by the most **
; ** popular alternate values for that parameter. **
; ** For a complete list of possible values; refer to MultiCam Studio and/or to **
; ** the MultiCam Reference documentation. **
; *****
;

; *****
*****
; ==Begin of "Camera properties Section"==
;
; -Camera Specification category-
;   Camera =           MyCameraLink;
;   CamConfig =        LxxxxRG;
;   Imaging =           LINE;
;   Spectrum =          BW;           <== BW COLOR ...
;
; -Camera Features category-
;   TapConfiguration =  BASE_1T8;      <== BASE_1T8 BASE_1T10 BASE_1T24 ...
;   TapGeometry =       1X;           <== 1X 1X2 2X ...
;   Expose is           WIDTH;
;   Readout is          INTCTL;
;   ColorMethod =       NONE;         <== NONE PRISM TRILINEAR RGB
;
; --Downstream signals--
;   FvalMode =          FN;
;   LvalMode =          LA;
;   DvalMode =          DN;           <== DN DG
;
; --Upstream signals--
;   ResetCtl =          DIFF;
;   ResetEdge =         GOHIGH;       <== GOHIGH GOLOW
;   CC1Usage =          RESET;        <== LOW HIGH RESET SOFT
;   CC2Usage =          LOW;          <== LOW HIGH RESET SOFT
;   CC3Usage =          LOW;          <== LOW HIGH RESET SOFT
;   CC4Usage =          LOW;          <== LOW HIGH RESET SOFT
```

```

;
; -Camera Timing category-
    Hactive_Px = 4096; <==
    HSyncAft_Tk = 0; <==
    LineRate_Hz = 5000; <== Max. line rate (= reciprocal of
readout duration)
    ExposeMin_us = 1; <== Min. exposure time (= RESET signal
pulse width)
    ExposeMax_us = 10000; <== Max. exposure time (= RESET signal
pulse width)
    ResetDur = 3000; <== Min. time interval, in ns, between
consecutive RESET pulses
;
; ==End of "Camera properties Section"==

;*****
; ==Begin of "System properties Section"==
;
; -Acquisition Control category-
    AcquisitionMode = WEB; <== WEB PAGE LONGPAGE
    TrigMode = IMMEDIATE; <== IMMEDIATE HARD SOFT COMBINED
    NextTrigMode = REPEAT; <== REPEAT (when AcquisitionMode = WEB or
LONGPAGE)
; <== SAME REPEAT HARD SOFT COMBINED (when
AcquisitionMode = PAGE)
    EndTrigMode = AUTO; <== AUTO HARD (when AcquisitionMode =
LONGPAGE)
; <== AUTO (when AcquisitionMode = WEB or
PAGE)
    BreakEffect = FINISH; <== FINISH ABORT
    SeqLength_Pg = -1; <== -1 1 .. 65534 (when AcquisitionMode =
PAGE)
    SeqLength_Ln = -1; <== -1 1 .. 65534 (when AcquisitionMode =
WEB or LONGPAGE)
    PageLength_Ln = 500; <== 1 .. 65535
;
; -Trigger Control category-
    TrigCtl = ISO; <== ISO DIFF
    TrigEdge = GOHIGH; <== GOHIGH GOLOW
    TrigFilter = MEDIUM; <== OFF ON MEDIUM STRONG
    TrigLine = NOM; <== NOM ...
;
; The following 4 parameters are relevant only when EndTrigMode = HARD!
    EndTrigCtl = ISO; <== ISO DIFF
    EndTrigEdge = GOLOW; <== GOHIGH GOLOW
    EndTrigFilter = MEDIUM; <== OFF ON MEDIUM STRONG
    EndTrigLine = NOM; <== NOM ...
;
; -Interleaved Acquisition category-
    InterleavedAcquisition = ON; <== Enable interleaved acquisition
; Define the exposure time for P1 and P2 (= RESET signal pulse width)
    ExposureTime_P1_us = 64.0; <== Float (0.16 up to 5000000)
    ExposureTime_P2_us = 64.0; <== Float (0.16 up to 5000000)
; Define the strobe duration for P1 and P2 (= STROBE1 and STROBE2 signals pulse
width)
    StrobeDuration_P1_us = 32.0; <== Float (0.16 up to 5000000)
    StrobeDuration_P2_us = 32.0; <== Float (0.16 up to 5000000)
; Define the strobe delay for P1 and P2 (relative time offset from RESET going
ON to STROBE going ON)
; The time offset can be positive, null or negative
    StrobeDelay_P1_us = 16.0; <== Float (-10000 up to 5000000)
    StrobeDelay_P2_us = 16.0; <== Float (-10000 up to 5000000)
; Select the Exposure delay control method
    ExposureDelayControl = MANUAL; <== MANUAL SAME_END_EXPOSURE SAME_START_

```



```

EXPOSURE
;      When ExposureDelayControl is MANUAL, select the minimum delay from the trigger
;      to the start of exposure (RESET signal going on)
ExposureDelay_MAN_P1_us = 0;          <== Float (0 up to 5000000)
ExposureDelay_MAN_P2_us = 0;          <== Float (0 up to 5000000)
StrobeLine_P1 = IOUT1;          <== IOUT1 IOUT2 NONE
StrobeLine_P2 = IOUT2;          <== IOUT1 IOUT2 NONE
StrobeOutput_P1 = ENABLE;        <== ENABLE DISABLE
StrobeOutput_P2 = ENABLE;        <== ENABLE DISABLE
;
; -Encoder Control category-
LineCaptureMode = ALL;
LineRateMode = PERIOD;          <== PERIOD PULSE CONVERT
;      The following 2 parameters are relevant only when LineRateMode = PERIOD:
Period_us = 1000;              <==
PeriodTrim = 0;                <==
;      The following 5 parameters are relevant only when LineRateMode = CONVERT:
LinePitch = 100;               <==
EncoderPitch = 100;            <==
ConverterTrim = 0;             <==
OnMinSpeed = IDLING;           <== IDLING MUTING
;      The following 4 parameters are relevant only when LineRateMode = PULSE or
CONVERT:
;      LineTrigCtl = DIFF_PAISED;          <== ISO DIFF ISO_PAISED DIFF_PAISED
;      If the default value is not adequate for your application, it is then
necessary to edit the value and
;      to remove the leading semi-column of the following line
;      LineTrigEdge = GOHIGH;              <== GOHIGH GOLOW RISING_A FALLING_A ALL_A
;      (when LineTrigCtl = ISO or DIFF)
;                                          <== ALL_A_B (when LineTrigCtl = ISO_PAISED
or DIFF_PAISED)
;                                          <== GOHIGH GOLOW (when LineTrigCtl = TTL
ITTL I12V or LVDS)
LineTrigFilter = MEDIUM;          <== OFF ON MEDIUM STRONG
LineTrigLine = NOM;               <== NOM ...
;      The following parameter controls the Rate divider circuit that is available
;      exclusively when LineRateMode = PULSE.
;      Remove the leading semi-column if the function is required!
;      RateDivisionFactor = 1;             <== 1..512
;      The following 2 parameters are controlling the Backward Motion Cancellation
circuit that is available
;      exclusively when LineTrigCtl = ISO_PAISED or DIFF_PAISED.
;      Remove the leading semi-column on both parameters if the function is required!
;      ForwardDirection = A_LEADS_B;       <== A_LEADS_B B_LEADS_A
;      BackwardMotionCancellationMode = OFF; <== OFF FILTERED COMPENSATE
;
; ==End of "System properties Section"==

;*****
; ==Begin of "Grabber properties Section"==
;
; -Grabber Configuration, Timing & Conditioning categories-
GrabWindow = NOBLACK;             <== NOBLACK MAN ...
;      The following 2 parameters are relevant only when GrabWindow = MAN:
WindowX_Px = 2048;               <==
OffsetX_Px = 0;                  <==
;
; -Look-Up Tables category-
;      LUT configuration parameters can be inserted here if required by the application
;
; -Cluster category-
ColorFormat = Y8;                 <== Y8 Y10 RGB24 RGB24PL ...
ImageFlipX = OFF;                 <== OFF ON
;

```

```
; End of "Grabber properties Section"

;*****
;*****
; End of File
;=====
```

Customizing Camera Parameters

As for any line-scan camera, the following camera parameters must be set according to the selected camera model:

Spectrum, TapConfiguration, TapGeometry, ColorMethod, DvalMode, ResetEdge, CC1Usage, CC2Usage, CC3Usage, CC4Usage, Hactive_Px and, HSyncAft_Tk.

For correct operation of the camera trigger overrun protection mechanism it is essential to carefully set the following parameters:

LineRate_Hz, ExposeMin_us, ExposeMax_us and, ResetDur.

Customizing Acquisition Control Parameters

As for any line-scan application, the following acquisition control parameters must be set according to the application needs: AcquisitionMode, TrigMode, NextTrigMode, EndTrigMode, BreakEffect, SeqLength_Pg, SeqLength_Ln and, PageLength_Ln.

Customizing Trigger Control Parameters

As for any line-scan application, the following trigger and end trigger control parameters must be set according to the application needs: TrigCtl, TrigEdge, TrigFilter, TrigLine, EndTrigCtl, EndTrigEdge, EndTrigFilter and, EndTrigLine.

Customizing Interleaved Acquisition parameters

Enable Interleaved Line-scan Acquisition by assigning the value **ON** to InterleavedAcquisition.

Customizing Interleaved Acquisition – Exposure and Strobe Timing Parameters

When Interleaved Line-scan Acquisition is enabled, the following exposure and strobe parameters are irrelevant:

Expose_us, ExposeTrim, StrobeMode, StrobeDur and, PreStrobe_us.

Instead, the exposure and strobe timings must be defined for P1 and P2 using the following parameter set:

ExposureTime_P1_us, ExposureTime_P2_us, StrobeDuration_P1_us, StrobeDuration_P2_us, StrobeDelay_P1_us and, StrobeDelay_P2_us.

Customizing Interleaved Acquisition – Exposure Delay Parameters

By default, MultiCam configures P1 and P2 with the smallest possible Exposure Delay value. This setting is satisfactory for the use cases where the exposure time is shorter than the readout time.

Optionally, keeping `ExposureDelayControl` set to `MANUAL`, you may manually change the minimum exposure delay value of P1 and/or P2 using the `ExposureDelay_MAN_P1_us` and `ExposureDelay_MAN_P2_us` parameters.

Alternatively, you may also change `ExposureDelayControl` to one of the automatic control methods: `SAME_START_EXPOSURE` or `SAME_END_EXPOSURE`.

With `SAME_START_EXPOSURE`, the start of exposure is delayed by the same amount of time for both programs: both exposure delay values are equal.

With `SAME_END_EXPOSURE` the end of exposure is delayed by the same amount of time for both programs.

In case of asymmetric exposure times, when at least one exposure time is greater than the readout time, the minimal line trigger period can be achieved when:

- Assigning the longest exposure time to P2
- Inserting an exposure delay prior to the lowest one

Customizing Encoder Control Parameters

As for any line-scan application, the following encoder control parameters must be set according to the application needs: `LineCaptureMode`, `LineRateMode`, `Period_us`, `PeriodTrim`, `LinePitch`, `EncoderPitch`, `ConverterTrim`, `OnMinSpeed`, `LineTrigCtl`, `LineTrigEdge`, `LineTrigFilter`, `LineTrigLine`, `RateDivisionFactor`, `ForwardDirection` and, `BackwardMotionCancellationMode`.

Customizing Interleaved Acquisition – Strobe Control Parameters

The `StrobeLine_P1` and `StrobeLine_P2` parameters designate the I/O lines used as strobe outputs for P1 and P2 respectively. The default values are `IOUT1` for P1 and `IOUT2` for P2.

Setting `StrobeLine_P2` to `NONE` disconnects the IOUT2 output from the P2 Strobe and makes it available for another usage (Software controlled I/O).

The `StrobeOutput_P1` and `StrobeOutput_P2` parameters control the delivery of the strobe pulse for P1 and P2 respectively. The delivery is enabled by default. Assigning the `DISABLE` value, inhibits the delivery of the strobe pulse.

Customizing Grabber Timing Parameters

As for any line-scan application, the following grabber configuration, timing and conditioning parameters must be set according to the application needs: `GrabWindow`, `WindowX_Px` and, `OffsetX_Px`.

Customizing Cluster Parameters

As for any line-scan application, the following cluster parameters must be set according to the application needs: `ColorFormat` and, `ImageFlipX`.

12.4. Two-line Synchronized Line-scan Acquisition

Applies to: Base DualBase Full FullXR

Acquisition, in a single scanning operation, of images from 2 (or more) Basler Sprint bi-linear Bayer CFA color line-scan cameras with 2 illumination devices turned on alternatively

Introduction

The *Two-line Synchronized Line-scan Acquisition* feature takes advantage of a specificity of the BASLER Sprint bilinear CMOS camera that, when operating in the so-called "Exsync controlled operation – Level controlled Mode – Enhanced Raw Line A" exposes light once every two *Exsync* cycles. For a full description of such camera cycle, refer to ["Two-line Camera Cycles" on the next page](#).

This feature extends the capability of Grablink cards to synchronize multiple line-scan acquisition channels using the 2-signal SyncBus. For an architectural description, refer to ["System Architecture" on page 223](#). For a description of the hardware layer and the SyncBus wiring, refer to ["SyncBus Wiring" on page 230](#).

This feature supports two line capture modes:

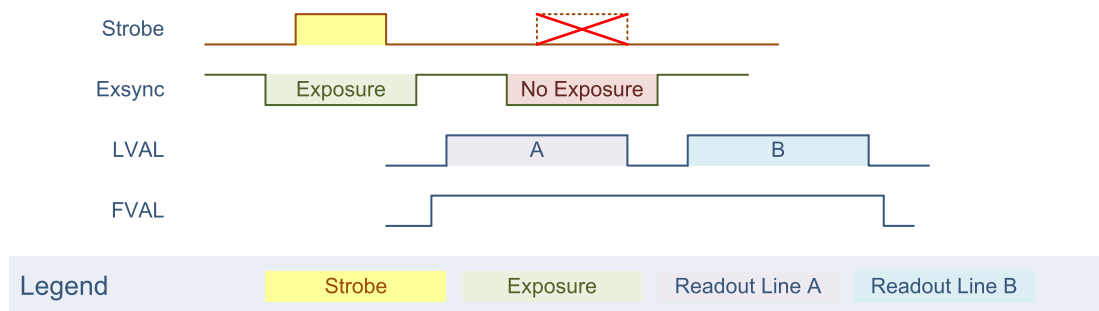
- **LineCaptureMode = ALL:** Take-All-Lines
- **LineCaptureMode = TAG:** Tag-A-Line

The "two-line synchronized acquisition" feature is available since MultiCam 6.9.8. The Tag-A-Line mode is available since MultiCam 6.12.

Two-line Camera Cycles

Basler Sprint Camera Cycle

A single camera cycle of a Basler Sprint CMOS bilinear line-scan camera operating in the "Exsync controlled operation – Level controlled Mode – Raw Line A" requires two consecutive Exsync pulses to be completed:



Basler Sprint camera cycle

The leading (=falling) edge of the first Exsync pulse initiates a new exposure.

The trailing (=rising) edge of the first Exsync terminates the exposure and initiates the readout of the first line (line A) of the sensor.

The leading (=falling) edge of the second Exsync pulse has no function.

The trailing (=rising) edge of the second Exsync pulse initiates the readout of the second line (line B) of the sensor.

The sensor integrates light for all pixels simultaneously during the time interval between the leading and the trailing edge of the first Exsync pulse. The strobe light must be fired during that time interval.

The sensor doesn't integrate light during the low period of the second Exsync pulse. Firing the strobe during that interval has no effect on the acquired data.

The camera qualifies each line of image data by the LVAL signal.

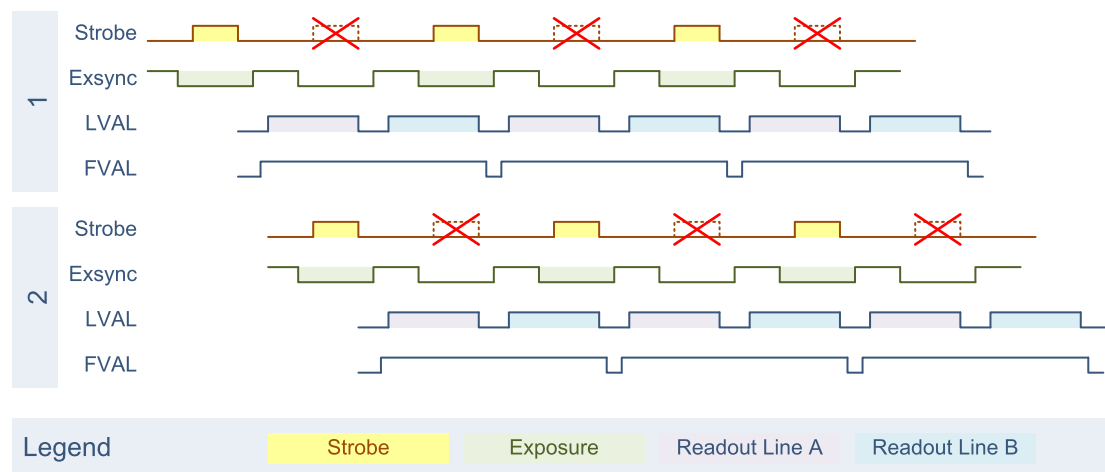
The camera delivers also an FVAL pulse surrounding the two LVAL pulses belonging to the same camera cycle. This allows the frame grabber to unambiguously identify the "line parity"(A or B).



NOTE

The FVAL Length CSR parameter of the Basler Sprint camera must be set to 2.

Phase-shifted Camera Cycles



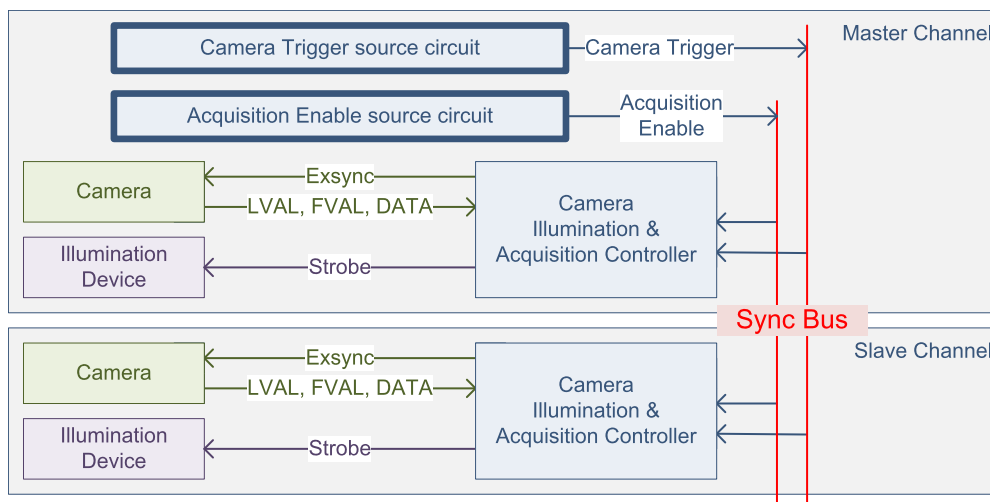
Phase-shifted camera cycles of two Basler Sprint cameras.

The drawing shows the camera cycles of two Basler Sprint cameras where the Exsync periods are synchronized with a phase shift of one period of the Exsync signal.

Notice that:

- The 2 cameras are never exposing simultaneously!
- Firing the illumination during the exposure time interval of a camera will not affect the other camera.

System Architecture



Two-line synchronized line-scan acquisition system

A two-line synchronized line-scan acquisition system is composed of at least 2 MultiCam acquisition channels:

- One "Master Channel"
- One or more "Slave Channels"

Each MultiCam acquisition channel includes:

- 1 Basler Sprint bilinear color line-scan camera
- 1 Strobed illumination device
- 1 Camera, Illumination and Acquisition controller (CIAC).

The Master channel includes:

- 1 Camera Trigger Source circuit that generates the **SyncBus: Camera Trigger** signal.
- 1 Acquisition Enable source circuit that generates the **SyncBus: Acquisition Enable** signal.

The SyncBus distributes the two signals to the Camera, Illumination and Acquisition Controller of all participating channels.

The leading edge of the **SyncBus: Camera Trigger** signal triggers simultaneously all camera and illumination controllers. Each controller sends an Exsync pulse (MultiCam reset signal) having a specified width to the camera. It generates also a strobe pulse once every two Exsync.

The leading edge of the **SyncBus: Acquisition Enable** signal initiates the image data capture on all channels. The image data capture effectively begins on the next occurrence of a line A ensuring that the image data capture begins always on a boundary of the 2 x 2 Bayer CFA pattern.

The falling edge of the **SyncBus: Acquisition Enable** signal terminates the image data capture on all channels. The image data capture effectively terminates after the next occurrence of a line B ensuring that the image data capture terminates always on a boundary of the 2 x 2 Bayer CFA pattern.

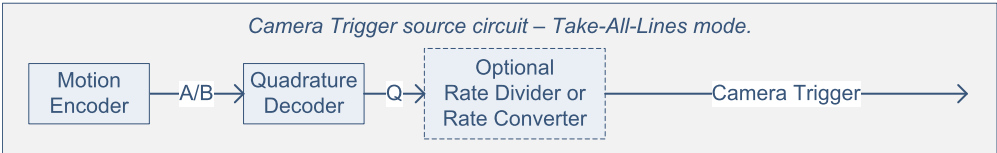
Line Capture Modes

Two line capture modes are available: Take-All-Lines and Tag-A-Line.

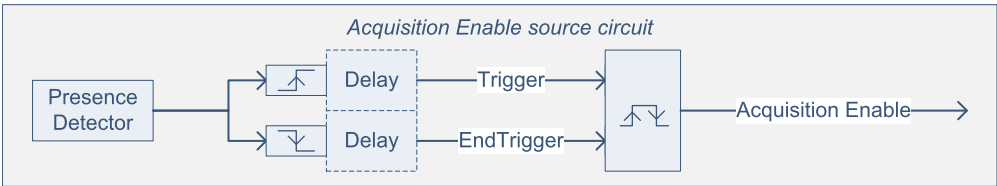
The following table summarizes the characteristics:

Characteristics	Take-All-Lines	Tag-A-Line
Camera line rate	Variable, linked to web speed	Fixed, defined by the frame grabber
Captured image data	All lines	All lines
Line tagging	No	Yes, linked to web speed
Image re-sampling	Not required	Required to be done by the application using tag metadata

Take-All-Lines Line Capture Mode

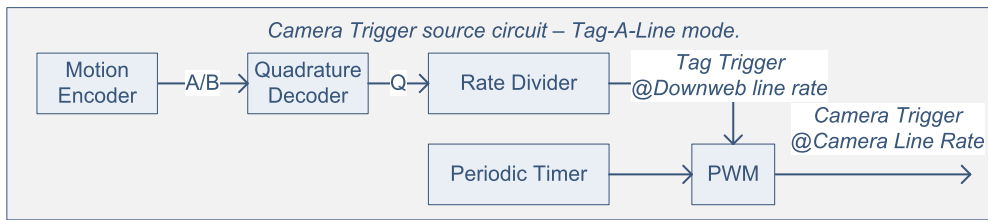


The **camera line rate** is proportional to the web speed to maintain a fixed pitch along the down-web web direction. Therefore, the Master Channel elaborates a **Camera Trigger** signal from the A/B signals using the Quadrature Decoder and, when necessary, the Rate Divider or the Rate Converter.



All lines are captured by the frame grabber when the acquisition is enabled. Therefore, the Master channel elaborates the **Acquisition Enable** signal from both edges of the position detector signal.

Tag-A-Line Line Capture Mode



The **camera line rate** is fixed. Therefore, the Master Channel elaborates a **Camera Trigger** signal using the Periodic Timer.

The **down-web line rate** is proportional to the web speed to obtain, after re-sampling by the application, a fixed pitch along the down-web web direction. The Master channel elaborates also a **Tag Trigger** signal from the A/B signals using the Quadrature Decoder and the Rate Divider. This signal will be used by all acquisition controllers to tag the lines of image data lines to be kept during the down-web re-sampling process.

The Master channel combines both **Tag Trigger** and **Camera Trigger** signals for transmission on the SyncBus.

All lines are tagged and captured by the frame grabber when the acquisition is enabled. Therefore, the Master channel elaborates the **Acquisition Enable** signal from both edges of the position detector signal in the same way as for the Take-All-Lines mode.

The RGB components data of the first pixel of each image line are replaced by a tag indicating if the line was preceded by a **Tag Trigger** or not.

All data bits of R,G and B components are set to 1 when a **Tag Trigger** occurred during the preceding line interval.

All data bits of R,G and B components are set to 0 when no **Tag Trigger** occurred during the preceding line interval.



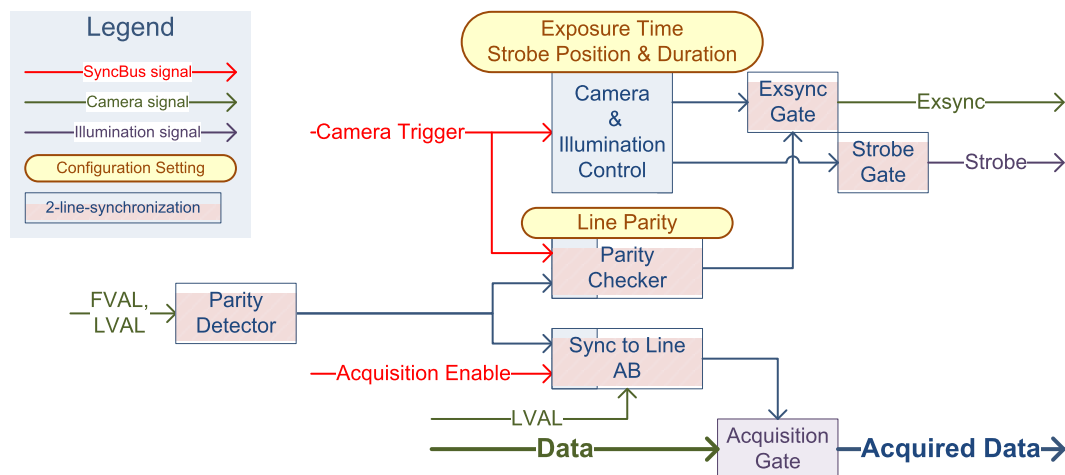
NOTE

When **ColorFormat** is set to **RGB32**, the alpha component is not tagged!

All image data lines are delivered to the application. The application has to perform down-web re-sampling using tags to obtain an undistorted image with a constant down-web line pitch.

Camera, Illumination and Acquisition controller

Camera and Illumination Controller



Camera and Illumination Controller block diagram

On every Camera Trigger event, the Camera and Illumination Controller generates:

- One single **Exsync** pulse, having a duration set by the MultiCam parameter **Expose_us**
- One single **Strobe** pulse, having a duration and a position set by the MultiCam parameters **StrobeDur** and **StrobePos**.

The width of the **Exsync** pulse determines the exposure time of the camera. The strobe pulse duration is entirely located within the exposure time interval. Its position, and its duration are defined as a percentage of the exposure time.

Two-line Synchronization Mode

The two-line synchronization mode of the CIAC must be enabled by setting the value **ENABLE** to the MultiCam Parameter **TwoLineSynchronization**.

This mode of synchronization ensures that the acquisition gate opens and closes at a line-pair boundary. It provides also the capability to control the "Line Parity" of the camera by means of the **TwoLineSynchronizationParity** parameter:

When set to **EVEN**, the camera line parity of the local camera is such that the camera cycle begins at an even line trigger count boundary.

When set to **ODD**, the camera line parity of the local camera is such that the camera cycle begins at an odd line trigger count boundary.

Line Parity Control

The line parity control is composed of three function blocks:

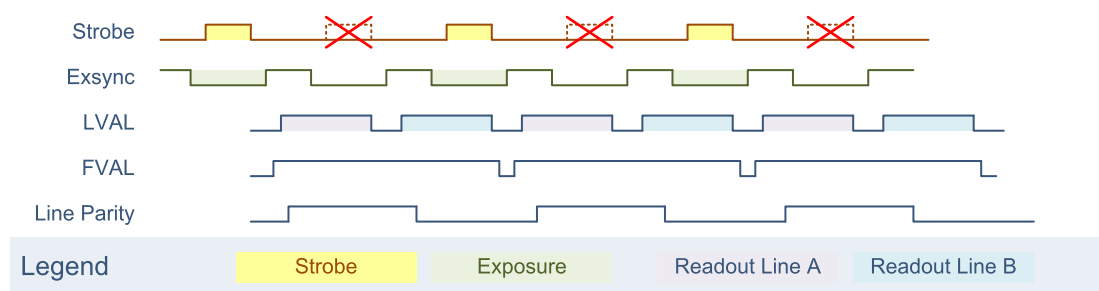
- Parity Detector
- Parity Checker
- Exsync Gate

The Parity Detector function block analyzes the **FVAL** and the **LVAL** signal of the camera and generates the **Camera Line Parity** signal. This signal identifies unambiguously the row A and the row B of the image sensor.

The Parity Checker function block checks whether the **Camera Line Parity** signal is as expected according to the **TwoLineSynchronizationParity** settings.

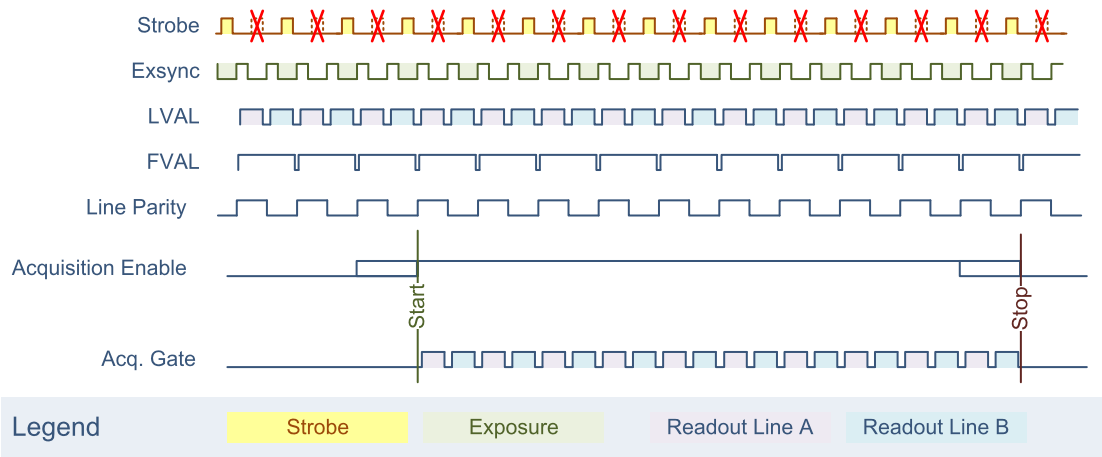
If the **Camera Line Parity** is incorrect, the Exsync Gate removes the next **Exsync** pulse. This action restores the appropriate line parity.

Strobe Gating



The Strobe Gate removes one strobe pulse every two. It keeps only, the strobe corresponding to the Exsync cycle where the local camera exposes.

Acquisition Gating



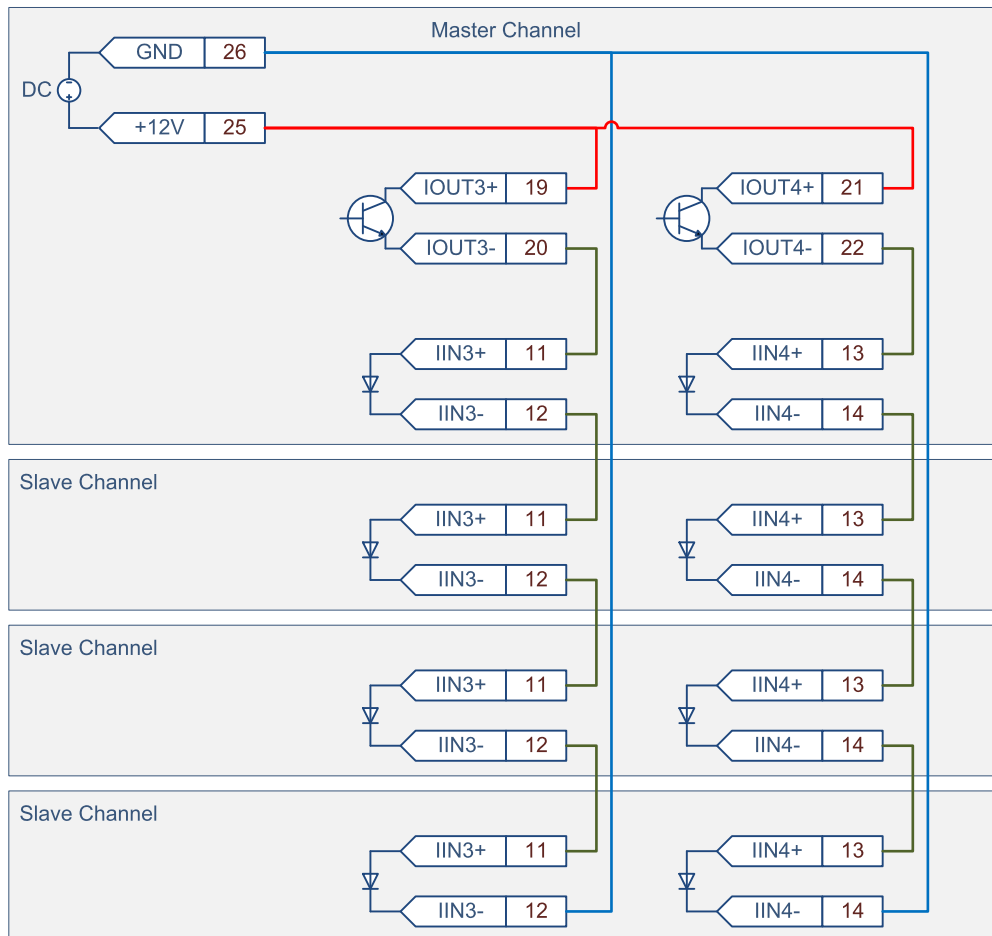
The acquisition gate opens and closes at line-pair boundaries to ensure that buffers always start with a line A and ends with a line B.

SyncBus Wiring

Isolated I/O SyncBus Wiring Scheme

The Isolated I/O SyncBus is implemented with a custom made wiring interconnecting a selected set of I/O pins of the internal I/O connector of each MultiCam Channel.

The following diagram shows the interconnections for a 4-channel SyncBus:



Isolated I/O SyncBus wiring diagram

Camfile Template – Take-All-Lines mode

The following section highlights the additions to the generic `MyCameraLink_LxxxxRG.cam` camfile for configuring the Master MultiCam Channel of a two-line synchronized line-scan acquisition system using the Take-All-Lines line capture mode.

```
;*****
*****
; Camera Manufacturer: Templates
; Camera Model: MyCameraLink
; Camera Configuration: Line-Scan, Grabber-Controlled Rate and Exposure
; Board: Grablink

;*****
*****
; This CAM file template is suitable for the following camera configuration:
;   - Line-scan camera
;   - Grabber-controlled rate
;   - Pulse-Width grabber-controlled exposure
; This CAM file template is suitable for the following system configuration:
;   - WEB, PAGE, or LONGPAGE Acquisition Modes
;   - Take all lines
;
; *****
; ** CAUTION: **
; ** This file is a template, it can be further customized! **
; ** The lines that can be edited are marked with an arrow followed by the most **
; ** popular alternate values for that parameter. **
; ** For a complete list of possible values; refer to MultiCam Studio and/or to **
; ** the MultiCam Reference documentation. **
; *****
;

;*****
*****
; ==Begin of "Camera properties Section"==
;
; -Camera Specification category-
;   Camera = MyCameraLink;
;   CamConfig = LxxxxRG;
;   Imaging = LINE;
;   Spectrum = BW; <== BW COLOR ...
;
; -Camera Features category-
;   TapConfiguration = BASE_1T8; <== BASE_1T8 BASE_1T10 BASE_1T24 ...
;   TapGeometry = 1X; <== 1X 1X2 2X ...
;   Expose is WIDTH;
;   Readout is INTCTL;
;   ColorMethod = NONE; <== NONE PRISM TRILINEAR RGB
;   TwoLineSynchronization = ENABLE;
;   TwoLineSynchronizationParity = EVEN; <== EVEN ODD
;
; --Downstream signals--
;   FvalMode = FN;
;   LvalMode = LA;
;   DvalMode = DN; <== DN DG
;
; --Upstream signals--
;   ResetCtl = DIFF;
;   ResetEdge = GOHIGH; <== GOHIGH GOLOW
;   CClUsage = RESET; <== LOW HIGH RESET SOFT
```

```

CC2Usage =          LOW;                <== LOW HIGH RESET SOFT
CC3Usage =          LOW;                <== LOW HIGH RESET SOFT
CC4Usage =          LOW;                <== LOW HIGH RESET SOFT
;
; -Camera Timing category-
Hactive_Px  =      4096;                <==
HSyncAft_Tk =        0;                <==
LineRate_Hz =     5000;                <==
ExposeMin_us =        1;                <==
ExposeMax_us =    10000;                <==
ResetDur    =      3000;                <==
;
; ==End of "Camera properties Section"==

;*****
;*****
; ==Begin of "System properties Section"==
;
; -Acquisition Control category-
    SynchronizedAcquisition = MASTER;    <== MASTER, SLAVE, LOCAL_MASTER, LOCAL_
SLAVE
    AcquisitionMode =    WEB;            <== WEB PAGE LONGPAGE
    TrigMode =          IMMEDIATE;       <== IMMEDIATE HARD SOFT COMBINED
    NextTrigMode =      REPEAT;          <== REPEAT (when AcquisitionMode = WEB or
LONGPAGE)
;                                     <== SAME REPEAT HARD SOFT COMBINED (when
AcquisitionMode = PAGE)
    EndTrigMode =      AUTO;             <== AUTO HARD (when AcquisitionMode =
LONGPAGE)
;                                     <== AUTO (when AcquisitionMode = WEB or
PAGE)
    BreakEffect =      FINISH;           <== FINISH ABORT
    SeqLength_Pg =     -1;               <== -1 1 .. 65534 (when AcquisitionMode =
PAGE)
    SeqLength_Ln =     -1;               <== -1 1 .. 65534 (when AcquisitionMode =
WEB or LONGPAGE)
    PageLength_Ln =    500;              <== 1 .. 65535
;
; -Trigger Control category-
;     The set of values and the default value assigned by MultiCam for the parameter
TrigCtl are board specific!
;     If the default value is not adequate for your application, it is then
necessary to edit the value and
;     to remove the leading semi-column of one of the following 3 lines
;     TrigCtl =        ISO;              <== Full/DualBase/Base: ISO DIFF
;     TrigCtl =        ITTL;             <== Avenue/Express: TTL ITTL I12V LVDS
;     TrigCtl =        ITTL;             <== Value/Expert 2: TTL ITTL I12V
    TrigEdge =          GOHIGH;          <== GOHIGH GOLOW
    TrigFilter =        MEDIUM;          <== OFF ON MEDIUM STRONG
    TrigLine =          NOM;             <== NOM ...
;
;     The following 4 parameters are relevant only when EndTrigMode = HARD!
;     The set of values and the default value assigned by MultiCam for the parameter
EndTrigCtl are board specific!
;     If the default value is not adequate for your application, it is then
necessary to edit the value and
;     to remove the leading semi-column of one of the following 3 lines
;     EndTrigCtl =     ISO;              <== Full/DualBase/Base: ISO DIFF
;     EndTrigCtl =     ITTL;             <== Avenue/Express: TTL ITTL I12V LVDS
;     EndTrigCtl =     ITTL;             <== Value/Expert 2: TTL ITTL I12V
    EndTrigEdge =       GOLOW;           <== GOHIGH GOLOW
    EndTrigFilter =     MEDIUM;          <== OFF ON MEDIUM STRONG
    EndTrigLine =       NOM;             <== NOM ...
;
; -Exposure & Strobe Control categories-

```



```

Expose_us =          90;                <==
ExposeTrim =         0;                <==
StrobeMode =          NONE;                <== To free the Strobe Output IO port
;
; -Encoder Control category-
LineCaptureMode =    ALL;
LineRateMode =       PERIOD;                <= PERIOD PULSE CONVERT
;   The following 2 parameters are relevant only when LineRateMode = PERIOD:
Period_us =          1000;                <=
PeriodTrim =         0;                <=
;   The following 5 parameters are relevant only when LineRateMode = CONVERT:
LinePitch =          100;                <=
EncoderPitch =       100;                <=
ConverterTrim =      0;                <=
OnMinSpeed =         IDLING;                <= IDLING MUTING
;   The following 4 parameters are relevant only when LineRateMode = PULSE or
CONVERT:
;   The set of values and the default value assigned by MultiCam for the parameter
LineTrigCtl are board specific!
;   If the default value is not adequate for your application, it is then
necessary to edit the value and
;   to remove the leading semi-column of one of the following 3 lines
;   LineTrigCtl =      DIFF_PAURED;                <== Full/DualBase/Base: ISO DIFF ISO_
PAURED DIFF_PAURED
;   LineTrigCtl =      ITTL;                <== Avenue/Express: TTL ITTL I12V LVDS
;   LineTrigCtl =      ITTL;                <== Value/Expert 2: TTL ITTL I12V
;   The set of values and the default value assigned by MultiCam for the parameter
LineTrigEdge are board specific!
;   If the default value is not adequate for your application, it is then
necessary to edit the value and
;   to remove the leading semi-column of the following line
;   LineTrigEdge =      GOHIGH;                <== GOHIGH GOLOW RISING_A FALLING_A ALL_A
(when LineTrigCtl = ISO or DIFF)
;   <== ALL_A_B (when LineTrigCtl = ISO_PAURED
or DIFF_PAURED)
;   <== GOHIGH GOLOW (when LineTrigCtl = TTL
ITTL I12V or LVDS)
LineTrigFilter =     MEDIUM;                <== OFF ON MEDIUM STRONG
LineTrigLine =       NOM;                <== NOM ...
;   The following parameter controls the Rate divider circuit that is available
;   exclusively on Grablink Full/DualBase/Base boards when LineRateMode = PULSE.
;   Remove the leading semi-column if the function is required!
RateDivisionFactor = 1;                <== 1..512
;   The following 2 parameters are controlling the Backward Motion Cancellation
circuit that is available
;   exclusively on Grablink Full/DualBase/Base boards when LineTrigCtl = ISO_
PAURED or DIFF_PAURED.
;   Remove the leading semi-column on both parameters if the function is required!
ForwardDirection =   A_LEADS_B;                <== A_LEADS_B B_LEADS_A
BackwardMotionCancellationMode = OFF; <== OFF FILTERED COMPENSATE
;
; ==End of "System properties Section"==

;*****
;*****
; ==Begin of "Grabber properties Section"==
;
; -Grabber Configuration, Timing & Conditioning categories-
GrabWindow =         NOBLACK;                <== NOBLACK MAN ...
;   The following 2 parameters are relevant only when GrabWindow = MAN:
WindowX_Px =         2048;                <==
OffsetX_Px =         0;                <==
;
; -Look-Up Tables category-
;   The Look-Up Table operator is available exclusively on Grablink Full/DualBase/Base

```

```
boards.
;   LUT configuration parameters can be inserted here if required by the application
;
; -Cluster category-
   ColorFormat =      Y8;                <== Y8 Y10 RGB24 RGB24PL ...
   ImageFlipX =      OFF;                <== OFF ON
;
; End of "Grabber properties Section"

;*****
*****
; End of File
;=====
```

Camfile Template – Tag-A-Line mode

The following section highlights the additions to the generic `MyCameraLink_LxxxxRG.cam` camfile for configuring the Master MultiCam Channel of a two-line synchronized line-scan acquisition using the Tag-A-Line line capture mode.

```
;*****
*****
; Camera Manufacturer: Templates
; Camera Model: MyCameraLink
; Camera Configuration: Line-Scan, Grabber-Controlled Rate and Exposure
; Board: Grablink

;*****
*****
; This CAM file template is suitable for the following camera configuration:
;   - Line-scan camera
;   - Grabber-controlled rate
;   - Pulse-Width grabber-controlled exposure
; This CAM file template is suitable for the following system configuration:
;   - WEB, PAGE, or LONGPAGE Acquisition Modes
;   - Take all lines
;
; *****
; ** CAUTION: **
; ** This file is a template, it can be further customized! **
; ** The lines that can be edited are marked with an arrow followed by the most **
; ** popular alternate values for that parameter. **
; ** For a complete list of possible values; refer to MultiCam Studio and/or to **
; ** the MultiCam Reference documentation. **
; *****
;

;*****
*****
; ==Begin of "Camera properties Section"==
;
; -Camera Specification category-
;   Camera = MyCameraLink;
;   CamConfig = LxxxxRG;
;   Imaging = LINE;
;   Spectrum = BW; <== BW COLOR ...
;
; -Camera Features category-
;   TapConfiguration = BASE_1T8; <== BASE_1T8 BASE_1T10 BASE_1T24 ...
;   TapGeometry = 1X; <== 1X 1X2 2X ...
;   Expose is WIDTH;
;   Readout is INTCTL;
;   ColorMethod = NONE; <== NONE PRISM TRILINEAR RGB
;   TwoLineSynchronization = ENABLE;
;   TwoLineSynchronizationParity = EVEN; <== EVEN ODD
;
; --Downstream signals--
;   FvalMode = FN;
;   LvalMode = LA;
;   DvalMode = DN; <== DN DG
;
; --Upstream signals--
;   ResetCtl = DIFF;
;   ResetEdge = GOHIGH; <== GOHIGH GOLOW
;   CClUsage = RESET; <== LOW HIGH RESET SOFT
```

```

CC2Usage =          LOW;                <== LOW HIGH RESET SOFT
CC3Usage =          LOW;                <== LOW HIGH RESET SOFT
CC4Usage =          LOW;                <== LOW HIGH RESET SOFT
;
; -Camera Timing category-
Hactive_Px  =      4096;                <==
HSyncAft_Tk =        0;                <==
LineRate_Hz =     5000;                <==
ExposeMin_us =        1;                <==
ExposeMax_us =    10000;                <==
ResetDur    =      3000;                <==
;
; ==End of "Camera properties Section"==

;*****
;*****
; ==Begin of "System properties Section"==
;
; -Acquisition Control category-
SynchronizedAcquisition = MASTER      <== MASTER, SLAVE, LOCAL_MASTER,
LOCAL_SLAVE
AcquisitionMode =  WEB;                <== WEB PAGE LONGPAGE
TrigMode =        IMMEDIATE;           <== IMMEDIATE HARD SOFT COMBINED
NextTrigMode =    REPEAT;              <== REPEAT (when AcquisitionMode = WEB or
LONGPAGE)
;                                     <== SAME REPEAT HARD SOFT COMBINED (when
AcquisitionMode = PAGE)
EndTrigMode =     AUTO;                <== AUTO HARD (when AcquisitionMode =
LONGPAGE)
;                                     <== AUTO (when AcquisitionMode = WEB or
PAGE)
BreakEffect =     FINISH;              <== FINISH ABORT
SeqLength_Pg =    -1;                  <== -1 1 .. 65534 (when AcquisitionMode =
PAGE)
SeqLength_Ln =    -1;                  <== -1 1 .. 65534 (when AcquisitionMode =
WEB or LONGPAGE)
PageLength_Ln =   500;                 <== 1 .. 65535
;
; -Trigger Control category-
; The set of values and the default value assigned by MultiCam for the parameter
TrigCtl are board specific!
; If the default value is not adequate for your application, it is then
necessary to edit the value and
; to remove the leading semi-column of one of the following 3 lines
; TrigCtl =        ISO;                <== Full/DualBase/Base: ISO DIFF
; TrigCtl =        ITTL;               <== Avenue/Express: TTL ITTL I12V LVDS
; TrigCtl =        ITTL;               <== Value/Expert 2: TTL ITTL I12V
TrigEdge =         GOHIGH;             <== GOHIGH GOLOW
TrigFilter =       MEDIUM;             <== OFF ON MEDIUM STRONG
TrigLine =         NOM;                <== NOM ...
;
; The following 4 parameters are relevant only when EndTrigMode = HARD!
; The set of values and the default value assigned by MultiCam for the parameter
EndTrigCtl are board specific!
; If the default value is not adequate for your application, it is then
necessary to edit the value and
; to remove the leading semi-column of one of the following 3 lines
; EndTrigCtl =     ISO;                <== Full/DualBase/Base: ISO DIFF
; EndTrigCtl =     ITTL;               <== Avenue/Express: TTL ITTL I12V LVDS
; EndTrigCtl =     ITTL;               <== Value/Expert 2: TTL ITTL I12V
EndTrigEdge =      GOLOW;              <== GOHIGH GOLOW
EndTrigFilter =    MEDIUM;             <== OFF ON MEDIUM STRONG
EndTrigLine =      NOM;                <== NOM ...
;
; -Exposure & Strobe Control categories-

```

```

Expose_us =          90;                <==
ExposeTrim =         0;                <==
StrobeMode =          NONE;                <== To free the Strobe Output IO port
;
; -Encoder Control category-
    LineCaptureMode =    TAG;
    LineRateMode =       PERIOD;          <= PERIOD
;    The following 2 parameters are relevant when LineCaptureMode = TAG:
    Period_us =          1000;           <=
    PeriodTrim =         0;              <=
;    The following 5 parameters are relevant when LineCaptureMode = TAG:
    LinePitch =          100;             <=
    EncoderPitch =       100;             <=
    ConverterTrim =      0;               <=
    OnMinSpeed =         IDLING;          <= IDLING MUTING
;    The following 4 parameters are relevant only when LineCaptureMode = TAG:
;    The set of values and the default value assigned by MultiCam for the parameter
LineTrigCtl are board specific!
;    If the default value is not adequate for your application, it is then
necessary to edit the value and
;    to remove the leading semi-column of one of the following 3 lines
;    LineTrigCtl =       DIFF_PAURED;     <== Full/DualBase/Base: ISO DIFF ISO_
PAURED DIFF_PAURED
;    LineTrigCtl =       ITTL;             <== Avenue/Express: TTL ITTL I12V LVDS
;    LineTrigCtl =       ITTL;             <== Value/Expert 2: TTL ITTL I12V
;    The set of values and the default value assigned by MultiCam for the parameter
LineTrigEdge are board specific!
;    If the default value is not adequate for your application, it is then
necessary to edit the value and
;    to remove the leading semi-column of the following line
;    LineTrigEdge =      GOHIGH;           <== GOHIGH GOLOW RISING_A FALLING_A ALL_A
(when LineTrigCtl = ISO or DIFF)
;    <== ALL_A_B (when LineTrigCtl = ISO_PAURED
or DIFF_PAURED)
;    <== GOHIGH GOLOW (when LineTrigCtl = TTL
ITTL I12V or LVDS)
    LineTrigFilter =     MEDIUM;          <== OFF ON MEDIUM STRONG
    LineTrigLine =       NOM;              <== NOM ...
;    The following parameter controls the Rate divider circuit that is available
when LineCaptureMode = TAG:
;    RateDivisionFactor = 1;                <== 1..512
;    The following 2 parameters are controlling the Backward Motion Cancellation
circuit that is available
;    exclusively on Grablink Full/DualBase/Base boards when LineTrigCtl = ISO_
PAURED or DIFF_PAURED.
;    Remove the leading semi-column on both parameters if the function is required!
;    ForwardDirection =  A_LEADS_B;         <== A_LEADS_B B_LEADS_A
;    BackwardMotionCancellationMode = OFF;  <== OFF FILTERED COMPENSATE
;
; ==End of "System properties Section"==

; *****
; *****
; ==Begin of "Grabber properties Section"==
;
; -Grabber Configuration, Timing & Conditioning categories-
    GrabWindow =         NOBLACK;          <== NOBLACK MAN ...
;    The following 2 parameters are relevant only when GrabWindow = MAN:
    WindowX_Px =         2048;             <==
    OffsetX_Px =          0;                <==
;
; -Look-Up Tables category-
;    The Look-Up Table operator is available exclusively on Grablink Full/DualBase/Base
boards.
;    LUT configuration parameters can be inserted here if required by the application

```

```
;
; -Cluster category-
    ColorFormat =      Y8;                <== Y8 Y10 RGB24 RGB24PL ...
    ImageFlipX =      OFF;                <== OFF ON
;
; End of "Grabber properties Section"

;*****
;*****
; End of File
;=====
```

Camfile Customization

Camera Parameters

As for any line-scan camera, the following camera parameters must be set according to the selected camera model:

Spectrum, TapConfiguration, TapGeometry, ColorMethod, DvalMode, ResetEdge, CC1Usage, CC2Usage, CC3Usage, CC4Usage, Hactive_Px and, HSyncAft_Tk.

For correct operation of the camera trigger overrun protection mechanism it is essential to carefully set the following parameters:

LineRate_Hz, ExposeMin_us, ExposeMax_us and, ResetDur.

To operate with bilinear line-scan cameras:

- The 2-line synchronization mode must be enabled by setting TwoLineSynchronization to **ENABLE**.
- The 2-line synchronization parity must be selected by setting TwoLineSynchronizationParity to **ODD** or **EVEN**.

There is a phase-shift of 1 Exsync cycle between cameras set to ODD and cameras set EVEN.

Acquisition Control Parameters

As for any line-scan application, the following acquisition control parameters must be set according to the application needs: AcquisitionMode, TrigMode, NextTrigMode, EndTrigMode, BreakEffect, SeqLength_Pg, SeqLength_Ln and, PageLength_Ln.

The synchronized acquisition feature must be enabled on all synchronized channels. Refer to "Synchronized Line-scan Acquisition" on page 156.

Trigger Control Parameters

As for any line-scan application, the following trigger and end trigger control parameters must be set according to the application needs: TrigCtl, TrigEdge, TrigFilter, TrigLine, EndTrigCtl, EndTrigEdge, EndTrigFilter and, EndTrigLine.

Exposure and Strobe Timing Parameters

As for any line-scan application, the following exposure and strobe control parameters must be set according to the application needs:

Expose_us, ExposeTrim, StrobeMode, StrobeDur and, PreStrobe_us.

Encoder Control Parameters

As for any line-scan application, the following encoder control parameters must be set according to the application needs: `LineCaptureMode`, `LineRateMode`, `Period_us`, `PeriodTrim`, `LinePitch`, `EncoderPitch`, `ConverterTrim`, `OnMinSpeed`, `LineTrigCtl`, `LineTrigEdge`, `LineTrigFilter`, `LineTrigLine`, `RateDivisionFactor`, `ForwardDirection` and, `BackwardMotionCancellationMode`.

Grabber Timing Parameters

As for any line-scan application, the following grabber configuration, timing and conditioning parameters must be set according to the application needs: `GrabWindow`, `WindowX_Px` and, `OffsetX_Px`.

Cluster Parameters

As for any line-scan application, the following cluster parameters must be set according to the application needs: `ColorFormat` and, `ImageFlipX`.

Basler spL4096-70kc Camfile for Tag-A-Line mode

This topic a customization of the generic `MyCameraLink_LxxxxRG.cam` camfile for a bilinear color *Basler spL4096-70kc* camera attached to the Master MultiCam Channel of a two-line synchronized line-scan acquisition using the Tag-A-Line line capture mode.

```
;*****
*****
; Camera Manufacturer: BASLER
; Camera Model: spL4096-70kc
; Camera Configuration: RAW Dual Line, 2048 pixels, Grabber-Controlled rate and
exposure
; Board: Grablink
; Minimum MultiCam Version: 6.5
; Last update: 25 Sept 2017

;*****
*****
;Disclaimer:
;
;These CAM-files are provided to you free of charge and "as is".
;You should not assume that these CAM-files are error-free or
;suitable for any purpose whatsoever.
;Nor should you assume that all functional modes of the camera are
;covered by these CAM files or that the associated documentation is complete.
;EURESYS does not give any representation or warranty that these CAM-files are
;free of any defect or error or suitable for any purpose.
;EURESYS shall not be liable, in contract, in torts or otherwise,
;for any damages, loss, costs, expenses or other claims for compensation,
;including those asserted by third parties, arising out of or in connection
;with the use of these CAM-files.

;*****
*****
;
; ==Begin of "Camera properties Section"==
;
; -Camera Specification category-
;   Camera =          MyCameraLink;
;   CamConfig =       LxxxxRG;
;   Imaging =         LINE;
;   Spectrum =        COLOR;          <== BW COLOR ...
;
; -Camera Features category-
;   TapConfiguration = BASE_2T8;      <== BASE_1T8 BASE_1T10 BASE_1T24 ...
;   TapGeometry =      1X2;           <== 1X 1X2 2X ...
;   Expose is          WIDTH;
;   Readout is         INTCTL;
;   ColorMethod =      BAYER;         <== NONE PRISM TRILINEAR RGB
;   ColorRegistration=  RG;
;   ColorRegistrationControl= FVAL;
;
; --Downstream signals--
;   FvalMode =         FN;
;   LvalMode =         LA;
;   DvalMode =         DN;           <== DN DG
;
; --Upstream signals--
;   ResetCtl =         DIFF;
;   ResetEdge =        GOLOW;        <== GOHIGH GOLOW
;   CClUsage =         RESET;        <== LOW HIGH RESET SOFT
```

```

CC2Usage =          LOW;          <== LOW HIGH RESET SOFT
CC3Usage =          LOW;          <== LOW HIGH RESET SOFT
CC4Usage =          LOW;          <== LOW HIGH RESET SOFT
;
; -Camera Timing category-
Hactive_Px =        2048;          <==
HSyncAft_Tk =        0;           <==
LineRate_Hz =       50000;         <==
ExposeMin_us =       2;           <==
ExposeMax_us =      10000;         <==
ResetDur =          3000;         <==

TwoLineSynchronization= ENABLE;
TwoLineSynchronizationParity= EVEN;
;
; ==End of "Camera properties Section"==

;*****
; ==Begin of "System properties Section"==
;
; -Acquisition Control category-
AcquisitionMode =    LONGPAGE;      <== WEB PAGE LONGPAGE

SynchronizedAcquisition= MASTER;

TrigMode =           HARD;          <== IMMEDIATE HARD SOFT COMBINED
NextTrigMode =       REPEAT;        <== REPEAT (when AcquisitionMode = WEB or
LONGPAGE)
;                               <== SAME REPEAT HARD SOFT COMBINED (when
AcquisitionMode = PAGE)
EndTrigMode =        HARD;          <== AUTO HARD (when AcquisitionMode =
LONGPAGE)
;                               <== AUTO (when AcquisitionMode = WEB or PAGE)
BreakEffect =        ABORT;         <== FINISH ABORT
SeqLength_Pg =       -1;            <== -1 1 .. 65534 (when AcquisitionMode =
PAGE)
SeqLength_Ln =       -1;            <== -1 1 .. 65534 (when AcquisitionMode = WEB
or LONGPAGE)
PageLength_Ln =      128;           <== 1 .. 65535
;
; -Trigger Control category-
; The set of values and the default value assigned by MultiCam for the parameter
TrigCtl are board specific!
; If the default value is not adequate for your application, it is then
necessary to edit the value and
; to remove the leading semi-column of one of the following 3 lines
TrigCtl =            ISO;           <== Full/DualBase/Base: ISO DIFF
; TrigCtl =           ITTL;         <== Avenue/Express: TTL ITTL I12V LVDS
; TrigCtl =           ITTL;         <== Value/Expert 2: TTL ITTL I12V
TrigEdge =            GOHIGH;       <== GOHIGH GOLOW
TrigFilter =          MEDIUM;       <== OFF ON MEDIUM STRONG
TrigLine =            IIN1;         <== NOM ...
;
; The following 4 parameters are relevant only when EndTrigMode = HARD!
; The set of values and the default value assigned by MultiCam for the parameter
EndTrigCtl are board specific!
; If the default value is not adequate for your application, it is then
necessary to edit the value and
; to remove the leading semi-column of one of the following 3 lines
EndTrigCtl =          ISO;          <== Full/DualBase/Base: ISO DIFF
; EndTrigCtl =         ITTL;         <== Avenue/Express: TTL ITTL I12V LVDS
; EndTrigCtl =         ITTL;         <== Value/Expert 2: TTL ITTL I12V
EndTrigEdge =          GOLOW;        <== GOHIGH GOLOW
EndTrigFilter =        MEDIUM;       <== OFF ON MEDIUM STRONG

```

```

    EndTrigLine =      IIN1;          <== NOM ...
;
; -Exposure & Strobe Control categories-
    Expose_us =        245;          <==
    ExposeTrim =        0;           <==
    StrobeMode =        AUTO;         <== To free the Strobe Output IO port
;
; -Encoder Control category-
    LineCaptureMode =   TAG;
    LineRateMode =      PERIOD;       <= PERIOD PULSE CONVERT
;    The following 2 parameters are relevant only when LineRateMode = PERIOD:
    Period_us =         250;         <=
    PeriodTrim =         0;           <=
;    The following 5 parameters are relevant only when LineRateMode = CONVERT:
    LinePitch =         100;         <=
    EncoderPitch =       100;         <=
    ConverterTrim =      0;           <=
    OnMinSpeed =         IDLING;      <= IDLING MUTING
;    The following 4 parameters are relevant only when LineRateMode = PULSE or
CONVERT:
;    The set of values and the default value assigned by MultiCam for the parameter
LineTrigCtl are board specific!
;    If the default value is not adequate for your application, it is then
necessary to edit the value and
;    to remove the leading semi-column of one of the following 3 lines
    LineTrigCtl =       ISO;          <== Full/DualBase/Base: ISO DIFF ISO_PAISED
DIFF_PAISED
;    LineTrigCtl =      ITTL;         <== Avenue/Express: TTL ITTL I12V LVDS
;    LineTrigCtl =      ITTL;         <== Value/Expert 2: TTL ITTL I12V
;    The set of values and the default value assigned by MultiCam for the parameter
LineTrigEdge are board specific!
;    If the default value is not adequate for your application, it is then
necessary to edit the value and
;    to remove the leading semi-column of the following line
;    LineTrigEdge =      GOHIGH;       <== GOHIGH GOLOW RISING_A FALLING_A ALL_A
(when LineTrigCtl = ISO or DIFF)
;
;                                     <== ALL_A_B (when LineTrigCtl = ISO_PAISED or
DIFF_PAISED)
;
;                                     <== GOHIGH GOLOW (when LineTrigCtl = TTL ITTL
I12V or LVDS)
    LineTrigFilter =     MEDIUM;      <== OFF ON MEDIUM STRONG
    LineTrigLine =       IIN2;         <== NOM ...
;    The following parameter controls the Rate divider circuit that is available
;    exclusively on Grablink Full/DualBase/Base boards when LineRateMode = PULSE.
;    Remove the leading semi-column if the function is required!
    RateDivisionFactor = 4;            <== 1..512
;    The following 2 parameters are controlling the Backward Motion Cancellation
circuit that is available
;    exclusively on Grablink Full/DualBase/Base boards when LineTrigCtl = ISO_
PAISED or DIFF_PAISED.
;    Remove the leading semi-column on both parameters if the function is required!
;    ForwardDirection =  A_LEADS_B;    <== A_LEADS_B B_LEADS_A
;    BackwardMotionCancellationMode = OFF;<== OFF FILTERED COMPENSATE
;
; ==End of "System properties Section"==

;*****
*****
; ==Begin of "Grabber properties Section"==
;
; -Grabber Configuration, Timing & Conditioning categories-
    GrabWindow =         NOBLACK;     <== NOBLACK MAN ...
;    The following 2 parameters are relevant only when GrabWindow = MAN:
    WindowX_Px =         4096;        <==
    OffsetX_Px =          0;           <==

```

```

;
; -Look-Up Tables category-
;   The Look-Up Table operator is available exclusively on Grablink Full/DualBase/Base
boards.
;   LUT configuration parameters can be inserted here if required by the application
;
; -Cluster category-
;   ColorFormat =          RGB24PL;          <== Y8 Y10 RGB24 RGB24PL ...
;   ImageFlipX =          OFF;              <== OFF ON
;
; End of "Grabber properties Section"

; *****
; *****
; End of File
; =====

```

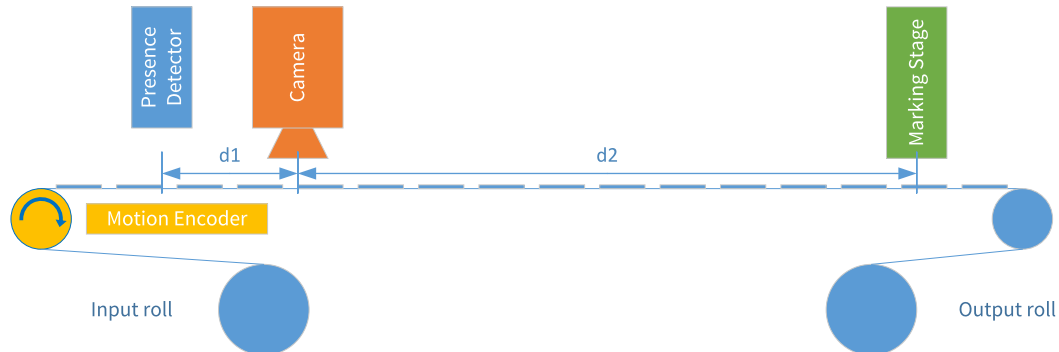
12.5. Machine Pipeline Control

Applies to: Base DualBase Full FullXR

Hardware controller for pipelined inspection machines

Pipe-lined Machine Description

Mechanical setup



The objects to inspect, for instance labels, are mounted on a tape. The tape is moving continuously from the **input roll** to the **output roll** at a variable speed. A **motion encoder**, typically a rotary quadrature motion encoder, is installed on a machine axis.

Every object successively passes under a **presence detector**, then under the **camera** and finally under a **marking stage** before exiting the machine.

The **presence detector** is installed upstream from the camera (distance **d1**).

The **camera** is a line-scan camera controlled by the frame grabber. It captures lines of image data at a rate proportional to the motion speed to avoid geometric distortions.

The **marking stage** is installed downstream from the camera (distance **d2**). It applies a "not good" mark on invalid objects. During the **d2** travel time, the application analyzes the object image and determines the action to be performed when it reaches the marking stage.

Normal operation mode

In normal operation mode, the machine executes the following operations for each object:

An object to inspect comes in from the input roll.

The presence detector delivers *trigger pulse* for each object

The Grablink frame grabber starts acquiring the first image line when the label reaches the camera field-of-view, i.e. after a position offset equal to **d1**.

Grablink stops acquiring after having captured a specified number of lines and wakes-up the application using the call-back mechanism when the image buffer are filled with the object image.

The application software analyzes the captured image and asserts a "set action" command specifying the action to execute when the object reaches the marking stage.

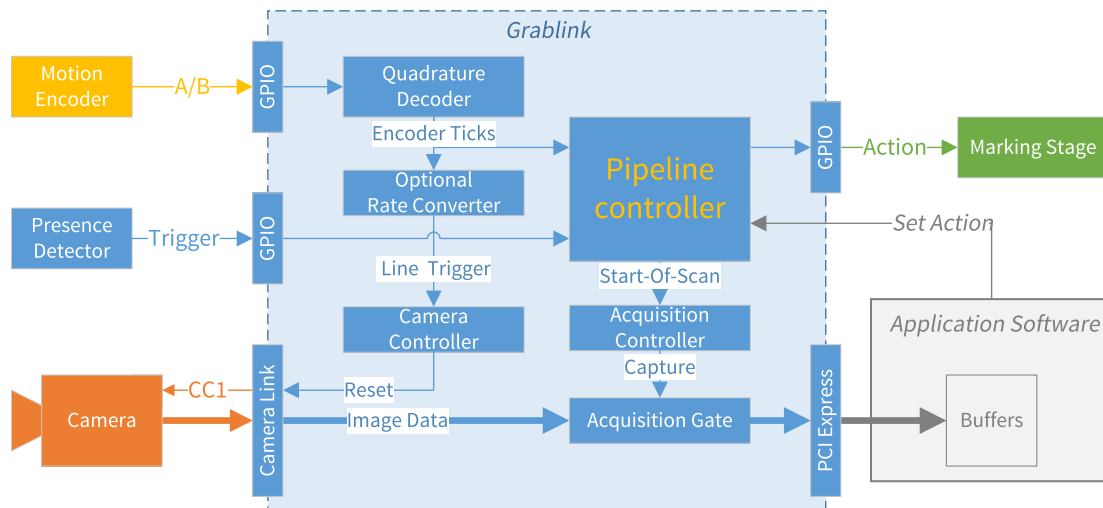
Grablink activates the marking stage, when an invalid object reaches the action stage, i.e. after a position offset equal to **d1** + **d2** .

The above operations are pipe-lined to optimize the machine throughput.

Pipeline Controller Description

Description of the Grablink implementation of the pipelined machine controller

Functional block diagram



System diagram for machine pipeline control

Camera controller

The camera controller is configured for line-scan grabber-controlled exposure mode: parameter **CamConfig** = **LxxxxRG**.

For every line trigger tick, the controller delivers a **RESET** pulse having a width equal to the exposure time set by parameter **Expose_us**. The **RESET** pulse is sent to the camera through the **Camera Link CC1** line.

Two options are applicable to select the line trigger source with the parameter **LineRateMode**:

- When set to **PULSE**, the encoder ticks are used directly as line triggers.
- When set to **CONVERT**, the rate converter output ticks are used as line triggers.

Rate converter

The rate converter is capable to multiply or divide the input rate of encoder ticks by a ratio of 2 integers defined by parameters **LinePitch** and **EncoderPitch**.

It operates within in a limited range of frequencies:

- The upper limit of the output frequency range is user configurable with parameter **MaxSpeed**. By default, it is set to the maximum line rate sustainable by the camera defined by **LineRate_Hz**.
- The lower limit of the output frequency takes is automatically set by the driver and reported to the application via the **MinSpeed** parameter. The **MaxSpeed/MinSpeed** ratio is typically greater than 100.

When the input rate drops below **MinSpeed**, the rate converter behaves according to the **OnMinSpeed** settings:

- When set to **MUTING**, it stops delivering line trigger ticks
- When set to **IDLING**, it continues delivering line trigger ticks at a constant frequency

To enlarge the usable speed range, it is mandatory to set **MaxSpeed** at a value slightly above the actual max camera line rate.

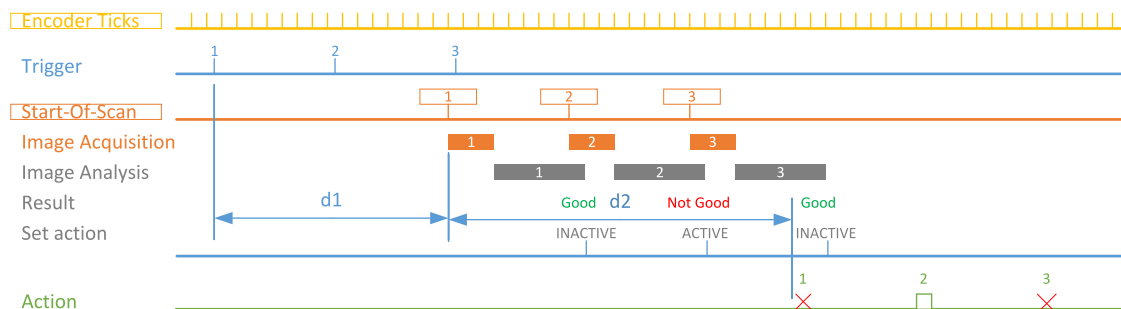
Pipeline controller

The pipeline controller is a piece of hardware responsible for the generation of two time-critical events:

- A **Start-of-Scan trigger** that initiates the image acquisition of one object when it reaches the camera.
- An **Action pulse** on a GPIO output port that triggers the marking stage.

The pipeline controller monitors the machine with two external sensors:

- A **motion encoder** that delivers encoder ticks at a rate proportional to the motion speed.
- A **presence detector** that delivers a trigger pulse for each object.



To generate the **Start-of-Scan trigger**, the pipeline controller delays the incoming **Trigger** by a configurable count of encoder ticks corresponding to the position offset between the detector and the camera field-of-view ($d1$).

Similarly, to generate the **Action pulse**, the pipeline controller delays the incoming **Trigger** by a configurable count of encoder ticks corresponding to the position offset between the detector and the marking stage ($d1 + d2$). The generation of the **Action pulse** is conditioned to the **Set action** command issued by the application software after image analysis.

The pipeline controller is capable of managing up to 32 objects in the pipeline between the detector and the marking stage.

Acquisition controller & acquisition gate

The acquisition controller is configured for PAGE or LONGPAGE operation modes using the **AcquisitionMode** parameter :

- When set to **PAGE**, one MultiCam surface is filled after each start-of-scan trigger.
- When set to **LONGPAGE**, several MultiCam surfaces are filled after each start-of-scan trigger.

In both cases, the acquisition controller is configured to start acquisition when the hardware trigger line senses a valid transition or on software command. Parameter **TrigMode** is set to **COMBINED** and parameter **NextTrigMode** is set to **COMBINED**).

Parameters **TrigCtl**, **TrigEdge** and **TrigFilter** specify the configuration of the hardware trigger input line. Parameter **TrigLine** specifies the location of a hardware trigger input line.

In **PAGE** mode, the acquisition controller stops after having acquired a number of lines specified by **PageLength_Ln**

In **LONGPAGE** mode, the acquisition controller is configured to stop acquisition automatically by setting parameter **EndTrigMode** to **AUTO**. Acquisition stops after having acquired a fixed number of lines specified by parameter **SeqLength_Ln**. The parameter **PageLength_Ln** specifies the maximum number of lines that can be stored in a buffer. The last buffer might be incompletely filled. The image data acquisition of an object starts always with a new buffer.

The **acquisition gate** opens and close at line boundaries according to the settings of the acquisition controller.

Packed RGB data acquired from the Camera Link interface are internally buffered and transmitted as soon as possible, by a DMA engine, to the application buffers in the Host PC memory via the PCI Express interface.

To minimize latencies, ensure that the PCI Express is capable of sustaining the image data rate with a comfortable margin.